Estimation of Random-Coefficient Dynamic Panel Data Models with a Fixed T*

Xun Tang† Pei Yu‡

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Abstract

We study dynamic linear panel data models in which the lagged outcomes and strictly exogenous covariates carry individual-specific coefficients and the time-varying errors have a flexible covariance structure. With a fixed number of time periods, we point-identify the joint distribution of the random coefficients and the structural errors under a distributional form of strict exogeneity, and propose a closed-form, multi-step estimator based on the inverse Radon transform. We establish a uniform convergence rate for the estimator of the random coefficient density, as well as uniform consistency of the estimator for the conditional density of the time-varying structural errors. Monte Carlo simulations demonstrate good finite-sample performance of the estimators.

Keywords: dynamic panel data; random coefficients; heterogeneous state dependence; fixed-T; inverse Radon transform

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†Department of Economics, Rice University. Email: xun.tang@rice.edu

‡Department of Economics, Rice University. Email: pei.yu@rice.edu

1 Introduction

Dynamic panel data models capture the persistence of economic outcomes, such as employment, earnings, or firm performance, by including lagged dependent variables among the covariates in structural equations. They are instrumental for disentangling "genuine" state dependence, or the direct impact of past outcomes on current ones, from "spurious" dependence caused by persistent unobserved heterogeneity across individuals. These distinct sources of dependence carry sharply different implications, because whether a transitory shock or a temporary policy intervention could leave a lasting imprint depends on the strength of true state dependence.

As such, dynamic panel models have become a workhorse in fields such as labor economics and industrial organization. Canonical applications include dynamic employment in Arellano and Bond (1991); Blundell and Bond (1998), production functions in Blundell and Bond (2000), corporate performance and competition in Nickell (1996), firm innovation in Aghion *et al.* (2005), market shares and stock valuation in Blundell *et al.* (1999), investments and financial factors in Bond and Meghir (1994); Bond *et al.* (2003).

Earlier econometrics literature had studied dynamic panel models with constant autoregressive coefficients where the state dependence is homogeneous across units. More recent works accommodated heterogeneous state dependence in workers, households, or firms by including random coefficients on the lagged dependent variables, and found it empirically salient in many settings, e.g., labor income dynamics in Fernández-Val *et al.* (2022) in a large-N, large-T setting.[1]

Bringing random coefficients to bear on dynamic panels raises two challenges that motivate this paper. The first is to let the lagged dependent variable carry a random coefficient in a short panel. Because the lagged outcome is predetermined rather than strictly exogenous, heterogeneous autoregressive dynamics are difficult to identify when the number of time periods T is small. Nonetheless, knowledge of the distribution of the autoregressive coefficient, rather than a single common value or its mean in the population, is needed for answering many questions. Examples include measuring the mass of units with non- or near-unit-root persistence (which governs long-run responses), and quantifying the spread and skewness of persistence (which affects the shape of the distribution of forecasts or policy targets). Whether large-T or fixed-T asymptotics is the

[1]Relatedly, Liu *et al.* (2020) and Lu *et al.* (2024) applied dynamic panel models with random coefficients on strictly exogenous covariates in the contexts of forecasting and program evaluation.

more suitable framework depends on the application and the data at hand, and neither approach dominates the other; our aim is to complement this developing literature by offering a methodological alternative that is valid when T is fixed.

The second challenge is to recover the joint distribution of the random coefficients *together with* the time-varying errors (random intercepts in the *structural* form), rather than selected moments or a marginal distribution. The distribution of random coefficients reveals how state dependence correlates with heterogeneous responses to the covariates, both of which are persistent; that of the random structural intercepts captures the distribution of time-varying shocks (permanent and transitory) underlying the process. Knowledge of their joint distribution is needed not only to separate and quantify the stochastic contribution by permanent vs transitory components, but also to simulate counterfactual trajectories in outcomes such as earnings or productivity. Means or low-order moments from their marginal distributions alone cannot answer these questions.

We study the dynamic linear panel data model

$$Y_{it} = \gamma_i Y_{it-1} + X_{it}'\beta_{it} + W_{it}'\delta_t + U_{it}, \qquad t = 1, \ldots, T,$$

where the autoregressive coefficient γ_i and the coefficients β_{it} on the strictly exogenous X_{it} are individual-specific, while the coefficients δ_t on the sequentially exogenous W_{it} are common across units i. The time-varying errors, or "random structural intercepts", U_{it}, absorb both fixed effects and transitory shocks (e.g., $U_{it} = \alpha_i + \varepsilon_{it}$). Our object of interest is the joint distribution of (γ_i, β_i, U_i) conditional on the initial Y_{i0}, along with δ_t.

The model identification requires a *distributional* form of strict exogeneity: (γ_i, β_i, U_i) are jointly independent of the exogenous covariate history conditional on Y_{i0} (Assumption 3.4). This condition permits flexible dependence among γ_i, β_i, and U_i, serial correlation in the errors, and dependence on Y_{i0}; it provides the source of variation to identify the full distribution of random coefficients and intercepts (rather than their means) in short panels with fixed T. We impose no restriction on the covariance structure of the time-varying errors, which may be serially correlated and heteroskedastic in the initial condition Y_{i0}. We show that both the joint distribution of (γ_i, β_i, U_i) given Y_{i0} and the coefficients for sequentially exogenous covariates δ_t are point identified.

Applying the analog principle to a constructive identification strategy, we propose a closed-form, multi-step estimator which does not involve any numerical optimization, simulation, or solution of an inverse problem beyond some deconvolution steps. We es-

tablish uniform convergence of our estimator for the joint density of (γ_i, β_i) and that of U_i conditional on the random coefficients.

Our analysis departs from the literature on three fronts, which we preview here and develop in Section 2. First, on the *model*: we let the coefficient on the lagged dependent variable be individual-specific and identify its distribution under a *fixed* T. This is the regime in which Arellano and Honoré (2001) showed that even the mean of a heterogeneous autoregressive coefficient is under-identified when the covariates are only sequentially exogenous. We circumvent this obstacle by bringing in strictly exogenous covariates under the distributional independence condition above, and we do so without any order condition relating T to the number of covariates, and without the large-T asymptotics that such panels often employ.

Second, on the *target*: we recover the joint distribution of the random coefficients and time-varying errors, rather than a mean, an average partial effect, or a finite set of moments of the random coefficients alone, which the nearest dynamic random-coefficient panel literature targets, often via second-moment conditions, averaging, or partial identification. In particular, we identify the distribution of the structural random intercepts, i.e., the time-varying errors in the *structural* equation, and not merely that of the slope coefficients. Triangular and simultaneous-equations models with random coefficients recover the coefficient distributions but leave that of the structural intercepts unidentified; we overcome this obstruction using a new argument that exploits the panel structure.

Third, on the *method*: we adapt the inverse Radon transform used to estimate random-coefficient densities in static cross-sections (Hoderlein *et al.*, 2010), coupled with a deconvolution step, and apply them to an intermediate reduced-form density implied by our panel. Recovering the structural objects from this reduced-form density is not immediate: it requires a Jacobian change of variables that extracts the joint density of (γ_i, β_i), and an original argument, central to the proof of Theorem 1, that identifies the distribution of U_i. These steps constitute a large part of our methodological contribution.

2 Related Literature

- **Dynamic linear panel data models with random coefficients.**

Several seminal papers estimated dynamic linear panel data models with *constant* coefficients under a fixed-T setting. These include Anderson and Hsiao (1982), Holtz-Eakin

et al. (1988), Arellano and Bond (1991), Arellano and Bover (1995), and Blundell and Bond (1998). When the autoregressive coefficient is instead heterogeneous, identification becomes markedly harder: Arellano and Honoré (2001) showed that with only sequentially exogenous covariates and fixed T, even the *mean* of that coefficient is under-identified. This motivates the strictly exogenous covariates and the distributional restriction we introduce in Section 3.2.

Some papers studied dynamic linear panel models where strictly exogenous covariates have random coefficients but predetermined and sequentially exogenous ones have *constant* coefficients. Arellano and Bonhomme (2012) identified distributional features of the random coefficients for strictly exogenous covariates using second-moment conditions on time-varying errors. In a forecasting framework, Liu *et al.* (2020) constructed point predictors for the posterior mean of heterogeneous coefficients of strictly exogenous covariates and deterministic trends, under the assumption of Gaussian innovations and using Tweedie's formula. We depart from these papers by letting the lagged dependent variable Y_{it-1} carry a random coefficient, by dispensing with the Gaussian or second-moment conditions on the time-varying errors, and by recovering the joint distribution of random coefficients and intercepts.

Chamberlain (2022) showed that in panel data models with sequentially exogenous covariates and a multi-dimensional vector of unobserved heterogeneity, the common finite-dimensional parameters (such as constant coefficients for other covariates) are not point identified. Lee (2026) studied the partial identification of a dynamic linear panel model with random coefficients for predetermined covariates and lagged dependent variables. In contrast, we obtain point (rather than partial) identification of the joint distribution of random coefficients and intercepts by exploiting the full distribution (rather than a finite set of moments) of the outcomes.

Within a framework of potential outcomes and treatment effects, Marx *et al.* (2024) obtain a causal interpretation of the classical IV/GMM estimands from the dynamic-panel literature (Anderson and Hsiao, 1982; Arellano and Bond, 1991) under a sequential exchangeability condition.[2] They also relax that condition and identify certain average causal effects by imposing homogeneous autoregressive dynamics on the *untreated* potential outcomes. Our paper differs from Marx *et al.* (2024) in two major ways. First, we

[2]Sequential exchangeability is a joint restriction on the dynamics of potential outcomes and on treatment selection. Marx *et al.* (2024) combine it with restrictions on treatment-effect heterogeneity to decompose the IV estimand into a convex-weighted aggregate of heterogeneous, history-dependent treatment effects.

study a distinct random-coefficient dynamic panel model and target a different parameter: the joint distribution of the random coefficients and time-varying errors in a model for *observed* outcomes, rather than aggregates of treatment effects in a *potential*-outcomes framework. Second, we leverage a different source of variation. Our main identification results (Section 3.2) hinge on a strong, distributional form of strict exogeneity of the covariates, whereas Marx *et al.* (2024) include no strictly exogenous covariate by design, and instead use lagged instruments to handle covariates that are not strictly exogenous (e.g., treatments that depend on past observed outcomes).

- **Multidimensional unobserved heterogeneity in linear panels.**

Other papers investigated linear panel models with vector, interactive, or correlated heterogeneity, typically focusing on the mean (instead of distribution) of such heterogeneity. Graham and Powell (2012) estimated average partial effects (or mean of random coefficients) in an "irregular" correlated random coefficient panel, using units whose covariates change little over time and requiring an order condition for just-identification (i.e., the number of periods T equals the dimension of covariates). Apart from the difference in the target parameter, their method does not apply in our case because the mean of composite random intercepts is not additive in a stationary function of the covariate history and a period-specific constant due to sequentially exogenous covariates. Moreover, our method does not require any order condition on the number of time periods.

Moon and Weidner (2017) estimated a dynamic linear panel model that has a lagged dependent variable with a constant coefficient and interactive fixed effects. Lu and Su (2023, 2025) studied linear panel models that allowed for random coefficients for sequentially exogenous covariates, but their theoretical frameworks do not include lagged dependent variables. Cao *et al.* (2024) allowed for random coefficients on lagged dependent variables, but required a "large-N, large-T" setting where both the cross-section and time dimensions go to infinity. In contrast with these papers, our method operates in a short panel "fixed-T" setting. Furthermore, while these papers focus on estimating the mean (or a homogeneous/average partial effect) of random slope coefficients, we recover the joint distribution of the random coefficients on lagged dependent variables alongside time-varying errors.

Earlier and wider literature on random-coefficient panel models likewise targeted averages or low-order moments in static designs. Examples include Swamy (1970), Chamberlain (1992), Wooldridge (2005), Murtazashvili and Wooldridge (2008), and Hsiao and

Pesaran (2008). Laage (2024) estimated a correlated random-coefficient panel model with time-varying endogeneity, showing identification of the mean of random coefficients through control variables. Her model does not accommodate random coefficients for a lagged dependent variable. Li (2026) estimated the average partial effect and the local average response in a correlated random coefficient panel data model, where regressors can be correlated with time-varying and individual-specific random coefficients.

- **Triangular and simultaneous systems with random coefficients.**

Our dynamic linear panel model shares the structure of a triangular system, because Y_{i1} feeds into Y_{i2} as a predetermined regressor. Yet our paper differs from those on the triangular system, in terms of the target and the identification strategy. We review the core differences here, and relegate details of the comparison to Appendix C.

Hoderlein *et al.* (2017) identified the distribution of the random coefficients in a triangular model under two key restrictions: mutual independence between the coefficients and the instruments and exogenous covariates, and independence of the first-stage slopes from the outcome-equation coefficients (their Theorem 7). As we show in Appendix C, these conditions are relaxed in our setting, and we additionally identify the distribution of random intercepts (time-varying errors) rather than that of the slope coefficients alone.

Masten and Torgovitsky (2016) identified an instrumental-variables correlated random coefficient model, relying on independence between the instruments and the coefficients together with a monotone relation between the endogenous covariate and a scalar latent control. Our model does not fit their control-function framework: the first-stage equation is itself of random-coefficient form, which, as Imbens (2007) and Kasy (2011) showed, admits no such reduced-form, monotone representation with a scalar control.

Masten (2018) treated triangular systems as a special case of simultaneous equations. He identified the joint distribution of the random coefficients for the endogenous regressor and the excluded instrument (his Proposition 4, Section 3.3), assuming the instrument is independent of all unobservables conditional on the shared covariates. His analysis did not identify the distribution of the time-varying errors (random intercepts), nor that of the random coefficients on the covariates shared in both equations.[3] In comparison, our panel setting avoids this obstruction, because each period supplies its own exogenous covariates: a suitable linear combination of the outcomes behaves as a single-equation

[3]Indeed, Masten (2018) showed the joint distribution of all structural unobservables is not point identified, because the reduced form is a seemingly-unrelated-regression system with common regressors (his Theorem 4).

random-coefficient model in *distinct* regressors. Building on this, and varying the combination weights, we obtain a new result—identification of the full joint distribution of all random coefficients *and* the time-varying errors. This requires an extended argument that exploits the conditional independence between the exogenous covariates and the random coefficients and intercepts given the initial condition.

- **Estimation of linear RC models via inverse Radon transform.**

Beran *et al.* (1996) and Hoderlein *et al.* (2010) estimated the joint density of the random coefficients in static cross-section regressions by inverting the Radon transform (which links the conditional density of the outcome to the coefficient density); the latter did so through a kernel estimator with a special Radon-transform kernel and sharp asymptotics. We use this method for the first step in our estimation.

3 The Model and Identification

We consider a dynamic panel data model:

$$Y_{it} = \gamma_i Y_{it-1} + X_{it}'\beta_{it} + W_{it}'\delta_t + U_{it} \quad \text{for } t = 1, 2, \ldots, T, \tag{1}$$

where $\gamma_i \in \mathbb{R}$ and $\beta_{it} \in \mathbb{R}^{J_x}$ are random coefficients, $U_{it} \in \mathbb{R}$ are random intercepts, while $\delta_t \in \mathbb{R}^{J_w}$ are non-zero constant coefficients. There are no constant intercepts in X_{it}, W_{it}. For a generic random array ζ_{it}, let $\zeta_i \equiv (\zeta_{i1}', \zeta_{i2}')'$ and $\zeta_i^t \equiv \{\zeta_{is} : s = 1, \ldots, t\}$ denote the history up to time t, and $\Delta\zeta_{it} \equiv \zeta_{it} - \zeta_{it-1}$.

Assumption 3.1. *(Strict and Sequential Exogeneity with Time Effects.) For* $t = 1, \ldots, T$*,*

$$E(U_{it} \mid X_i^T, W_i^t, Y_{i0}, \gamma_i) = \mu_t(Y_{i0}) < \infty \quad \textit{almost surely,}$$

where $\mu_t(\cdot)$ *is an unknown, unrestricted function.*

Under this condition, X_{it} is strictly exogenous while W_{it} is sequentially exogenous. It permits period-specific fixed effects $\mu_t(Y_{i0})$. Arellano and Bonhomme (2012) investigated a similar specification where the coefficients for strictly exogenous covariates are random but those for predetermined or sequentially exogenous covariates are all *constant*, and $\mu_t(Y_{i0}) = 0$ for all t. In comparison, our specification differs by allowing a predetermined regressor, namely, the lagged dependent variable Y_{it-1}, to have a random coefficient.

While Assumption 3.1 restricts the conditional mean of U_{it} given γ_i, Y_{i0}, it does allow dependence between U_i and γ_i through higher moments (as in the correlated and scale designs in our simulation study).

The first part of our identification results (Section 3.1) does not require conditions on the second moments of time-varying errors used by Arellano and Bonhomme (2012). On the other hand, our method does rely on a stronger notion of strict exogeneity, i.e., the distributional independence in Assumption 3.4 instead of mean independence (Assumption 1) in Arellano and Bonhomme (2012).

The parameters of interest are the joint distribution of random coefficients and intercepts $(\gamma_i, \beta_{i1}', \ldots, \beta_{iT}', U_{i1}, \ldots, U_{iT})$ and the constant coefficients $(\delta_1, \ldots, \delta_T)$. We prove point identification of these parameters via sequential steps.

3.1 The conditional mean of random coefficients

First, we identify the constant coefficients δ_t for $t = 1, \ldots, T$ and the conditional mean of random coefficients. To fix ideas, focus on the case with $T = 2$. By recursive substitution and first-differencing,

$$\begin{aligned}\Delta Y_{i2} &\equiv Y_{i2} - Y_{i1} \\ &= \gamma_i(\gamma_i - 1)Y_{i0} + (\gamma_i - 1)X_{i1}'\beta_{i1} + X_{i2}'\beta_{i2} + (\gamma_i - 1)W_{i1}'\delta_1 + W_{i2}'\delta_2 + \varpi_i,\end{aligned}$$

where $\varpi_i \equiv \Delta U_{i2} + \gamma_i U_{i1}$. Let H_i be shorthand for the covariates that are exogenous with respect to $U_i \equiv (U_{i1}, U_{i2})'$ conditional on the initial condition Y_{i0}. That is:

$$H_i \equiv (X_{i1}', X_{i2}', W_{i1}')'.$$

We also maintain a conditional mean independence condition on the random coefficients. Let $\theta_i \equiv (\gamma_i, \beta_i')'$ with $\beta_i \equiv (\beta_{i1}', \beta_{i2}')'$.

Assumption 3.2. *$E(\theta_i \mid H_i, Y_{i0}) = E(\theta_i \mid Y_{i0})$ and $E(\theta_i\gamma_i \mid H_i, Y_{i0}) = E(\theta_i\gamma_i \mid Y_{i0})$.*

Under Assumptions 3.1 and 3.2,

$$E(\varpi_i \mid H_i, Y_{i0}) = E\big[\gamma_i\mu_1(Y_{i0}) + \Delta\mu(Y_{i0}) \mid H_i, Y_{i0}\big] = m(Y_{i0}),$$

where $m(Y_{i0}) \equiv E(\gamma_i \mid Y_{i0})\mu_1(Y_{i0}) + \Delta\mu(Y_{i0})$ with $\Delta\mu \equiv \mu_2 - \mu_1$, and

$$E(Y_{i1} \mid H_i, Y_{i0}) = \pi_1(Y_{i0}) + X_{i1}' E(\beta_{i1} \mid Y_{i0}) + W_{i1}'\delta_1, \tag{2}$$

with $\pi_1(Y_{i0}) \equiv E(\gamma_i \mid Y_{i0})Y_{i0} + \mu_1(Y_{i0})$, and

$$E(\Delta Y_{i2} \mid H_i, Y_{i0}) = Z_i'\Psi(Y_{i0}), \tag{3}$$

where $Z_i \equiv (\,1, H_i', E(W_{i2}' \mid H_i, Y_{i0})\,)'$ and

$$\Psi(Y_{i0}) \equiv \begin{pmatrix} \psi_1(Y_{i0}) \\ \psi_2(Y_{i0}) \\ \psi_3(Y_{i0}) \\ \psi_4(Y_{i0}) \\ \psi_5(Y_{i0}) \end{pmatrix} \equiv \begin{pmatrix} E(\gamma_i^2 - \gamma_i \mid Y_{i0})Y_{i0} + m(Y_{i0}) \\ E[\beta_{i1}(\gamma_i - 1) \mid Y_{i0}] \\ E(\beta_{i2} \mid Y_{i0}) \\ \delta_1 E(\gamma_i - 1 \mid Y_{i0}) \\ \delta_2 \end{pmatrix}. \tag{4}$$

That is, the means of Y_{i1} and ΔY_{i2} conditional on H_i, Y_{i0} are both linear in Z_i. We maintain the following condition on the (conditional) support of Z_i.

Assumption 3.3. *(Non-singularity.) $E(\,\|Z_i\|^2 \mid Y_{i0}\,) < \infty$, and the support of Z_i conditional on Y_{i0} is not contained in any proper linear subspace of $\mathbb{R}^{2(J_x+J_w)+1}$ almost surely.*

A sufficient condition for Assumption 3.3 is that for any nonzero $c \in \mathbb{R}^{J_w}$, $c'E(W_{i2} \mid H_i, Y_{i0})$ is not almost surely affine in H_i given Y_{i0}. That is, nonlinearity in the conditional mean of W_{i2} is essential. Under Assumption 3.3, both $E(Z_iZ_i' \mid Y_{i0})$ and $E(R_iR_i' \mid Y_{i0})$ are non-singular almost surely, where $R_i \equiv (1, X_{i1}', W_{i1}')'$ is a sub-vector of Z_i. Therefore, we can identify $\pi_1(Y_{i0})$, $E(\beta_{i1} \mid Y_{i0})$, and δ_1 from (2) using variation in R_i given Y_{i0}, and identify $\Psi(Y_{i0})$ from (3) using variation in Z_i given Y_{i0}. It then follows from (4) that

$$\begin{aligned} &E(\beta_{i2} \mid Y_{i0}) = \psi_3(Y_{i0}), \quad \delta_2 = \psi_5(Y_{i0}), \quad E(\gamma_i \mid Y_{i0}) = \delta_1'\psi_4(Y_{i0})/\|\delta_1\|^2 + 1, \\ &\mu_1(Y_{i0}) = \pi_1(Y_{i0}) - E(\gamma_i \mid Y_{i0})Y_{i0}, \quad E(\gamma_i\beta_{i1} \mid Y_{i0}) = \psi_2(Y_{i0}) + E(\beta_{i1} \mid Y_{i0}). \end{aligned} \tag{5}$$

The proposition below collects these identification results based on the mean of Y_{it}.

Proposition 1. *Under Assumptions 3.1, 3.2 and 3.3, δ_1, δ_2, $E(\beta_{i1} \mid Y_{i0})$, $E(\beta_{i2} \mid Y_{i0})$, $E(\gamma_i \mid Y_{i0})$, $E(\gamma_i\beta_{i1} \mid Y_{i0})$ and $\mu_1(Y_{i0})$ are identified from $E(Y_{it} \mid H_i, Y_{i0})$ for $t = 1, 2$.*

Remark 1. *The coefficients for sequentially exogenous covariates δ_1, δ_2 are all we need*

for the next step in Section 3.2. Nevertheless, we note the identification of the other parameters in Proposition 1 is also useful, as they are obtained from conditional mean outcomes without invoking the distributional form of strict exogeneity in Assumption 3.4.

Remark 2. *The second moment $E(\gamma_i^2 \mid Y_{i0})$ and $\mu_2(Y_{i0})$ enter additively in $\psi_1(Y_{i0})$, and cannot be separately identified from the conditional means without further assumptions. We propose two approaches to identify them separately. The first is to use a stationarity condition that $\mu_t(Y_{i0})$ is identical over $t = 1, 2$ almost surely. This allows us to recover $E(\gamma_i^2 \mid Y_{i0})$ from $\psi_1(Y_{i0})$ using knowledge of parameters identified in Proposition 1. The second approach is to strengthen the mean independence of random intercepts and coefficients in Assumptions 3.1 and 3.2 to a stronger form of distributional independence (Assumption 3.4), which allows us to recover the distribution (and the second moment) of γ_i given Y_{i0} as in Theorem 1. We can then recover $\mu_2(Y_{i0})$ from $\psi_1(Y_{i0})$, again using results from Proposition 1.*

Remark 3. *If we strengthen Assumption 3.2 with $E(\theta_i \mid Y_{i0}) = E(\theta_i)$, $E(\gamma_i^2 \mid Y_{i0}) = E(\gamma_i^2)$, and $\mu_t(Y_{i0}) = \mu_t$, then the conditional means of Y_{i1} and ΔY_{i2} are both* unconditionally *linear in H_i* and *Y_{i0}. The unconditional expectations of U_i, γ_i, β_i as well as δ_1, δ_2 are identified from simple regressions of $Y_{i1}, \Delta Y_{i2}$ that pool over Y_{i0} and include it as a regressor along with H_i.*

Remark 4. *Our method does not need T to be at least as large as the dimension of strictly exogenous covariates; this differs from Arellano and Bonhomme (2012) (with T strictly larger than the latter) and Graham and Powell (2012) (with T equal to the latter). Furthermore, our model differs from these two papers by allowing the distribution of the random coefficients for strictly exogenous covariates to vary over time, and not to be additive in a stationary function of covariate history. (See Appendix C for details.)*

Proposition 1 uses conditional mean outcomes for identification, and invites comparison with Section 3 of Marx *et al.* (2024). Beyond the difference in target parameters, their identifying assumptions and ours are non-nested. To compare the two, map their treatment D_{it} into our sequentially exogenous covariate W_{it}, and their composite error $\theta_t + \alpha_i + \varepsilon_{it}$ into our random intercept U_{it}. First, Marx *et al.* (2024) focus on the IV/GMM estimand and include *no* strictly exogenous covariate, whereas we require at least one. Second, sequential exchangeability in Marx *et al.* (2024) permits past outcomes to feed into the current treatment, carrying information about the fixed effects beyond Y_{i0} and resulting in $E(U_{i2} \mid D_{i1}, D_{i2}, Y_{i0}) \neq \mu_2(Y_{i0})$ in general, thus violating our Assumption 3.1.

Moreover, the two approaches draw on different variation: we exploit the (conditional) linearity of $E(Y_{it} \mid H_i, Y_{i0})$ in exogenous covariates, whereas Marx *et al.* (2024) use lagged outcomes as instruments in a first-differenced equation.

3.2 Distribution of random coefficients and intercept

Next, we identify the joint distribution of random coefficients and intercepts, using a stronger form of conditional independence. Throughout this section, we treat δ_1, δ_2 as known, having been identified in Proposition 1.

Assumption 3.4. *$(\gamma_i, \beta_i, U_i) \perp H_i \mid Y_{i0}$, and the conditional distribution of (γ_i, β_i, U_i) given Y_{i0} admits a Lebesgue density almost surely.*

Assumption 3.4 allows flexible dependence among γ_i, β_i and U_i, and is compatible with Assumptions 3.1 and 3.2. It also allows the distribution of random coefficients and intercepts to be heterogeneous in the initial condition Y_{i0}. In the special case with additive fixed effects $U_{it} = \alpha_i + \varepsilon_{it}$, Assumption 3.4 accommodates general dependence between $Y_{i0}, \gamma_i, \beta_i, \alpha_i$ and $\varepsilon_i \equiv (\varepsilon_{i1}, \varepsilon_{i2})$, as well as serial correlation of ε_{it}.

Define $\widetilde{Y}_{it} \equiv Y_{it} - W_{it}'\delta_t$ for $t = 1, 2$. For any $d \equiv (d_1, d_2) \in \mathbb{R}^2$,

$$d_1\widetilde{Y}_{i1} + d_2\widetilde{Y}_{i2} = C_i(d) + X_{i1}'\underbrace{(\beta_{i1}d_1 + \beta_{i1}\gamma_i d_2)}_{S_{i1}(d)} + X_{i2}'\underbrace{\beta_{i2}d_2}_{S_{i2}(d)} + W_{i1}'\underbrace{\delta_1\gamma_i d_2}_{S_{i3}(d)}, \tag{6}$$

where

$$C_i(d) \equiv (d_1\gamma_i + d_2\gamma_i^2)Y_{i0} + (d_1 + d_2\gamma_i)U_{i1} + d_2U_{i2}.$$

Let $S_i(d) \equiv (S_{ik}(d) : k = 1, 2, 3)$. Under Assumption 3.4,

$$(\, C_i(d), S_i(d)\,) \perp H_i \mid Y_{i0} \text{ for any } d \in \mathbb{R}^2. \tag{7}$$

The *first* step in identifying the distribution of random coefficients is to recover the joint distribution of $(\, C_i(d), S_i(d)\,)$ conditional on Y_{i0}.

Assumption 3.5. *(i) The support of $H_i \mid Y_{i0}$ contains an open ball in $\mathbb{R}^{2J_x+J_w}$ almost surely.*

(ii) For every $d \in \mathbb{R}^2$, the conditional distribution of $(C_i(d), S_i(d))$ given Y_{i0} is uniquely determined by its moments, and has finite absolute moments of all orders almost

surely.

For linear regressions with random coefficients and intercept in reduced form, Masten (2018) (Lemma 2) showed that the moment-determinacy and finite-moment conditions such as (ii) are sufficient for point-identification, provided these are independent of the regressors whose joint support satisfies an open-ball condition as in (i).[4]

A primitive sufficient condition for (ii) is that, conditional on Y_{i0}, the components of (γ_i, β_i, U_i) have sub-Gaussian tails, which permits them to have non-compact support. Every linear combination of $(C_i(d), S_i(d))$, whose entries include the quadratic term $\gamma_i^2 Y_{i0}$ as well as the products $\gamma_i U_{i1}$ and $\gamma_i \beta_{i1}$, is then sub-exponential and hence satisfies Carleman's condition, ensuring moment determinacy of the joint conditional distribution for every $d \in \mathbb{R}^2$ (Petersen, 1982).

The following lemma follows immediately from the conditional independence we established in (7), Assumption 3.5, and Lemma 2 of Masten (2018).

Lemma 3.1. *Suppose Assumptions 3.4 and 3.5 hold. For all $d \in \mathbb{R}^2$, the distribution of $(C_i(d), S_i(d))$ given Y_{i0} is identified from that of $(d_1\widetilde{Y}_{i1} + d_2\widetilde{Y}_{i2}, H_i')$ given Y_{i0}.*

The *second* step is to recover the joint density of $(C_i(d), \gamma_i, \beta_i)$ given Y_{i0} from that of $(C_i(d), S_i(d))$ using Jacobian transformation. Specifically, choose $j \in \{1, \ldots, J_w\}$ so that the j-th component of δ_1, denoted by $\delta_{1,j}$, is nonzero, and define a Jacobian matrix:

$$J_i(d) \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & d_2\beta_{i1} & (d_1 + d_2\gamma_i)I & 0 \\ 0 & 0 & 0 & d_2 I \\ 0 & d_2\delta_{1,j} & 0 & 0 \end{pmatrix}, \quad \text{where } I \text{ is a } J_x\text{-by-}J_x \text{ identity matrix.} \tag{8}$$

Suppose for d with $d_2 \neq 0$, the determinant of $J_i(d)$ is non-zero almost surely. (Because the conditional distribution of γ_i given Y_{i0} is atomless under Assumption 3.4, $\Pr\{d_1 + d_2\gamma_i = 0 \mid Y_{i0}\} = 0$ for any $d_2 \neq 0$ almost surely.) Recover the joint density of $C_i(d), \gamma_i, \beta_i$

[4]The generic pair (A, B) and regressors Z in Masten (2018) correspond to $(C_i(d), S_i(d))$ and H_i in our case, respectively. Masten (2018) also showed that if $\text{supp}(H_i \mid Y_{i0})$ is bounded, then the conditions in (ii) are necessary for identification.

conditional on $Y_{i0} = y_0$ as:

$$\begin{aligned} &f_{C_i(d),\gamma_i,\beta_{i1},\beta_{i2}|Y_{i0}=y_0}(c,\gamma,b_1,b_2) \\ =&f_{C_i(d),S_{i1}(d),S_{i2}(d),S_{i3,j}(d)|Y_{i0}=y_0}(c,b_1(d_1+d_2\gamma),b_2d_2,d_2\gamma\delta_{1,j}) \times |\delta_{1,j}d_2^{J_x+1}(d_1+d_2\gamma)^{J_x}|. \end{aligned} \tag{9}$$

This identifies the joint distribution of (γ_i, β_i) given Y_{i0}.

In the *third* step, for any $a \equiv (a_1, a_2) \in \mathbb{R}^2$ and realization of random coefficients γ, define the map $\bar{d}(a, \gamma) \equiv (a_1 - a_2\gamma,\, a_2)$, with components $\bar{d}_1 \equiv a_1 - a_2\gamma$ and $\bar{d}_2 \equiv a_2$. Then for any $b \equiv (b_1', b_2')'$ and initial condition y_0,

$$\begin{aligned} &C_i(\bar{d}) \mid \gamma_i = \gamma, \beta_i = b, Y_{i0} = y_0 \\ \sim\ &(\bar{d}_1 + \bar{d}_2\gamma)U_{i1} + \bar{d}_2U_{i2} + \psi^* \mid \gamma_i = \gamma, \beta_i = b, Y_{i0} = y_0 \\ \sim\ &a_1U_{i1} + a_2U_{i2} + \psi^* \mid \gamma_i = \gamma, \beta_i = b, Y_{i0} = y_0, \end{aligned} \tag{10}$$

where $\psi^* \equiv (\bar{d}_1\gamma + \bar{d}_2\gamma^2)y_0 = \gamma a_1 y_0$ is a known constant.

Lastly, combine the three steps above and note for any non-zero $a_1, a_2 \in \mathbb{R}$, the distribution of $a_1U_{i1} + a_2U_{i2}$ conditional on $(\gamma_i, \beta_i, Y_{i0})$ is identified from that of $(\widetilde{Y}_{i1}, \widetilde{Y}_{i2}, H_i')$ given Y_{i0}. Therefore, the characteristic function (and the distribution) of $U_i = (U_{i1}, U_{i2})'$ given $(\gamma_i, \beta_i, Y_{i0})$ is identified.[5] In particular, the intercept means $\mu_t(Y_{i0}) = E(U_{it} \mid Y_{i0})$ and $E(\gamma_i^2 \mid Y_{i0})$—left non-separable by the conditional-mean step—are recovered here as features of this joint law. The next theorem formalizes the identification result.

Theorem 1. *Suppose Assumptions 3.1, 3.2, 3.3, 3.4 and 3.5 hold. Then δ_1, δ_2 and the joint distribution of (U_i, γ_i, β_i) given Y_{i0} are identified.*

Remark 5. *Theorem 1 uses Assumption 3.1–3.3 only to identify δ_1, δ_2 via Proposition 1. Given δ_1, δ_2, the distributional argument rests on Assumption 3.4–3.5 alone. Assumption 3.2 is implied by Assumption 3.4 (as long as the conditional means exist).*

The identification strategy above is constructive; we apply the analog principle to define closed-form estimators in the next section.

In Appendix B, we extend this identification strategy to any fixed $T \geq 3$ even without variation in sequentially exogenous W_{it}.

[5] We show identification of the characteristic function over nonzero (a_1, a_2). Recovery of the characteristic function at a s.t. $a_1a_2 = 0$ follows from uniform continuity of characteristic functions.

4 Closed-Form Estimators

We propose multi-step estimators for δ_1, δ_2 and the joint density of (γ_i, β_i, U_i). The identification results are conditional on the initial condition Y_{i0}; we fix it at a constant $y_0 \neq 0$ throughout this section, and suppress it from the notation. As in Section 3, we focus on the case with $T = 2$.

4.1 Moments of random coefficients and δ_t

Let $T = 2$ and use lower-case letters to denote the realizations in the sample, e.g., $h_i \equiv (x_{i1}', x_{i2}', w_{i1}')'$. Let $\widehat{w}_{i2}$ denote a Nadaraya-Watson estimator for the mean of W_{i2} conditional on h_i; let $\widehat{z}_i \equiv (1, h_i', \widehat{w}_{i2}')'$, and

$$\widehat{\Psi} \equiv \left(\sum\nolimits_i \widehat{z}_i \widehat{z}_i'\right)^{-1} \left(\sum\nolimits_i \widehat{z}_i \Delta y_{i2}\right),$$

where the k-th component $\widehat{\psi}_k$ is an estimator of ψ_k in (4), with the initial condition y_0 fixed and suppressed in notation. Let $r_i \equiv (1, x_{i1}', w_{i1}')'$ and estimate the period-1 level regression (2) by

$$\widehat{\Phi} \equiv \left(\sum\nolimits_i r_i r_i'\right)^{-1} \left(\sum\nolimits_i r_i \, y_{i1}\right) = (\widehat{\pi}_1, \ \widehat{E}(\beta_{i1})', \ \widehat{\delta}_1')'. \tag{11}$$

Together with $\widehat{\Psi}$ from the regression on ΔY_{i2}, we also estimate

$$\widehat{E}(\beta_{i2}) \equiv \widehat{\psi}_3, \quad \widehat{\delta}_2 \equiv \widehat{\psi}_5, \quad \widehat{E}(\gamma_i) \equiv 1 + \widehat{\delta}_1' \widehat{\psi}_4 / \|\widehat{\delta}_1\|^2. \tag{12}$$

We can also estimate the following parameters (even though they are not needed for subsequent estimation of the joint density of $(\gamma_i, \beta_i', U_i')'$):

$$\widehat{\mu}_1 \equiv \widehat{\pi}_1 - \widehat{E}(\gamma_i)\, y_0, \quad \widehat{E}(\gamma_i^2) \equiv \int \gamma^2 \, \widehat{f}_{\gamma_i}(\gamma)\, d\gamma, \quad \widehat{E}(\gamma_i \beta_{i1}) \equiv \widehat{\psi}_2 + \widehat{E}(\beta_{i1}),$$
$$\widehat{\mu}_2 \equiv \widehat{\psi}_1 - \big(\widehat{E}(\gamma_i^2) - \widehat{E}(\gamma_i)\big) y_0 - \widehat{E}(\gamma_i)\widehat{\mu}_1 + \widehat{\mu}_1,$$

where $\widehat{f}_{\gamma_i}(\gamma) \equiv \int \widehat{f}_{\gamma_i, \beta_i}(\gamma, b)\, db$ uses the density estimator of Section 4.2.

With $\delta_1 \neq 0$, the probability limit of the denominator $\|\widehat{\delta}_1\|^2$ in $\widehat{E}(\gamma_i)$ is strictly bounded away from zero. Consequently, a standard argument using Slutsky's Theorem ensures that division by $\|\widehat{\delta}_1\|^2$ does not affect the $\sqrt{n}$-consistency and asymptotic normality of the estimators in (11) and (12).

4.2 The joint density of random coefficients

For the rest of this section and Section 5, assume $W_{i1} \in \mathbb{R}$ for simplicity. Fix $d = (0, 1)$, so that (6) reduces to

$$\widetilde{Y}_{i2} = \underbrace{\gamma_i^2 y_0 + \gamma_i U_{i1} + U_{i2}}_{C_i^*} + X_{i1}' \underbrace{\beta_{i1}\gamma_i}_{S_{i1}^*} + X_{i2}' \underbrace{\beta_{i2}}_{S_{i2}^*} + W_{i1}' \underbrace{\delta_1 \gamma_i}_{S_{i3}^*}.$$

Following Hoderlein *et al.* (2010), we define an (inverse) Radon transform estimator (RTE) for the density of (C_i^*, S_i^*).

Recall $H_i \equiv (X_{i1}', X_{i2}', W_{i1}')'$. For $i = 1, \ldots, n$, define

$$Q_i \equiv \|(1, H_i')\|^{-1}(1, H_i')' \in \mathbb{S}_+^{l-1} \quad \text{and} \quad V_i \equiv \|(1, H_i')\|^{-1}\left(Y_{i2} - W_{i2}'\widehat{\delta}_2\right) \in \mathbb{R},$$

where $\|\cdot\|$ denotes the Euclidean norm, and $\mathbb{S}_+^{l-1} = \{z \in \mathbb{R}^l : z_1 > 0, \|z\| = 1\}$ is the upper hemisphere of the unit sphere in $\mathbb{R}^l$ with $l \equiv \dim[(1, H_i')] = 2J_x + 2$.

Define $\mathbb{S}(\underline{q}_n) \equiv \{q \in \mathbb{S}_+^{l-1} : q_1 \geq \underline{q}_n\}$, where $\underline{q}_n \to 0$ as $n \to \infty$. Estimate the density of (C_i^*, S_i^*) by

$$\widehat{f}_{C_i^*, S_i^*}(c, s) \equiv \frac{2}{n}\sum\nolimits_i \frac{\mathbf{1}\{q_{i1} \geq \underline{q}_n\}}{\widehat{f}_{Q_i}(q_i)} K_\nu(q_i'(c, s')' - v_i), \tag{13}$$

where $\mathbf{1}(\cdot)$ is the indicator function, and K_ν is the kernel defined by Equations (7)–(8) of Bissantz *et al.* (2014).[6] The denominator is a kernel estimator of the density of Q_i on its spherical support:

$$\widehat{f}_{Q_i}(\tilde{q}) \equiv \frac{1}{n}\sum\nolimits_i \varkappa(\tau)\widetilde{K}\left(\tau^{-2}(1 - \tilde{q}'q_i)\right),$$

where $\widetilde{K}(\cdot)$ is a kernel function, $\tau > 0$ is a smoothing parameter, and $\varkappa(\tau)$ is a normalization constant defined in Hoderlein *et al.* (2010).

Estimate the density of $(C_i^*, \gamma_i, \beta_i)$ by

$$\widehat{f}_{C_i^*, \gamma_i, \beta_i}(c, \gamma, b) \equiv \widehat{f}_{C_i^*, S_i^*}(c, b_1\gamma, b_2, \widehat{\delta}_1\gamma)|\widehat{\delta}_1\gamma^{J_x}|,$$

[6] The kernel is defined through its Fourier transform $\mathcal{F}K_\nu(t) = \frac{1}{2}(2\pi)^{-l+1}|t|^{l-1}\mathcal{L}(\nu|t|)$, which by symmetry of $\mathcal{L}$ admits an explicit form: $K_\nu(w) \equiv (2\pi)^{-l}\int_0^\infty \cos(tw)\, t^{l-1}\mathcal{L}(\nu t)\, dt$. The symmetric filter is $\mathcal{L}(t) \equiv (1 - |t|^r)\mathbf{1}_{[-1,1]}(t)$, with an order parameter $0 < r < \infty$. The kernel K_ν can be written, via a change of variable $\tilde{t} = \nu t$, as $K_\nu(w) = \nu^{-l}(2\pi)^{-l}\int_0^\infty \cos\left(\tilde{t}\,\frac{w}{\nu}\right)\tilde{t}^{l-1}\mathcal{L}(\tilde{t})\, d\tilde{t} = \nu^{-l}K(\tilde{w})\big|_{\tilde{w}=w/\nu}$, where $K(\tilde{w}) \equiv (2\pi)^{-l}\int_0^\infty \cos(t\tilde{w})\, t^{l-1}\mathcal{L}(t)\, dt$.

and estimate the density of (γ_i, β_i) by

$$\widehat{f}_{\gamma_i,\beta_i}(\gamma, b) \equiv \int_{\mathcal{C}_n} \widehat{f}_{C_i^*,\gamma_i,\beta_i}(c, \gamma, b)\, dc,$$

where $\mathcal{C}_n \equiv [-\bar{c}_n, \bar{c}_n] \subset \mathbb{R}$, with $\bar{c}_n \to \infty$ as $n \to \infty$.

4.3 The joint density of U_i conditional on random coefficients

For any $a = (a_1, a_2) \in \mathbb{R}^2$ and any given γ on the support of γ_i, define $\bar{d} \equiv (a_1 - a_2\gamma, a_2)$. With slight abuse of notation, write $C_i(\bar{d}) = C_i(a, \gamma)$ and $S_i(\bar{d}) = S_i(a, \gamma)$. Define:

$$V_i(a, \gamma) \equiv \|(1, H_i')\|^{-1}[\ (a_1 - \gamma a_2)\widehat{Y}_{i1} + a_2\widehat{Y}_{i2}\ ],$$

where $\widehat{Y}_{it} \equiv Y_{it} - W_{it}'\widehat{\delta}_t$ for $t = 1, 2$. Estimate the density of $(C_i(a, \gamma), S_i(a, \gamma))$ by replacing V_i with $V_i(a, \gamma)$ in (13), and denote that estimator by $\widehat{f}_{C_i(a,\gamma),S_i(a,\gamma)}$. For simplicity, we suppress the arguments (a, γ) for the rest of this subsection.

Estimate the density of (C_i, γ_i, β_i) at (c, γ, b) by

$$\widehat{f}_{C_i,\gamma_i,\beta_i}(c, \gamma, b) \equiv \widehat{f}_{C_i,S_{i1},S_{i2},S_{i3}}(c, b_1 a_1, b_2 a_2, \widehat{\delta}_1 a_2 \gamma)|\widehat{\delta}_1 a_1^{J_x} a_2^{J_x+1}|,$$

and estimate the conditional density of C_i given $\gamma_i = \gamma, \beta_i = b$ as

$$\widehat{f}_{C_i|\gamma_i=\gamma,\beta_i=b}(c) = \frac{\widehat{f}_{C_i,\gamma_i,\beta_i}(c, \gamma, b)}{\widehat{f}_{\gamma_i,\beta_i}(\gamma, b)}.$$

For any $a \in \mathbb{R}^2$, let $\widetilde{U}_i(a) \equiv C_i - \psi^* = a_1 U_{i1} + a_2 U_{i2}$, where $\psi^* = a_1\gamma y_0$ as defined in (10). Estimate the conditional density of $\widetilde{U}_i(a)$ by

$$\widehat{f}_{\widetilde{U}_i(a)|\gamma_i=\gamma,\beta_i=b}(\tilde{u}) = \widehat{f}_{C_i|\gamma_i=\gamma,\beta_i=b}(\tilde{u} + a_1\gamma y_0).$$

Estimate the joint density of $U_i = (U_{i1}, U_{i2})'$ conditional on $\gamma_i = \gamma, \beta_i = b$ following steps similar to those in Masten (2018). First, estimate the conditional characteristic function of U_i by

$$\widehat{\phi}_{U_{i1},U_{i2}|\gamma_i=\gamma,\beta_i=b}(a_1, a_2) = \int_{\mathcal{U}_n} \exp(i\tilde{u})\, \widehat{f}_{\widetilde{U}_i(a)|\gamma_i=\gamma,\beta_i=b}(\tilde{u})\, d\tilde{u},$$

where $\mathcal{U}_n$ is the interval implied by $\bar{c}_n$, which expands as $n \to \infty$; see the proof of Theorem 2 in Appendix A.

Then, use the inverse Fourier transform to estimate the conditional density of U_i as

$$\begin{aligned}&\widehat{f}_{U_{i1},U_{i2}|\gamma_i=\gamma,\beta_i=b}(u_1,u_2)\\ =&\mathrm{Re}\left(\frac{1}{(2\pi)^2}\int_{\mathcal{A}_n}\exp[-i(a_1u_1+a_2u_2)]\widehat{\phi}_{U_{i1},U_{i2}|\gamma_i=\gamma,\beta_i=b}(a_1,a_2)da_1da_2\right),\end{aligned}$$

where $\mathrm{Re}(\cdot)$ is the real part of a complex number and

$$\mathcal{A}_n \equiv \{a \in \mathbb{R}^2 : \ \underline{\vartheta}_n \leq |a_1| \leq \bar{\vartheta}_n, \ \underline{\vartheta}_n \leq |a_2| \leq \bar{\vartheta}_n\},$$

with tuning sequences $\bar{\vartheta}_n \to \infty$ and $\underline{\vartheta}_n \to 0$ as $n \to \infty$. Truncating the Fourier inversion to a bounded region is standard in deconvolution (Fan, 1991; Alexander, 2009); Masten (2018) applied the same technique in estimation.

5 Asymptotic Properties

Define a Sobolev space

$$\mathcal{W}^m(\mathbb{R}^l) \equiv \{f \in L^2(\mathbb{R}^l) : (1+\|\cdot\|^2)^{m/2}\mathcal{F}f(\cdot) \in L^2(\mathbb{R}^l)\}$$

with a semi-norm

$$\|f\|_m^2 \equiv \int_{\mathbb{R}^l}(1+\|\xi\|^2)^m|\mathcal{F}f(\xi)|^2d\xi,$$

where $\mathcal{F}f(\xi) \equiv \int_{\mathbb{R}^l} f(x)\exp(-ix'\xi)dx$ is the Fourier transform of $f(\cdot)$, and $m > 0$ is the *order of smoothness* of the Sobolev space. For any constant $B > 0$, define

$$\mathcal{W}^m(\mathbb{R}^l; B) \equiv \{f \in \mathcal{W}^m(\mathbb{R}^l) : \|f\|_m \leq B\}.$$

We treat δ_t as known. The first-step estimators $\widehat{\delta}_1, \widehat{\delta}_2$ in Section 4.1 are $\sqrt{n}$-consistent (with Y_{i0} fixed at a constant y_0 in the sample). With $m > l/2$, the uniform convergence rate of the intermediate density estimator is strictly slower than $n^{-1/2}$ (Lemma A.1). The plug-in error from replacing δ_t by $\widehat{\delta}_t$ in $\widetilde{Y}_{it}$ is asymptotically negligible relative to the inverse Radon transform estimation.

For any $d = (d_1, d_2) \in \mathbb{R}^2$, define a Radon transform estimator $\widehat{f}_{C_i(d),S_i(d)}$ for the joint

density of $(C_i(d), S_i(d))$ by replacing v_i in (13) with $v_i(d) \equiv \|(1, h_i')\|^{-1}(d_1\tilde{y}_{i1} + d_2\tilde{y}_{i2})$, where $\tilde{y}_{it} \equiv y_{it} - w_{it}'\delta_t$.

We establish uniform convergence of our estimators for the joint density of (γ_i, β_i) and the conditional density of U_i given (γ_i, β_i). Recall these are all conditional on Y_{i0}, which is fixed at a constant value y_0 in the sample, and suppressed in notation.

We maintain the conditions for identification in Theorem 1 throughout, and do not reiterate them below for brevity. In addition, we maintain the following assumptions.

Assumption 5.1. *Let $\mathcal{T}$ be a compact subset of the joint support of (γ_i, β_i) such that*

(i) the Jacobian determinant $|J_i(d)|$ in (8) is bounded away from zero for any fixed $d \in \mathbb{R}^2$ with $d_2 \neq 0$.

(ii) $\sup_{(\gamma,b)\in\mathcal{T}} f_{\gamma_i,\beta_i}(\gamma, b) < \infty$.

(iii) there exists a constant $M_b > 0$ such that $\inf_{(\gamma,b)\in\mathcal{T}} f_{\gamma_i,\beta_i}(\gamma, b) \geq M_b$.

For a fixed d, the Jacobian $|J_i(d)| = |\delta_1 d_2^{J_x+1}(d_1 + d_2\gamma)^{J_x}|$ vanishes at $\gamma = -d_1/d_2$. Assumption 5.1(i) excludes this value from the compact set $\mathcal{T}$. Assumption 5.1(iii) bounds f_{γ_i,β_i} away from zero on $\mathcal{T}$, so that $\widehat{f}_{\gamma_i,\beta_i}$ stays away from zero asymptotically.

Assumption 5.2. *For any $d \in \mathbb{R}^2$ with $d_2 \neq 0$, the joint density $f_{C_i(d),S_i(d)}$ satisfies*

(i) $f_{C_i(d),S_i(d)} \in \mathcal{W}^m(\mathbb{R}^l; B)$ for some $B > 0$ and smoothness order $m > l/2$, and

(ii) there exist constants $M_c > 0$ and $\kappa > l - 1$ such that $f_{C_i(d),S_i(d)}(z) \leq M_c\,(1 + \|z\|)^{-\kappa}$ for all $z \in \mathbb{R}^l$.

The Sobolev smoothness in Assumption 5.2(i) controls the bias of the Radon transform estimator, and $m > l/2$ ensures that $f_{C_i(d),S_i(d)}$ is bounded and continuous. This coincides with the condition of Bissantz *et al.* (2014) for the uniform bound on the bias. The decay exponent $\kappa > l-1$ controls the variance of the Radon transform estimator. Holzmann and Meister (2020) impose a similar polynomial tail bound that determines the optimal rate of their Fourier-based estimator. Goldenshluger and Kim (2021) impose a comparable tail bound on the target density.

Assumption 5.3. *(i) The density f_{Q_i} is uniformly bounded.*

(ii) For some even integer $\zeta > (l-1)^2/(2m - l + 1 - 2\varsigma)$ with some constant $\varsigma \in (0, m - l/2)$, the partial derivatives of f_{Q_i} up to order ζ exist and are bounded.

(iii) *There exist constants $M_q > 0$ and $k > 1$ such that $f_{Q_i}(q) \geq M_q\, q_1^k$ for any $q \in \mathbb{S}_+^{l-1}$.*

(iv) *The smoothing parameter of $\widehat{f}_{Q_i}$ satisfies $\tau \asymp (\ln n/n)^{1/(2\zeta+l-1)}$.*

(v) *The spherical kernel $\widetilde{K}$ is bounded, Hölder continuous, and of order at least ζ.*

Assumption 5.3(i) corresponds to the upper boundedness condition in Assumption 3 of Hoderlein *et al.* (2010); (iii) adapts the lower bound in their Assumption 4 to the trimmed setting, which yields $M_{q,n} \equiv \inf_{q \in \mathbb{S}(\underline{q}_n)} f_{Q_i}(q) \geq M_q \underline{q}_n^k$. The smoothness condition (ii), together with the spherical kernel condition (v), gives the uniform rate of the spherical kernel estimator $\widehat{f}_{Q_i}$ under $\tau \asymp (\ln n/n)^{1/(2\zeta+l-1)}$; see Hoderlein *et al.* (2010, Appendix B).

The next assumption is on the kernel function $K_\nu(\cdot) = \nu^{-l} K(\cdot/\nu)$ in Footnote 6.

Assumption 5.4. *(i) The kernel K is uniformly bounded.*

(ii) *There exists a constant $M_k > 0$ such that, for all $z, z' \in \mathbb{R}$, $|K(z) - K(z')| \leq M_k \|z - z'\|$.*

(iii) *The smoothing parameter ν is of the order $\nu \asymp (\underline{q}_n^{1-k} \ln n/n)^{1/(2m+l-1-2\varsigma)}$.*

(iv) *The order parameter r of the kernel K_ν satisfies $r \geq m - l/2$.*

Assumption 5.4(i) and (ii) require the kernel K to be bounded and Lipschitz continuous. They are standard for uniform convergence of kernel density estimators, as in Li and Racine (2007) and Masten (2018). Assumption 5.4(iii) and (iv) specify the rate of the bandwidth ν and the order of the kernel. Following Lemma 1 of Bissantz *et al.* (2014), the order $r \geq m - l/2$ ensures the kernel does not limit the bias rate of the intermediate Radon transform estimator $f_{C_i(d),S_i(d)}$, which is then determined by the smoothness m.

Assumption 5.5. *For any (γ, b) on the joint support of (γ_i, β_i),*

(i) *there exists $p > 1$ such that $\sup_{(\gamma,b)\in\mathcal{T}} E(|U_{ij}|^p \mid \gamma_i = \gamma, \beta_i = b) < \infty$ for $j = 1, 2$.*

(ii) $\max\{\int \sup_{u_2 \in \mathbb{R}} f_{U_i|\gamma_i=\gamma,\beta_i=b}(u_1, u_2)\, du_1, \int \sup_{u_1 \in \mathbb{R}} f_{U_i|\gamma_i=\gamma,\beta_i=b}(u_1, u_2)\, du_2\} < \infty$.

(iii) *The conditional density $f_{U_i|\gamma_i=\gamma,\beta_i=b}$ is continuously differentiable, and both $\{\int \sup_{u_1}(1 + |u_2|)\, |\partial_{u_1} f_{U_i|\gamma,b}(u_1, u_2)|\, du_2$ and $\int \sup_{u_2}(1 + |u_1|)\, |\partial_{u_2} f_{U_i|\gamma,b}(u_1, u_2)|\, du_1\}$ are finite.*

(iv) *There exist constants $M_\phi > 0$ and $\rho > 2$ such that $|\phi_{U_i|\gamma_i=\gamma,\beta_i=b}(a)| \leq M_\phi(1 + \|a\|)^{-\rho}$ for all $a \in \mathbb{R}^2$, where $\phi_{U_i|\gamma_i,\beta_i}(\cdot)$ denotes the characteristic function.*

Assumption 5.5(i) and (iv) bound the conditional moments of the random intercepts

and the tail of their characteristic function. Conditions (i), (ii), and (iv) are instrumental for dealing with the trimming devices $\mathcal{C}_n$, $\mathcal{U}_n$, and $\mathcal{A}_n$ in asymptotics. The smoothness condition (iii) adapts Assumption E1.3 of Masten (2018) to non-compact support, yielding a bound on $\|\nabla_a f_{\widetilde{U}_i(a)|\gamma,b}\|$ that is analogous to his Lemma S3 under compact support.

Assumption 5.5(iv) is the ordinary-smooth condition of the deconvolution literature (Fan, 1991; Alexander, 2009). Since $(1+\|a\|)^{-\rho}$ is integrable on $\mathbb{R}^2$ for $\rho > 2$, the conditional characteristic function is absolutely integrable, which is Assumption E1.4 of Masten (2018) for the Fourier inversion. The decay helps to account for the bias from truncating the inversion on $\mathcal{A}_n$.

Define $r_{1n} \equiv (\underline{q}_n^{1-k} \ln n/n)^{\frac{m-l/2-\varsigma}{2m+l-1-2\varsigma}}$, $r_{2n} \equiv (\ln n/n)^{\frac{\zeta}{2\zeta+l-1}}$, and let

$$r_n \equiv r_{1n} + \underline{q}_n^{1/2} + \underline{q}_n^{1-k} r_{2n} r_{1n}^{-\frac{2(l-1)}{2m-l-2\varsigma}}.$$

Assumption 5.6. *$\underline{q}_n \to 0$, $\bar{c}_n \asymp r_n^{-1/(p+1)}$, and*

$$\underline{q}_n^{-1} (\ln n/n)^{\min\left\{ \frac{2m+l-2-2\varsigma}{k(4m+2l-3-4\varsigma)+1},\ \frac{\zeta}{k(2\zeta+l-1)},\ \frac{\zeta(2m-l+1-2\varsigma)-(l-1)^2}{2(k-1)(m-\varsigma)(2\zeta+l-1)} \right\}} \to 0.$$

The sequence r_n characterizes the uniform convergence rate of an intermediate density estimator $\widehat{f}_{C_i(d),S_i(d)}$ for a fixed index d (Lemma A.1). Assumption 5.6 restricts the trimming sequence $\underline{q}_n$ and the truncation sequence $\bar{c}_n$, and ensures $r_n \to 0$.

The next proposition establishes uniform convergence of our estimator for the density of (γ_i, β_i). Recall that $l = 2J_x + 2$ denotes the dimension of $(1, H_i')'$ (with $W_{i1} \in \mathbb{R}$).

Proposition 2. *Fix $d \in \mathbb{R}^2$ with $d_2 \neq 0$, and suppose Assumptions 5.1*(i)–(ii)*, 5.2, 5.3, 5.4, 5.5*(i) *and 5.6 hold. Then, for any moment order p in Assumption 5.5*(i)*,*

$$\sup_{(\gamma,b)\in\mathcal{T}} \left| \widehat{f}_{\gamma_i,\beta_i}(\gamma,b) - f_{\gamma_i,\beta_i}(\gamma,b) \right| = O_p\big(r_n^{p/(p+1)}\big).$$

Remark 6. *The adjustment factor $p/(p+1)$ is due to the accommodation of non-compact support of random intercepts U_i. The convergence rate is r_n under compact support, or arbitrarily close to r_n if the conditional moments are finite for all orders $p > 1$ (e.g., a Gaussian, a Gaussian mixture, or a logistic distribution).*

Remark 7. *Hoderlein et al. (2010) estimate the joint density of random coefficients in*

a cross-sectional linear model using a Radon transform estimator, including a trimmed version for models with an intercept. In our setting, the Radon transform estimator is applied to the intermediate reduced-form density $f_{C_i(d),S_i(d)}$. Proposition 2 provides a sup-norm rate required for the conditional density of U_i in Theorem 2.

We next establish uniform convergence of our estimator for the joint density of $U_i \equiv (U_{i1}, U_{i2})'$ conditional on (γ_i, β_i), which we recover by Fourier inversion of the characteristic function estimator $\widehat{\phi}_{U_{i1},U_{i2}|\gamma_i,\beta_i}$. The result is obtained under Assumptions S.1 and S.2 in Online Appendix S2. Assumption S.1 imposes tail and smoothness bounds on the density of $(C_i(\bar{d}), S_i(\bar{d}))$ uniformly over $a \in \mathcal{A}_n$ almost everywhere. Assumption S.2 imposes restrictions on the tuning sequences $\bar{\vartheta}_n$ and $\underline{\vartheta}_n$.

Theorem 2. *Suppose Assumptions 5.1, 5.2, 5.3, 5.4, 5.5, 5.6, S.1, S.2 hold, and that $E(|\widetilde{Y}_{i1}|) < \infty$ and $E(|\widetilde{Y}_{i2}|) < \infty$. Then, for any (γ, b) on the support of (γ_i, β_i),*

$$\sup_{(u_1,u_2)\in\mathbb{R}^2} \left| \widehat{f}_{U_{i1},U_{i2}|\gamma_i=\gamma,\beta_i=b}(u_1,u_2) - f_{U_{i1},U_{i2}|\gamma_i=\gamma,\beta_i=b}(u_1,u_2) \right| = o_p(1).$$

Remark 8. *Masten (2018, Section C.4) proves sup-norm consistency of this Fourier inversion under compact support; Theorem 2 permits non-compact support. The proof bounds the error of the Fourier inversion of $\widehat{\phi}$ over $\mathbb{R}^2$ by decomposing it into $R_1+R_2+R_3$, where*

$$R_1 = \frac{1}{(2\pi)^2} \int_{\mathcal{A}_n} |\widehat{\phi}(a) - \phi(a)| \, da, \qquad R_2 + R_3 = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2 \setminus \mathcal{A}_n} |\phi(a)| \, da,$$

where R_2 and R_3 decompose further over $[-\bar{\vartheta}_n, \bar{\vartheta}_n]^2 \setminus \mathcal{A}_n$ and $\mathbb{R}^2 \setminus [-\bar{\vartheta}_n, \bar{\vartheta}_n]^2$, respectively. We derive the convergence rates of these components. The estimation error R_1 is due to three sources: the inverse Radon transform estimation, the trimming over $\mathcal{A}_n$, and the truncation of the non-compact intercept support. The truncation errors satisfy $R_2 + R_3 = O(\underline{\vartheta}_n + \bar{\vartheta}_n^{-(\rho-2)})$, which follows from the characteristic function decay in Assumption 5.5(iv)*. Theorem 2 then follows from Assumption S.2.*

6 Monte Carlo Simulations

In this section, we evaluate the finite-sample performance of the multi-step estimators for the joint density of (γ_i, β_i) and the conditional density of $U_i \equiv (U_{i1}, U_{i2})'$ given (γ_i, β_i).

We focus on the case with $T = 2$ and fix the initial condition at $y_0 = 1$ throughout this simulation study.

The data-generating process (DGP) follows (1), with $(\delta_1, \delta_2) = (1, 1)$. The regressors are generated independently of (γ_i, β_i, U_i) as follows:

$$X_i \equiv (X_{i1}, X_{i2})' \sim \mathcal{N}(0, \Sigma_x), \qquad \Sigma_x = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

and $W_{i2} = 0.5\, W_{i1} + 0.3\, W_{i1}^2 + \varepsilon_{w,i}$, where $W_{i1} \sim \mathcal{N}(0, 2.5)$ and $\varepsilon_{w,i} \sim \mathcal{N}(0, 2.5)$. The autoregressive coefficient is generated as $\gamma_i \sim \text{Beta}(6, 3)$. The marginal distributions of β_i and U_i are given by

$$\beta_i \sim \mathcal{N}(\mu_\beta, \Sigma_\beta), \qquad U_i \sim \mathcal{N}(0, \Sigma_u), \tag{14}$$

where

$$\mu_\beta = (1, 1)', \qquad \Sigma_\beta = \begin{pmatrix} 0.5 & 0.25 \\ 0.25 & 0.5 \end{pmatrix}, \qquad \Sigma_u = \begin{pmatrix} 0.3 & 0.15 \\ 0.15 & 0.3 \end{pmatrix}.$$

We consider the following four designs.

Baseline. The random coefficients and intercepts (γ_i, β_i, U_i) are mutually independent, with marginals as specified above.

Correlated. This design introduces dependence among (γ_i, β_i, U_i) while preserving the conditional mean restriction in Assumption 3.1. While γ_i has the same Beta marginal as in the baseline design, β_i and U_i are constructed via

$$\beta_i = \mu_\beta + \exp(\lambda_\beta \gamma_i)\, L_\beta\, \eta_i^\beta, \qquad U_i = \exp(\lambda_u \gamma_i)\, L_u\, \eta_i^u,$$

where L_β and L_u are the lower-triangular matrices from the Cholesky decompositions of Σ_β and Σ_u, respectively. The shocks $\eta_i = (\eta_i^\beta, \eta_i^u)'$ are drawn independently of γ_i from $\mathcal{N}(0, \Sigma_{\text{corr}})$, and the 4–by–4 correlation matrix Σ_{corr} has identity diagonal blocks and cross-block matrix

$$\Sigma^{\beta,u} = \begin{pmatrix} 0.25 & 0.10 \\ 0.10 & 0.25 \end{pmatrix}.$$

We set $\lambda_\beta = \lambda_u = 0.3$. Under this design, conditional on γ_i, the vectors β_i and U_i have the same means as in the baseline design but covariance matrices that scale with γ_i:

$$\beta_i \mid \gamma_i \sim \mathcal{N}\left(\mu_\beta,\, \exp(2\lambda_\beta \gamma_i)\, \Sigma_\beta\right), \qquad U_i \mid \gamma_i \sim \mathcal{N}\left(0,\, \exp(2\lambda_u \gamma_i)\, \Sigma_u\right).$$

Scale-dependence. This design is simpler than the preceding correlated design in that there is no direct correlation between β_i and U_i; their dependence is solely driven by γ_i through the scale:

$$\beta_i \mid \gamma_i \sim \mathcal{N}\left(\mu_\beta,\, v(\gamma_i)\, \Sigma_\beta\right), \qquad U_i \mid \gamma_i \sim \mathcal{N}\left(0,\, w(\gamma_i)\, \Sigma_u\right),$$

where $v(\gamma) = \exp\left[2\lambda_\beta(\gamma - \mathbb{E}[\gamma])\right]/Z(\lambda_\beta)$ and $w(\gamma) = \exp\left[2\lambda_u(\gamma - \mathbb{E}[\gamma])\right]/Z(\lambda_u)$, with $Z(\cdot)$ being a normalization constant such that $\mathbb{E}[v(\gamma_i)] = \mathbb{E}[w(\gamma_i)] = 1$.

Bimodal. The random coefficients γ_i and β_i are independent with the same marginal distributions as in the baseline design, but U_i follows a symmetric Gaussian mixture:

$$U_i \sim \tfrac{1}{2}\,\mathcal{N}(\mu_u, \Sigma_u) + \tfrac{1}{2}\,\mathcal{N}(-\mu_u, \Sigma_u), \qquad \mu_u = (0.7, -0.7)'.$$

Hoderlein *et al.* (2010) and Masten (2018) use similar Gaussian-mixture designs to investigate estimator performance for multi-modal densities.

For each design, we vary the sample sizes $n \in \{500, 2000, 8000\}$ and generate $N_{\text{rep}} = 500$ Monte Carlo replications. The estimators for the joint density $\widehat{f}_{\gamma_i,\beta_i}$ and for the conditional density $\widehat{f}_{U_{i1},U_{i2}|\gamma_i=\gamma,\beta_i=b}$ are defined as in Sections 4.2 and 4.3. Following Hoderlein *et al.* (2010), we select the bandwidths (τ, ν) by minimizing the mean density-weighted integrated squared error. Following Masten (2018), we obtain an optimal ν for a single $a \in [-\bar{\vartheta}_n, \bar{\vartheta}_n]^2$ and then rescale it to the others. In the baseline design at $n = 8000$, the average ν used over all grid points a is 0.65. As the asymptotic theory prescribes, the cutoff $\bar{\vartheta}_n$ increases with the sample size, and we scale the bandwidths for the other sample sizes accordingly.

We first present the performance of $\widehat{f}_{\gamma_i,\beta_i}$ through one-dimensional conditional densities. Table 1 reports the mean integrated squared error (MISE) of $\widehat{f}_{\gamma_i|\beta_i}$, $\widehat{f}_{\beta_{i1}|\gamma_i,\beta_{i2}}$, and $\widehat{f}_{\beta_{i2}|\gamma_i,\beta_{i1}}$. For every design and conditional density reported, the MISE declines with n, confirming the consistency of our estimator.

Table 1: Conditional densities of the estimated joint density of (γ_i, β_i)

		MISE		
	n	$\widehat{f}_{\gamma_i\|\beta_i}$	$\widehat{f}_{\beta_{i1}\|\gamma_i,\beta_{i2}}$	$\widehat{f}_{\beta_{i2}\|\gamma_i,\beta_{i1}}$
Baseline	500	0.2359	0.1054	0.0654
	2000	0.2001	0.0790	0.0476
	8000	0.1749	0.0655	0.0369
Correlated	500	0.2699	0.0664	0.0414
	2000	0.2372	0.0468	0.0306
	8000	0.2052	0.0374	0.0227
Scale	500	0.2417	0.1031	0.0607
	2000	0.2046	0.0792	0.0459
	8000	0.1747	0.0644	0.0355
Bimodal	500	0.2434	0.1066	0.0639
	2000	0.2096	0.0816	0.0483
	8000	0.1860	0.0695	0.0395

Notes: The table reports, across $N_{\text{rep}} = 500$ Monte Carlo replications, the mean integrated squared error (MISE) of the estimated joint density $\widehat{f}_{\gamma_i,\beta_i}$, summarized through its three one-dimensional conditional densities; each varies one coordinate of $(\gamma_i, \beta_{i1}, \beta_{i2})$ with the other two held at their medians.

Figures 1 and 2 show the estimated conditional density of U_i along each axis and as contour plots, and Table 2 reports its MISE and the bias, standard deviation (SD), and mean squared error (MSE) of the estimated interquartile range (IQR) of its two marginals. For the three unimodal designs (Baseline, Correlated, and Scale), the MISE falls with n and the Monte Carlo mean density converges toward the true density, with the bias small in large samples. The conditional location is recovered accurately: along each axis the estimated density stays centered on the true density even in small samples. Masten (2018, Figure 2 and Table 2) likewise finds the location well recovered even when the full shape is hard to estimate.

The IQR in Table 2 is estimated less precisely in small samples but approaches its true value as n grows. For the three unimodal designs it tends to be over-estimated, possibly due to over-smoothing at small n which blurs the estimate: it moves probability mass away from the peak into the sides, so the estimated density is too flat and too wide, and

its IQR too large (Figure 1). Masten (2018, Table 2) reports the same upward IQR bias for his unimodal designs, where the estimated density is more spread out than the true density.

The Bimodal design is the most demanding. The MISE declines slowly (0.054 to 0.035). Along each axis the estimate for the marginal density does not register a clear bimodal pattern even at $n = 8000$ (Figure 1), although the contour separates its two modes by $n = 2000$ (Figure 2). Relatedly, the spread and hence the IQR are *under*-estimated. Again, this pattern is consistent with Masten (2018)'s observation that the shape is harder to estimate for multi-modal densities.

Table 2: Estimator for the conditional density of U_i

			$\widehat{\text{IQR}}(U_{i1} \mid \gamma_i, \beta_i)$			$\widehat{\text{IQR}}(U_{i2} \mid \gamma_i, \beta_i)$		
	n	MISE	Bias	SD	MSE	Bias	SD	MSE
Baseline	500	0.0841	0.4571	0.0306	0.2099	0.4353	0.0294	0.1903
	2000	0.0277	0.2313	0.0171	0.0538	0.2157	0.0172	0.0468
	8000	0.0053	0.1039	0.0099	0.0109	0.0903	0.0099	0.0082
Correlated	500	0.0516	0.4449	0.0293	0.1988	0.4256	0.0297	0.1820
	2000	0.0163	0.2117	0.0171	0.0451	0.1920	0.0180	0.0372
	8000	0.0044	0.0602	0.0125	0.0038	0.0435	0.0137	0.0021
Scale	500	0.0833	0.4552	0.0311	0.2082	0.4331	0.0292	0.1884
	2000	0.0271	0.2297	0.0182	0.0531	0.2130	0.0179	0.0457
	8000	0.0055	0.1024	0.0097	0.0106	0.0871	0.0107	0.0077
Bimodal	500	0.0538	−0.2158	0.0738	0.0520	−0.2236	0.0775	0.0560
	2000	0.0421	−0.2136	0.0626	0.0496	−0.2129	0.0734	0.0507
	8000	0.0346	−0.1057	0.0489	0.0136	−0.1246	0.0594	0.0191

Notes: The table reports, across $N_{\text{rep}} = 500$ Monte Carlo replications, the mean integrated squared error (MISE) of the estimated conditional density $\widehat{f}_{U_{i1},U_{i2}|\gamma_i,\beta_i}$; and the bias, standard deviation (SD), and mean squared error (MSE) of the estimated interquartile range $\widehat{\text{IQR}}(U_{ij} \mid \gamma_i, \beta_i)$, $j = 1, 2$, of the two marginal distributions. All quantities are evaluated at the median value of (γ_i, β_i).

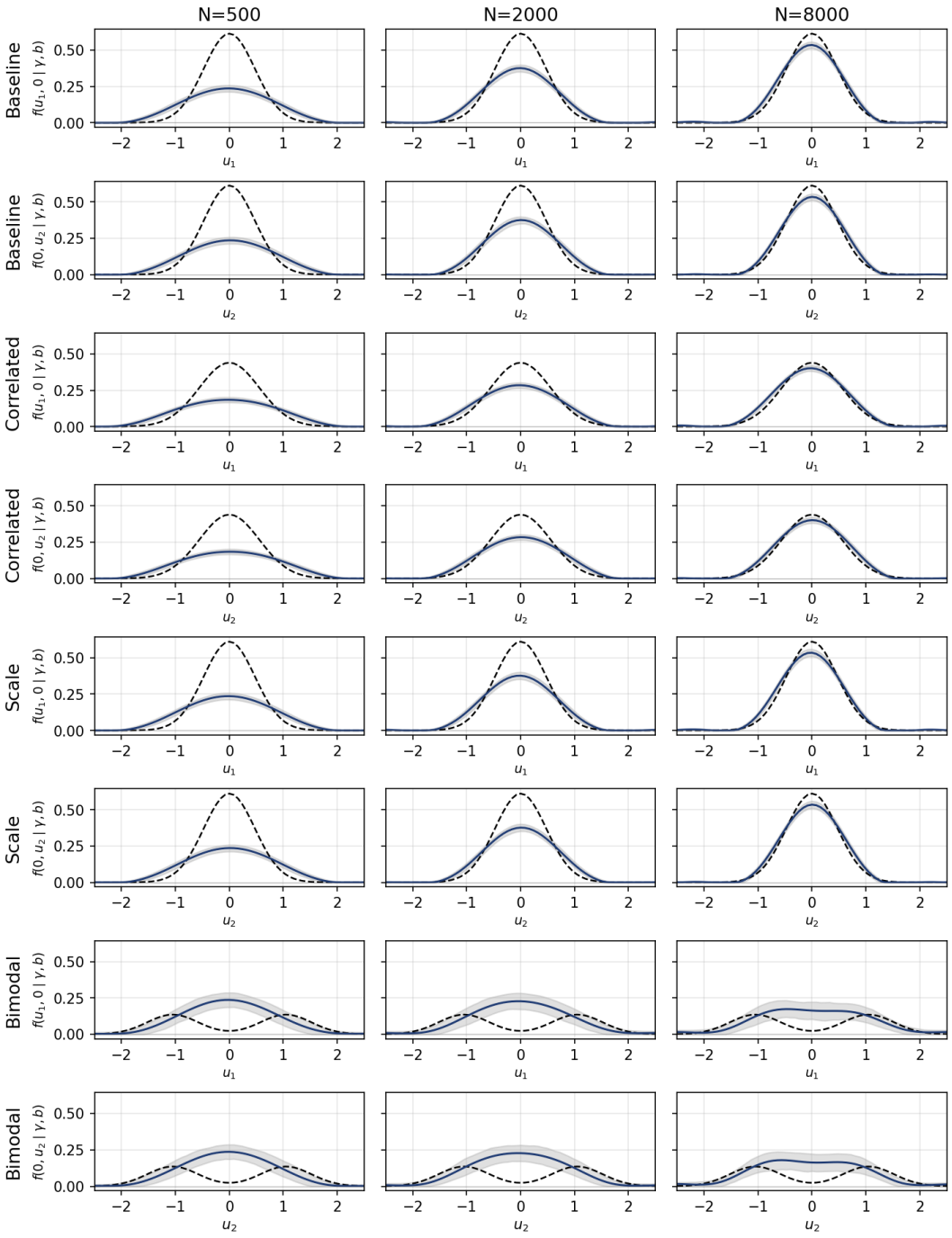


Figure 1: Estimated conditional density of U_i along each axis

Notes: $\widehat{f}_{U_{i1},U_{i2}|\gamma_i,\beta_i}$ along u_1 (at $u_2 = 0$) and along u_2 (at $u_1 = 0$), at the median value of (γ_i, β_i). Each row shows these two curves for one design; columns are the sample sizes $n \in \{500, 2000, 8000\}$. Black dashed: the true density; navy solid: the Monte Carlo mean across $N_{\text{rep}} = 500$ replications; shaded band: the 95% pointwise Monte Carlo confidence band.

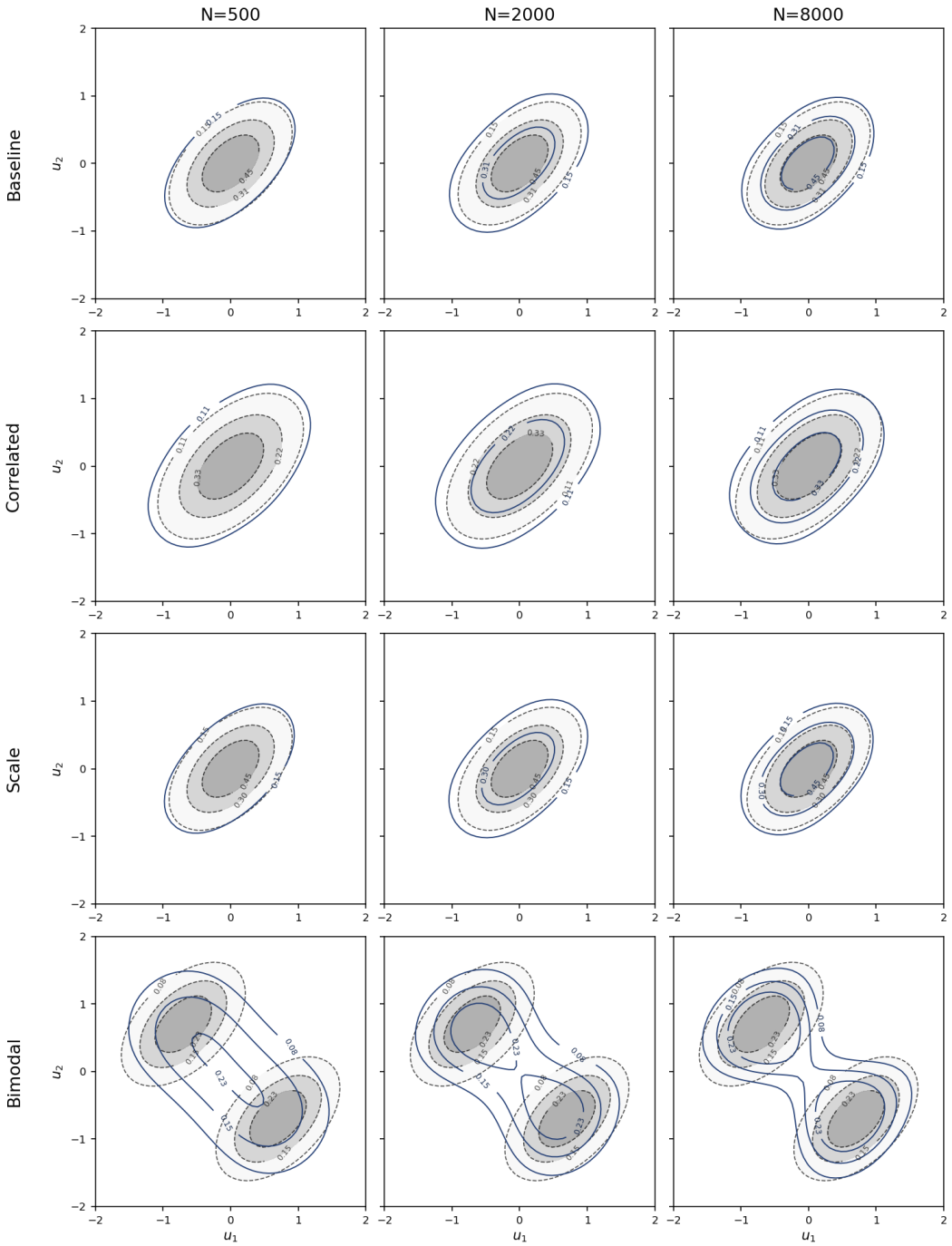


Figure 2: Contour plots of the estimated conditional density of U_i

Notes: $\widehat{f}_{U_{i1},U_{i2}|\gamma_i,\beta_i}$ for the four designs (rows) at sample sizes $n \in \{500, 2000, 8000\}$ (columns), conditioning on the median value of (γ_i, β_i). Gray fill with black dashed contours: the true density; navy solid contours: the Monte Carlo mean across $N_{\text{rep}} = 500$ replications. The contours of the true density enclose 25%, 50%, and 75% of its probability mass, and the Monte Carlo contours use the same levels.

7 Conclusion

We recover the entire joint distribution of the random coefficients and the time-varying errors. This is new relative to the literature on random-coefficient triangular and simultaneous equation models which left the distribution of time-varying errors unidentified. This is possible because a linear combination of the outcome history behaves as a single-equation random-coefficient regression in distinct regressors: inverting a Radon transform recovers a reduced-form density, and varying the combination and using deconvolution recover the distribution of the time-varying errors. We apply the analog principle to this constructive identification strategy, and propose a closed-form, multi-step estimator.

The identifying condition, a distributional form of strict exogeneity under which the covariates are independent of the coefficients and errors given the initial condition, is both the source of this reach and the main limitation of the approach: it permits arbitrary dependence among the coefficients and errors, but lets the covariates and the unobserved heterogeneity be related only through the initial condition. Several questions remain for future research, including inference, data-driven bandwidth choices, and a continuously distributed initial condition.

Appendix

A Proofs in Section 5

We maintain the conditions for identification in Theorem 1 throughout, and do not reiterate them separately below for brevity. Under the conditions of Theorem 1, the conditional density $f_{V_i|Q_i}$, which equals the Radon transform of $f_{C_i(d),S_i(d)}$ over the directions on the support of Q_i, uniquely determines $f_{C_i(d),S_i(d)}$ within the class of distributions satisfying Assumption 3.5.

The following lemma establishes uniform convergence of the Radon transform estimator $\widehat{f}_{C_i(d),S_i(d)}$ for any $d \in \mathbb{R}^2$ with $d_2 \neq 0$. Hoderlein *et al.* (2010) proved unweighted L^2 consistency of the Radon transform estimator; we strengthen this to sup-norm convergence by combining the variance analysis as in Li and Racine (2007) with the uniform bias bound of Bissantz *et al.* (2014).

Lemma A.1 (Sup-norm convergence of the RTE)**.** *Fix $d \in \mathbb{R}^2$ with $d_2 \neq 0$, and let $\mathcal{S} \subset \mathbb{R}^l$*

be a compact subset of the joint support of $(C_i(d), S_i(d))$. *Suppose Assumptions 5.2, 5.3, 5.4, and 5.6 hold. Then,*

$$\sup_{(c,s)\in\mathcal{S}} \left| \widehat{f}_{C_i(d),S_i(d)}(c,s) - f_{C_i(d),S_i(d)}(c,s) \right| = O_p(r_n)\,,$$

where $r_n \equiv r_{1n} + \underline{q}_n^{1/2} + \underline{q}_n^{1-k} r_{2n}\, r_{1n}^{-2(l-1)/(2m-l-2\varsigma)}$.

Proof of Lemma A.1. Define an infeasible estimator

$$\bar{f}_{C_i(d),S_i(d)}(c,s) = \frac{2}{n}\sum\nolimits_i \frac{\mathbf{1}\{Q_{i1} \geq \underline{q}_n\}}{f_{Q_i}(Q_i)} K_\nu(Q_i'(c,s')' - V_i(d)),$$

and decompose $\widehat{f}_{C_i(d),S_i(d)} - f_{C_i(d),S_i(d)}$ as

$$\underbrace{\widehat{f}_{C_i(d),S_i(d)} - \bar{f}_{C_i(d),S_i(d)}}_{I_1} + \underbrace{\bar{f}_{C_i(d),S_i(d)} - E(\bar{f}_{C_i(d),S_i(d)})}_{I_2} + \underbrace{E(\bar{f}_{C_i(d),S_i(d)}) - f_{C_i(d),S_i(d)}}_{I_3}.$$

The Radon transform of $f : \mathbb{R}^l \to \mathbb{R}$ is $(Rf)(\varphi,\pi) \equiv \int_{\{z:\varphi' z=\pi\}} f(z)\,dz$, where $\pi \in \mathbb{R}$ and $\varphi \in \mathbb{S}^{l-1}$ with $\mathbb{S}^{l-1} \equiv \{z \in \mathbb{R}^l : \|z\| = 1\}$ being the unit sphere in $\mathbb{R}^l$. A regularized inverse operator of the Radon transform, denoted by A_ν, is a mapping from $\{g : \mathbb{S}_+^{l-1} \times \mathbb{R} \to \mathbb{R}\}$ to $\{f : \mathbb{R}^l \to \mathbb{R}\}$ defined as

$$[A_\nu g](z) \equiv 2\int_{\mathbb{S}_+^{l-1}} \int_{-\infty}^{\infty} K_\nu(\varphi' z - \pi) g(\varphi,\pi)\,d\pi\,d\mu(\varphi),$$

where K_ν is a kernel function defined in Section 4.2 and μ denotes the Lebesgue measure on $\mathbb{S}_+^{l-1}$. Define $[A_{\nu,n}\,g](z) \equiv 2\int_{\mathbb{S}(\underline{q}_n)} \int_{-\infty}^{\infty} K_\nu(\varphi' z - \pi) g(\varphi,\pi)\,d\pi\,d\mu(\varphi)$. We suppress d in $C_i(d)$ and $S_i(d)$. By construction,

$$\begin{aligned}
E(\bar{f}_{C_i,S_i}(c,s)) &= \frac{2}{n}\sum\nolimits_i E\left[\frac{\mathbf{1}\{Q_{i1} \geq \underline{q}_n\}}{f_{Q_i}(Q_i)} E(K_\nu(Q_i'(c,s')' - V_i) \mid Q_i)\right] \\
&= \frac{2}{n}\sum\nolimits_i E\left[\frac{\mathbf{1}\{Q_{i1} \geq \underline{q}_n\}}{f_{Q_i}(Q_i)} \int_{-\infty}^{\infty} K_\nu(Q_i'(c,s')' - v)(Rf_{C_i,S_i})(Q_i,v)\,dv\right] \\
&= 2\int_{\mathbb{S}(\underline{q}_n)} \left[\frac{1}{f_{Q_i}(q)} \int_{-\infty}^{\infty} K_\nu(q'(c,s')' - v)(Rf_{C_i,S_i})(q,v)\,dv\right] f_{Q_i}(q)\,d\mu(q) \\
&= [A_{\nu,n}(Rf_{C_i,S_i})](c,s),
\end{aligned}$$

where the first equality uses the law of iterated expectations (with the outer expectation

taken with respect to Q_i), the second holds because $f_{V_i|Q_i}(v \mid q) = (Rf_{C_i,S_i})(q,v)$ which is implied by $(C_i, S_i) \perp Q_i$ under Assumption 3.4, the third holds because the observations Q_i are i.i.d. draws from the same density f_{Q_i}.

First, consider the bias term $I_3 = A_{\nu,n}(Rf_{C_i,S_i}) - f_{C_i,S_i}$, and decompose it as

$$I_3 = \underbrace{(A_{\nu,n} - A_\nu)\,(Rf_{C_i,S_i})}_{I_{3,1}} + \underbrace{A_\nu(Rf_{C_i,S_i}) - f_{C_i,S_i}}_{I_{3,2}}.$$

Lemma S.1 bounds the two terms over $\mathbb{R}^l$, and since $\mathcal{S} \subset \mathbb{R}^l$,

$$\sup_{(c,s)\in\mathcal{S}} |I_{3,1}| \le M_T\, B\, \underline{q}_n^{1/2}, \qquad \sup_{(c,s)\in\mathcal{S}} |I_{3,2}| \le M_A\, B\, \nu^{m-l/2-\varsigma},$$

for some constants $M_T > 0$ and $M_A > 0$. Then, by Lemma S.2, $\sup_{(c,s)\in\mathcal{S}} |I_2| = O_p(\Xi_{q,n}^{1/2}\, \nu^{-l+1/2}\sqrt{\ln n/n})$, where $\Xi_{q,n} \equiv \int_{\mathbb{S}(\underline{q}_n)} f_{Q_i}^{-1}(q)\, d\mu(q)$. By Assumption 5.3(iii),

$$\begin{aligned} \Xi_{q,n} &\le M_q^{-1} \int_{\mathbb{S}(\underline{q}_n)} q_1^{-k}\, d\mu(q) = M_q^{-1}\omega_{l-1} \int_{\underline{q}_n}^1 q_1^{-k}(1-q_1^2)^{(l-3)/2}\, dq_1 \\ &\le M_q^{-1}\omega_{l-1} \int_{\underline{q}_n}^1 q_1^{-k}\, dq_1 = \frac{\omega_{l-1}}{M_q\,(k-1)}\,(\underline{q}_n^{1-k} - 1) = O\left(\underline{q}_n^{1-k}\right), \end{aligned} \tag{15}$$

where ω_{l-1} is the surface area of $\mathbb{S}^{l-2}$, the first equality follows from Equation (1.5.4) in Dai (2013), the second inequality uses $(1-q_1^2)^{(l-3)/2} \le 1$ on $[0,1]$ as $l > 3$, and the last equality uses $k > 1$.Then,

$$\sup_{(c,s)\in\mathcal{S}} |I_2| = O_p\left(\underline{q}_n^{(1-k)/2}\nu^{-l+1/2}\sqrt{\ln n/n}\right).$$

Balance the rates of I_2 and $I_{3,2}$ by setting $\underline{q}_n^{(1-k)/2}\nu^{-l+1/2}\sqrt{\ln n/n} \asymp \nu^{m-l/2-\varsigma}$; this yields $\nu \asymp (\underline{q}_n^{1-k}\ln n/n)^{1/(2m+l-1-2\varsigma)}$ (Assumption 5.4(iii)). Substituting back and applying the triangle inequality, $\sup_{(c,s)\in\mathcal{S}} |I_2 + I_{3,2}| = O_p(r_{1n})$. By Lemma S.3 and (15),

$$\sup_{(c,s)\in\mathcal{S}} |I_1| = O_p(\nu^{-l+1}\Xi_{q,n} r_{2n}) = O_p(\underline{q}_n^{1-k} r_{2n}\, r_{1n}^{-2(l-1)/(2m-l-2\varsigma)}).$$

Combining these with the bound on the trimming bias $I_{3,1}$,

$$\sup_{(c,s)\in\mathcal{S}} \left|\widehat{f}_{C_i,S_i} - f_{C_i,S_i}\right| = O_p\left(r_{1n} + \underline{q}_n^{1/2} + \underline{q}_n^{1-k} r_{2n}\, r_{1n}^{-2(l-1)/(2m-l-2\varsigma)}\right).$$

■

Proof of Proposition 2. To simplify notation, we suppress d in $C_i(d), S_i(d)$ and $J_i(d)$ throughout this proof. Recall $\mathcal{C}_n \equiv [-\bar{c}_n, \bar{c}_n]$ with $\bar{c}_n \to \infty$ as $n \to \infty$. The estimator $\widehat{f}_{\gamma_i,\beta_i}(\gamma, b)$ integrates over $\mathcal{C}_n$, whereas $f_{\gamma_i,\beta_i}(\gamma, b)$ integrates over the support of C_i conditional on (γ, b). By the triangle inequality,

$$\begin{aligned}\sup_{(\gamma,b)\in\mathcal{T}} \left|\widehat{f}_{\gamma_i,\beta_i}(\gamma,b) - f_{\gamma_i,\beta_i}(\gamma,b)\right| \leq & \underbrace{\sup_{(\gamma,b)\in\mathcal{T}} \int_{\mathcal{C}_n} \left|\widehat{f}_{C_i,\gamma_i,\beta_i}(c,\gamma,b) - f_{C_i,\gamma_i,\beta_i}(c,\gamma,b)\right| dc}_{T_1} \\ & + \underbrace{\sup_{(\gamma,b)\in\mathcal{T}} \int_{|c|>\bar{c}_n} f_{C_i,\gamma_i,\beta_i}(c,\gamma,b)\, dc}_{T_2}.\end{aligned} \tag{16}$$

For T_1, we have

$$\begin{aligned} T_1 &\leq 2\bar{c}_n \sup_{(c,\gamma,b)\in\mathcal{C}_n\times\mathcal{T}} \left|\widehat{f}_{C_i,\gamma_i,\beta_i}(c,\gamma,b) - f_{C_i,\gamma_i,\beta_i}(c,\gamma,b)\right| \\ &= 2\bar{c}_n \sup_{(c,\gamma,b)\in\mathcal{C}_n\times\mathcal{T}} \left\{\left|\widehat{f}_{C_i,S_i}(c,\tilde{s}) - f_{C_i,S_i}(c,\tilde{s})\right| \, |J_i|\right\} \\ &\leq 2\bar{c}_n \sup_{(\gamma,b)\in\mathcal{T}} |J_i| \sup_{(c,\gamma,b)\in\mathcal{C}_n\times\mathcal{T}} \left|\widehat{f}_{C_i,S_i}(c,\tilde{s}) - f_{C_i,S_i}(c,\tilde{s})\right|, \end{aligned}$$

where $\tilde{s} \equiv (b_1(d_1 + d_2\gamma), b_2 d_2, d_2\delta_1\gamma)$, the second step holds under Assumption 5.1(i), and $\sup_{(\gamma,b)\in\mathcal{T}} |J_i| < \infty$ because $|J_i|$ is continuous on the compact set $\mathcal{T}$. For any $(c, \gamma, b) \in \mathcal{C}_n \times \mathcal{T}$, $(c, \tilde{s})$ belongs to the set $\mathcal{S}_n \equiv \mathcal{C}_n \times \{\tilde{s} : (\gamma, b) \in \mathcal{T}\}$, where the second component is compact because $\tilde{s}$ is continuous on $\mathcal{T}$.

Lemma A.1 establishes the uniform convergence rate of $\widehat{f}_{C_i,S_i}(c, s)$ over a compact set $\mathcal{S}$; its proof applies to $\mathcal{S}_n$ because the set enters the proof (via Lemmas S.2 and S.3) only through the number of cubes that cover $\mathcal{S}_n$, $N_n \asymp |\mathcal{S}_n|\, \ell_n^{-l}$, where $|\mathcal{S}_n|$ denotes the Lebesgue measure of $\mathcal{S}_n$ and ℓ_n is the side length of the cubes. By Assumption 5.6, $\bar{c}_n \asymp r_n^{-1/(p+1)}$ and the restriction on $\underline{q}_n$ imply $\bar{c}_n = O(n^C)$ for some finite constant $C > 0$, so $N_n = o(n^{2l+C}(\ln n)^{-2l})$. Thus, by an argument similar to that in the proof of Lemma S.2, applying the Borel–Cantelli lemma and a summability condition, the rate of Lemma A.1 holds uniformly over $\mathcal{S}_n$. Hence

$$T_1 = O_p\left(\bar{c}_n\, r_n\right). \tag{17}$$

For T_2, write

$$T_2 = \sup_{(\gamma,b)\in\mathcal{T}} f_{\gamma_i,\beta_i}(\gamma,b) \int_{|c|>\bar{c}_n} f_{C_i|\gamma_i,\beta_i}(c,\gamma,b)\,dc$$
$$\leq \bar{M} \sup_{(\gamma,b)\in\mathcal{T}} \Pr(|C_i| > \bar{c}_n \mid \gamma_i = \gamma, \beta_i = b) \leq \bar{M}\,\bar{c}_n^{-p} \sup_{(\gamma,b)\in\mathcal{T}} E(|C_i|^p \mid \gamma_i = \gamma, \beta_i = b),$$

where $\bar{M} \equiv \sup_{(\gamma,b)\in\mathcal{T}} f_{\gamma_i,\beta_i}(\gamma,b) < \infty$ by Assumption 5.1(ii) and the last inequality follows from Markov's inequality with moment order p of Assumption 5.5(i). Since $C_i \mid \gamma, b = (d_1\gamma + d_2\gamma^2)y_0 + (d_1 + d_2\gamma)U_{i1} + d_2U_{i2}$ and $p > 1$, the c_r-inequality gives

$$E(|C_i|^p \mid \gamma, b) \leq 3^{p-1}\Big(|(d_1\gamma + d_2\gamma^2)y_0|^p + |d_1 + d_2\gamma|^p\, E(|U_{i1}|^p \mid \gamma, b) + |d_2|^p\, E(|U_{i2}|^p \mid \gamma, b)\Big), \tag{18}$$

where $(d_1\gamma + d_2\gamma^2)y_0$ and $d_1 + d_2\gamma$ are continuous in γ and hence bounded on the compact set $\mathcal{T}$. Together with Assumption 5.5(i), the bound in (18) yields $\sup_{(\gamma,b)\in\mathcal{T}} E(|C_i|^p \mid \gamma, b) < \infty$. Then, we have

$$T_2 = O(\bar{c}_n^{-p}). \tag{19}$$

Combining (17) and (19), $\sup_{(\gamma,b)\in\mathcal{T}} |\widehat{f}_{\gamma_i,\beta_i} - f_{\gamma_i,\beta_i}| = O_p(\bar{c}_n r_n) + O(\bar{c}_n^{-p})$. Setting $\bar{c}_n \asymp r_n^{-1/(p+1)}$ balances the two terms. Then

$$\sup_{(\gamma,b)\in\mathcal{T}} |\widehat{f}_{\gamma_i,\beta_i} - f_{\gamma_i,\beta_i}| = O_p\left(r_n^{p/(p+1)}\right),$$

where $r_n \to 0$ under Assumption 5.6. ∎

Proof of Theorem 2. The proof is an adaptation of Theorem S2 in Masten (2018). By the construction in Section 4.3,

$$\widehat{f}_{U_{i1},U_{i2}|\gamma_i=\gamma,\beta_i=b}(u_1,u_2) - f_{U_{i1},U_{i2}|\gamma_i=\gamma,\beta_i=b}(u_1,u_2)$$
$$= \mathrm{Re}\left[\frac{1}{(2\pi)^2}\left(\int_{\mathcal{A}_n} \exp[-i(a_1u_1 + a_2u_2)]\,\widehat{\phi}(a)\,da - \int_{\mathbb{R}^2} \exp[-i(a_1u_1 + a_2u_2)]\,\phi(a)\,da\right)\right],$$

where $\phi(a) \equiv \phi_{U_{i1},U_{i2}|\gamma_i=\gamma,\beta_i=b}(a)$ is the conditional characteristic function of U_i and $\widehat{\phi}$ its estimator. Using $|\mathrm{Re}(z)| \leq |z|$ and applying the triangle inequality to complex-valued

integrands, we have

$$\begin{aligned}
&\left|\widehat{f}_{U_{i1},U_{i2}|\gamma_i=\gamma,\beta_i=b}(u_1,u_2) - f_{U_{i1},U_{i2}|\gamma_i=\gamma,\beta_i=b}(u_1,u_2)\right| \\
&\leq \frac{1}{(2\pi)^2}\left|\int_{\mathcal{A}_n} \exp[-i(a_1u_1+a_2u_2)]\,\widehat{\phi}(a)\,da - \int_{\mathbb{R}^2} \exp[-i(a_1u_1+a_2u_2)]\,\phi(a)\,da\right| \\
&\leq \underbrace{\frac{1}{(2\pi)^2}\int_{\mathcal{A}_n} |\widehat{\phi}(a)-\phi(a)|\,da}_{R_1(\bar{\vartheta}_n,\underline{\vartheta}_n)} + \underbrace{\frac{1}{(2\pi)^2}\int_{[-\bar{\vartheta}_n,\bar{\vartheta}_n]^2\setminus\mathcal{A}_n} |\phi(a)|\,da}_{R_2(\underline{\vartheta}_n)} + \underbrace{\frac{1}{(2\pi)^2}\int_{\mathbb{R}^2\setminus[-\bar{\vartheta}_n,\bar{\vartheta}_n]^2} |\phi(a)|\,da}_{R_3(\bar{\vartheta}_n)},
\end{aligned}$$

where the second inequality follows from $|\exp[-i(a_1u_1+a_2u_2)]| = 1$ for any (a,u).

Bound on R_1**.** Recall that $\widehat{\phi}(a) = \int_{\mathcal{U}_n} \exp(i\tilde{u})\,\widehat{f}_{\widetilde{U}_i(a)|\gamma_i=\gamma,\beta_i=b}(\tilde{u})\,d\tilde{u}$, while the true characteristic function $\phi(a)$ integrates $f_{\widetilde{U}_i(a)|\gamma_i=\gamma,\beta_i=b}(\tilde{u})$ over $\mathbb{R}$. Let $C_i(a,\gamma) \equiv \widetilde{U}_i(a) + a_1\gamma y_0$, and $\mathcal{U}_n \equiv \{\tilde{u}\in\mathbb{R} : |\tilde{u}+a_1\gamma y_0| \leq \bar{c}_n\}$. By the triangle inequality and $|\exp(i\tilde{u})| = 1$,

$$\begin{aligned}
|\widehat{\phi}(a)-\phi(a)| &\leq \int_{\mathcal{U}_n} \left|\widehat{f}_{\widetilde{U}_i(a)|\gamma,b}(\tilde{u}) - f_{\widetilde{U}_i(a)|\gamma,b}(\tilde{u})\right|\,d\tilde{u} + \int_{\mathbb{R}\setminus\mathcal{U}_n} f_{\widetilde{U}_i(a)|\gamma,b}(\tilde{u})\,d\tilde{u} \\
&= \int_{\mathcal{C}_n} \left|\widehat{f}_{C_i(a,\gamma)|\gamma,b}(c) - f_{C_i(a,\gamma)|\gamma,b}(c)\right|\,dc + \int_{\mathbb{R}\setminus\mathcal{C}_n} f_{C_i(a,\gamma)|\gamma,b}(c)\,dc \\
&\leq 2\bar{c}_n \sup_{c\in\mathcal{C}_n} \left|\widehat{f}_{C_i(a,\gamma)|\gamma,b}(c) - f_{C_i(a,\gamma)|\gamma,b}(c)\right| + \Pr(|C_i(a,\gamma)| > \bar{c}_n \mid \gamma,b) \\
&\leq 2\bar{c}_n \sup_{c\in\mathcal{C}_n} \left|\widehat{f}_{C_i(a,\gamma)|\gamma,b}(c) - f_{C_i(a,\gamma)|\gamma,b}(c)\right| + \bar{c}_n^{-p}\,E(|C_i(a,\gamma)|^p \mid \gamma,b)\,,
\end{aligned}$$

where the equality follows from the change of variables $c = \tilde{u} + a_1\gamma y_0$, which maps $\mathcal{U}_n$ onto $\mathcal{C}_n$, the second inequality uses that the Lebesgue measure of $\mathcal{C}_n$ is $2\bar{c}_n$, and the last inequality applies Markov's inequality at moment order p of Assumption 5.5(i). By (18) in the proof of Proposition 2, for $a\in\mathcal{A}_n$ and $p>1$,

$$E(|C_i(a,\gamma)|^p \mid \gamma,b) \leq 3^{p-1}\Big(|a_1\gamma y_0|^p + |a_1|^p\,E(|U_{i1}|^p \mid \gamma,b) + |a_2|^p\,E(|U_{i2}|^p \mid \gamma,b)\Big) = O(\bar{\vartheta}_n^p),$$

where the equality uses $|a_1|, |a_2| \leq \bar{\vartheta}_n$ on $\mathcal{A}_n$. Then,

$$\sup_{a\in\mathcal{A}_n} |\widehat{\phi}(a)-\phi(a)| \leq 2\bar{c}_n \sup_{a\in\mathcal{A}_n}\sup_{c\in\mathcal{C}_n} \left|\widehat{f}_{C_i(a,\gamma)|\gamma,b}(c) - f_{C_i(a,\gamma)|\gamma,b}(c)\right| + O(\bar{c}_n^{-p}\bar{\vartheta}_n^p).$$

Using $|\mathcal{A}_n| \leq 4\bar{\vartheta}_n^2$, Lemma A.2, and $\bar{c}_n \asymp r_n^{-1/(p+1)}$, we have

$$R_1 = O_p\big(\bar{\vartheta}_n^2\bar{c}_n\,\varepsilon_n^*\big) + O\big(\bar{c}_n^{-p}\bar{\vartheta}_n^{p+2}\big) = O_p\big(\bar{\vartheta}_n^2 r_n^{-1/(p+1)}\varepsilon_n^*\big) + O\big(\bar{\vartheta}_n^{p+2} r_n^{p/(p+1)}\big)\,,$$

where ε_n^* denotes the uniform convergence rate of the estimator $\widehat{f}_{C_i(a,\gamma)|\gamma,\beta}$ over $\mathcal{A}_n$, as defined in Assumption S.2.

Bound on R_2**.** The region $[-\bar{\vartheta}_n, \bar{\vartheta}_n]^2 \setminus \mathcal{A}_n$ lies in $\{a : |a_1| < \underline{\vartheta}_n\} \cup \{a : |a_2| < \underline{\vartheta}_n\}$. Since $\|a\| \geq |a_2|$, Assumption 5.5(iv) gives $|\phi(a)| \leq M_\phi(1 + \|a\|)^{-\rho} \leq M_\phi(1 + |a_2|)^{-\rho}$ for $\rho > 2$. Thus, integrating $|\phi(a)|$ over the set $\{a : |a_1| < \underline{\vartheta}_n\}$ yields

$$\frac{1}{(2\pi)^2} \int_{|a_1|<\underline{\vartheta}_n} \int_{\mathbb{R}} |\phi(a)| \, da_2 \, da_1 \leq \frac{2M_\phi \, \underline{\vartheta}_n}{(2\pi)^2} \int_{\mathbb{R}} (1 + |a_2|)^{-\rho} \, da_2 = \frac{M_\phi \, \underline{\vartheta}_n}{\pi^2(\rho - 1)} = O(\underline{\vartheta}_n),$$

and the set $\{a : |a_2| < \underline{\vartheta}_n\}$ is bounded symmetrically using $\|a\| \geq |a_1|$. Hence $R_2 = O(\underline{\vartheta}_n)$.

Bound on R_3**.** Since $\mathbb{R}^2 \setminus [-\bar{\vartheta}_n, \bar{\vartheta}_n]^2 \subseteq \{a : \|a\| > \bar{\vartheta}_n\}$, under Assumption 5.5(iv),

$$\begin{aligned} R_3 &\leq \frac{M_\phi}{(2\pi)^2} \int_{\|a\|>\bar{\vartheta}_n} (1 + \|a\|)^{-\rho} \, da = \frac{M_\phi}{(2\pi)^2} \int_0^{2\pi} \int_{\bar{\vartheta}_n}^{\infty} (1 + t)^{-\rho} \, t \, dt \, d\theta \leq \frac{M_\phi}{2\pi} \int_{\bar{\vartheta}_n}^{\infty} t^{1-\rho} \, dt \\ &= \frac{M_\phi \, \bar{\vartheta}_n^{-(\rho-2)}}{2\pi(\rho - 2)} = O\big(\bar{\vartheta}_n^{-(\rho-2)}\big), \end{aligned}$$

where the first equality uses polar coordinates $a = (t\cos\theta, t\sin\theta)$, the second inequality uses $\int_0^{2\pi} d\theta = 2\pi$ and $(1 + t)^{-\rho} \, t \leq t^{1-\rho}$ for $t \geq \bar{\vartheta}_n > 0$, and the second equality holds for $\rho > 2$. Because the bounds on R_1, R_2, and R_3 do not depend on (u_1, u_2),

$$\begin{aligned} &\sup_{(u_1,u_2)\in\mathbb{R}^2} \left| \widehat{f}_{U_{i1},U_{i2}|\gamma_i=\gamma,\beta_i=b}(u_1, u_2) - f_{U_{i1},U_{i2}|\gamma_i=\gamma,\beta_i=b}(u_1, u_2) \right| \\ &= O_p\big(\bar{\vartheta}_n^2 r_n^{-1/(p+1)} \varepsilon_n^*\big) + O\big(\bar{\vartheta}_n^{p+2} r_n^{p/(p+1)}\big) + O(\underline{\vartheta}_n) + O\big(\bar{\vartheta}_n^{-(\rho-2)}\big) = o_p(1), \end{aligned}$$

where $\rho > 2$, the first two terms vanish under Assumption S.2, and the last two vanish because $\underline{\vartheta}_n \to 0$ and $\bar{\vartheta}_n \to \infty$. ■

B Extension: Long Horizons with $T \geq 3$

Our identification extends to panels with longer horizons ($T \geq 3$). To fix ideas, consider a model with only strictly exogenous covariates X_{it}. Suppose

$$Y_{it} = \gamma_i Y_{it-1} + X_{it}'\beta_{it} + U_{it}, \qquad t = 1, \ldots, T,$$

and let $H_i \equiv (X_{i1}', \ldots, X_{iT}')'$. Recursive substitution gives

$$\sum_{t=1}^{T} d_t Y_{it} = \underbrace{\Big(\sum_{t=1}^{T} d_t \gamma_i^t\Big) Y_{i0} + \sum_{s=1}^{T} p_s(\gamma_i; d) U_{is}}_{C_i(d)} + \sum_{s=1}^{T} X_{is}' \underbrace{p_s(\gamma_i; d)\, \beta_{is}}_{S_{is}(d)},$$

where $p_s(\gamma; d) \equiv \sum_{t=s}^{T} d_t \gamma^{t-s}$. This reduces to (6) without the W_{i1} term when $T = 2$.

B.1 Time-invariant random slopes

Suppose that the random coefficients for strictly exogenous covariates are time-invariant ($\beta_{it} = \beta_i \in \mathbb{R}^{J_x}$ for all $t = 1, \ldots, T$). Then

$$S_{i,T-1}(d) = (d_{T-1} + d_T \gamma_i)\beta_i, \qquad S_{iT}(d) = d_T \beta_i.$$

In addition to the analogs of Assumption 3.4 and 3.5 maintained throughout, we assume $\Pr\{\beta_{i,k} = 0 \mid Y_{i0}\} = 0$ for some k. For any d with $d_T \neq 0$ and any k with $\Pr\{\beta_{i,k} = 0 \mid Y_{i0}\} = 0$, the map $(\gamma_i, \beta_i) \mapsto (S_{i,T-1,k}(d), S_{iT}(d))$ is invertible with Jacobian determinant $|d_T^{J_x+1} \beta_{i,k}|$. The joint density of $(C_i(d), \gamma_i, \beta_i)$ given Y_{i0} then follows from a change of variables similar to (9), with the remaining slopes providing over-identifying restrictions.

It remains to recover the conditional distribution of time-varying errors U_{it}. For any $a \equiv (a_1, \ldots, a_T)$ and realized coefficient $\gamma_i = \gamma$, define

$$d_s(a, \gamma) = a_s - \gamma a_{s+1} \text{ for } s = 1, \ldots, T, \text{ with } a_{T+1} \equiv 0,$$

so that $p_s(\gamma; d(a, \gamma)) = a_s$. The characteristic function of $(U_{i1}, \ldots, U_{iT})$ conditional on $(\gamma_i, \beta_i, Y_{i0})$ is then recovered exactly as in (10); continuity handles the boundary case $a_T = 0$. The random coefficients and intercepts are thus identified under moment determinacy and support conditions as stated in the main text. A sufficient condition for the moment determinacy condition at $T \geq 3$ is that, conditional on Y_{i0}, γ_i has bounded support while the components of (β_i, U_i) have sub-Gaussian (or merely sub-exponential) tails.

B.2 Time-varying random slopes

If β_{it} are time-varying, then for any fixed d, the mapping from $(\gamma_i, \beta_{i1}, \ldots, \beta_{iT}) \in \mathbb{R}^{TJ_x+1}$ to $(S_{i1}(d), \ldots, S_{iT}(d)) \in \mathbb{R}^{TJ_x}$ is not invertible. Nevertheless, varying d jointly with

the direction in which $(S_{i1}(d), \ldots, S_{iT}(d))$ is evaluated helps to identify the full joint distribution of $(\gamma_i, \beta_{i1}, \ldots, \beta_{iT})$. We maintain the following conditions, which are analogs of Assumption 3.4 and 3.5.

Assumption B.1. *(i)* $(\gamma_i, \beta_i, U_i) \perp H_i \mid Y_{i0}$*, and* $(\gamma_i, \beta_i, U_i) \mid Y_{i0}$ *admits a Lebesgue density almost surely.*

(ii) The support of $H_i \mid Y_{i0}$ *contains an open ball in* $\mathbb{R}^{TJ_x}$ *almost surely; and for every* $d \in \mathbb{R}^T$*, the conditional distribution of* $(C_i(d), S_{i1}(d), \ldots, S_{iT}(d))$ *given* Y_{i0} *is uniquely determined by its moments and has finite absolute moments of all orders almost surely.*

Under Assumption B.1, the joint distribution of $(C_i(d), S_{i1}(d), \ldots, S_{iT}(d))$ given Y_{i0} is identified from that of $\left(\sum_{t=1}^T d_t Y_{it},\ H_i'\right)$ given Y_{i0} for every $d \in \mathbb{R}^T$. This is the general-T analog of Lemma 3.1 and follows from Lemma 2 of Masten (2018) by the same argument, with $H_i \in \mathbb{R}^{TJ_x}$ playing the role of the regressors.

Proposition 3. *Suppose Assumption B.1 holds. Then, for any fixed* $T \geq 2$ *and* $J_x \geq 1$*, the joint distribution of* $(\gamma_i, \beta_{i1}, \ldots, \beta_{iT})$ *given* Y_{i0} *is identified.*

Proof of Proposition 3. The proof is conditional on Y_{i0}, which we suppress in notation. The target is the joint characteristic function (c.f.) of $(\gamma_i, \beta_{i1}', \ldots, \beta_{iT}')$:

$$\Phi(g, t) \equiv E\Big[\exp\Big\{i\Big(g\gamma_i + \sum\nolimits_{s=1}^T t_s'\beta_{is}\Big)\Big\}\Big], \qquad g \in \mathbb{R}, \qquad t = (t_1', \ldots, t_T')' \in \mathbb{R}^{TJ_x}.$$

Since Φ is uniformly continuous, it suffices to identify $\Phi(g, t)$ for every g and every t in the open dense set $\mathcal{T}_0 \equiv \{t : t_s \neq 0 \text{ for all } s\}$. Fix such (g, t) for the remainder of the proof, and define the scalar variables $V_{is}(t_s) \equiv t_s'\beta_{is}$, $s = 1, \ldots, T$. By construction, $\Phi(g, t)$ is the c.f. of $(\,\gamma_i, V_{i1}(t_1), \ldots, V_{iT}(t_T)\,)$ evaluated at $(g, 1, \ldots, 1)$, so it suffices to identify the joint density of $(\gamma_i, V_{i1}(t_1), \ldots, V_{iT}(t_T))$.[7]

(Step 1.) For the fixed vector $t \in \mathbb{R}^{TJ_x}$, and for any $\lambda \in \mathbb{R}$, we can construct a directional vector

$$\tilde{t}(\lambda) \equiv (t_1', t_2', \ldots, t_{T-1}', \lambda t_T')',$$

and evaluate the vector $(S_{i1}(d), \ldots, S_{iT}(d))$ in the direction $\tilde{t}(\lambda)$. Recall that $S_{is}(d) =$

[7] This density exists under Assumption B.1(i), because $V(t)$ is a linear function of $(\gamma_i, \beta_{i1}', \ldots, \beta_{iT}')$.

$p_s(\gamma_i; d)\beta_{is}$ is a scalar multiple of β_{is}, and $p_T(\gamma; d) = d_T$. Therefore,

$$\tilde{t}_s(\lambda)' S_{is}(d) = p_s(\gamma_i; d)\, \tilde{t}_s(\lambda)'\beta_{is} = \begin{cases} p_s(\gamma_i; d)\, V_{is}(t_s), & s = 1, \ldots, T-1, \\ d_T\, \lambda V_{iT}(t_T), & s = T. \end{cases}$$

To simplify notation, we suppress t_s in $V_{is}(t_s)$ for $s = 1, \ldots, T$ below.

Summing over s and substituting $p_s(\gamma; d) = \sum_{r=s}^{T} d_r \gamma^{r-s}$,

$$\begin{aligned} \sum_{s=1}^{T} \tilde{t}_s(\lambda)' S_{is}(d) &= \sum_{s=1}^{T-1} \sum_{r=s}^{T} d_r\, \gamma_i^{r-s} V_{is} \;+\; d_T\, \lambda V_{iT} = \sum_{r=1}^{T} d_r \sum_{s=1}^{\min(r, T-1)} \gamma_i^{r-s} V_{is} \;+\; d_T\, \lambda V_{iT} \\ &= \sum_{r=1}^{T-1} d_r \underbrace{\sum_{s=1}^{r} \gamma_i^{r-s} V_{is}}_{\equiv\, W_r} + d_T \underbrace{\sum_{s=1}^{T-1} \gamma_i^{T-s} V_{is}}_{\equiv\, \gamma_i W_{T-1}} + d_T\, \lambda V_{iT}, \end{aligned}$$

where the second equality interchanges the order of summation over $\{(s, r) : 1 \le s \le T-1,\ s \le r \le T\} = \{(s, r) : 1 \le r \le T,\ 1 \le s \le \min(r, T-1)\}$, and the third separates the term $r = T$, whose inner sum is $\sum_{s=1}^{T-1} \gamma_i^{T-s} V_{is} = \gamma_i \left(\sum_{s=1}^{T-1} \gamma_i^{T-1-s} V_{is} \right) \equiv \gamma_i W_{T-1}$. Note $W_r(\cdot)$ is indexed by $(t_1, \ldots, t_r)$ through $(V_{i1}, \ldots, V_{ir})$, which we suppress in notation.

Collecting terms,

$$\sum\nolimits_{s=1}^{T} \tilde{t}_s(\lambda)' S_{is}(d) = \sum\nolimits_{r=1}^{T-1} d_r\, W_r + d_T \big(\gamma_i W_{T-1} + \lambda V_{iT} \big), \qquad d \in \mathbb{R}^T. \tag{20}$$

As noted above, the joint distribution of $(S_{i1}(d), \ldots, S_{iT}(d))$ given Y_{i0} is identified for all $d \in \mathbb{R}^T$ under Assumption B.1. Therefore, for any fixed $t \in \mathbb{R}^{TJ_x}$, the distribution of the left-hand side of (20) is identified for every $(d, \lambda) \in \mathbb{R}^{T+1}$. (Recall that $\tilde{t}(\lambda)$ is constructed from the direction vector t in the c.f. of interest, $\Phi(g, t)$.)

Hence, the quantity

$$E\Big[\exp\Big\{ i \Big(\sum_{r=1}^{T-1} d_r W_r + d_T \gamma_i W_{T-1} + d_T \lambda V_{iT} \Big) \Big\} \Big] = \phi_{\Xi}(d_1, \ldots, d_{T-1},\, d_T,\, d_T \lambda) \tag{21}$$

is identified for all $(d, \lambda) \in \mathbb{R}^{T+1}$, where $\phi_{\Xi}(\cdot)$ is the characteristic function of

$$\Xi \equiv (W_1, W_2, \ldots, W_{T-1}, \gamma_i W_{T-1}, V_{iT}) \in \mathbb{R}^{T+1}.$$

Note Ξ is a random array indexed by the fixed vector $t \in \mathbb{R}^{TJ_x}$ alone, and *not* by the vector (d, λ); the latter only determines the direction argument entering $\phi_\Xi(d_1, \ldots, d_T, \lambda d_T)$.

By varying d over the set $\{d \in \mathbb{R}^T : d_T \neq 0\}$ and λ over $\mathbb{R}$, the last argument $z \equiv d_T\lambda$ in (21) ranges over $\mathbb{R}$. So, we can use (21) to recover the c.f. of Ξ over the set $\{(d_1, \ldots, d_T, z) \in \mathbb{R}^{T+1} : d_T \neq 0\}$, which is dense in $\mathbb{R}^{T+1}$. Characteristic functions are uniformly continuous, so ϕ_Ξ is identified on $\mathbb{R}^{T+1}$. By Lévy's uniqueness theorem, this determines the distribution of Ξ.

(*Step 2.*) Define $\psi(\gamma, v_1, \ldots, v_T) \equiv (w_1, \ldots, w_{T-1}, \gamma w_{T-1}, v_T)$ with $w_r \equiv \sum_{s=1}^r \gamma^{r-s} v_s$, so that $\psi(\gamma_i, V_{i1}, \ldots, V_{iT}) = \Xi$. Over $\{w_{T-1} \neq 0\}$, ψ is injective: γ is recovered as the ratio of the T-th to the $(T-1)$-th coordinate, and then $v_1 = w_1$, $v_r = w_r - \gamma w_{r-1}$ for $r = 2, \ldots, T-1$, while v_T is the last coordinate. Decompose the mapping ψ as the composite of two: $\psi_2 \circ \psi_1$, where $\psi_1(\gamma, v_1, \ldots, v_T) \equiv (\gamma, w_1, \ldots, w_{T-1}, v_T)$ is triangular with unit Jacobian, and $\psi_2(\gamma, w_1, \ldots, w_{T-1}, v_T) \equiv (w_1, \ldots, w_{T-1}, \gamma w_{T-1}, v_T)$ has Jacobian determinant $(-1)^{T+1} w_{T-1}$. Hence the Jacobian of ψ satisfies $|\det \mathrm{D}\psi| = |w_{T-1}|$. Under Assumption B.1(i), $\Pr\{W_{T-1} = 0\} = 0$. Therefore, for almost every $(\gamma, v) \in \mathbb{R}^{T+1}$,

$$f_{\gamma_i, V_{i1}, \ldots, V_{iT}}(\gamma, v_1, \ldots, v_T) = f_\Xi\big(w_1, \ldots, w_{T-1}, \gamma w_{T-1}, v_T\big)\,\big|w_{T-1}\big|,$$

where $w_r = \sum_{s=1}^r \gamma^{r-s} v_s$. This identifies the joint distribution of $(\gamma_i, V_{i1}, \ldots, V_{iT})$.

(*Step 3.*) By construction, for any fixed vector (g, t),

$$\Phi(g, t) = \int_{\mathbb{R}^{T+1}} \exp\Big\{i\Big(g\gamma + \sum\nolimits_{s=1}^T v_s\Big)\Big\}\, f_{\gamma_i, V_{i1}, \ldots, V_{iT}}(\gamma, v)\, d\gamma\, dv,$$

where t is the fixed, directional vector defining $(V_{i1}, \ldots, V_{iT})$. (Recall that such dependence is suppressed in notation.) With the distribution of $(\gamma_i, V_{i1}, \ldots, V_{iT})$ recovered in Step 2, $\Phi(g, t)$ is identified for every $g \in \mathbb{R}$ and $t \in \mathcal{T}_0$, and by continuity on all of $\mathbb{R}^{1+TJ_x}$. Since Φ is the joint characteristic function of $(\gamma_i, \beta_{i1}, \ldots, \beta_{iT})$, the conclusion follows. ■

C Relation to Existing Literature

Graham and Powell (2012) study a correlated random coefficient panel data model:

$$Y_t = \chi_t' \mathcal{B}_t, \text{ where } E(\mathcal{B}_t \mid \chi^T) = \mathcal{B}_0(\chi^T) + \mathcal{D}_t,$$

where $\mathcal{B}_0(\cdot)$ is a deterministic function that does not vary over time and $\mathcal{D}_t$ is a vector of constants that may vary over $t = 1, 2, \ldots, T$. Mapping their specification into the reduced form of our model with $T = 2$, we have $\chi_1 \equiv (1, Y_0, X_1, W_1)'$, $\chi_2 \equiv (1, Y_0, X_1, X_2, W_1, W_2)'$ and $\mathcal{B}_1 \equiv (U_1, \gamma, \beta_1, \delta_1)'$, $\mathcal{B}_2 \equiv (\gamma U_1 + U_2, \gamma^2, \gamma\beta_1, \beta_2, \gamma\delta_1, \delta_2)'$.

Our model is not nested within Graham and Powell (2012), because the mean of the random intercept in $\mathcal{B}_2$, i.e., $E(\gamma U_1 + U_2 \mid \chi^T)$, is not additive in any stationary function $\mathcal{B}_0(\chi^T)$ and a constant $\mathcal{D}_2$ due to the sequential exogeneity of W_t in χ_t. Besides, Graham and Powell (2012) study the (ir)regular identification of average partial effects $E(\mathcal{B}_t)$ when the number of time periods is no smaller than the dimension of covariates, using cross-section units whose covariates change little over time. In contrast, we assume the coefficients for sequentially exogenous covariates are constant, and are interested in recovering the joint distribution of the random coefficients and intercepts. Our method does not require the number of time periods to be strictly larger than the dimension of exogenous covariates.

Masten and Torgovitsky (2016) study the identification of correlated random coefficient models in the following form:

$$Y^* = B_0^* + B_1^* X^* + B_2^* Z_2^*, \qquad X^* = h(Z^*, V^*), \tag{22}$$

where $h(\cdot)$ is an unknown function with a *scalar* noise V^*, and $Z^* \equiv (Z_1^*, Z_2^*)$ is a vector of instruments. The identifying conditions include:

(a) $Z^* \perp (V^*, B^*)$, where $B^* \equiv (B_0^*, B_1^*, B_2^*)$; (b) $h(Z^*, V^*)$ is increasing in $V^* \in \mathbb{R}$.

For simplicity, let Y^*, X^*, Z_2^* all be scalars, and suppress the individual subscript i below. To fit our dynamic panel data model with random coefficients into the model of Masten and Torgovitsky (2016) in (22), one may relate the variables in our model to those in Masten and Torgovitsky (2016) as follows:

$$Y^* \equiv Y_2,\ X^* \equiv Y_1,\ Z_2^* \equiv X_2,\ Z_1^* \equiv X_1;\ B_1^* \equiv \gamma,\ B_2^* \equiv \beta_2,\ B_0^* \equiv U_2.$$

To simplify comparison, let the initial condition Y_0 in our model be degenerate at zero so that the structural equation in the first period is simplified to $Y_1 = X_1\beta_1 + U_1$, where β_1 is a random coefficient. However, it is known from Imbens (2007) and Kasy (2011) that this cannot be fitted into the control function framework above.

Hoderlein *et al.* (2017) consider a triangular model with random coefficients:

$$\widetilde{Y} = \widetilde{B}_0 + \widetilde{B}_1\widetilde{X} + \widetilde{B}_2'\widetilde{W},$$
$$\widetilde{X} = \widetilde{A}_0 + \widetilde{A}_1'\widetilde{Z} + \widetilde{A}_2'\widetilde{W},$$

where $\widetilde{X}, \widetilde{Y}$ are scalars. Let $\widetilde{A} \equiv (\widetilde{A}_0, \widetilde{A}_1', \widetilde{A}_2')'$; likewise for $\widetilde{B}$. The main identifying conditions in Hoderlein *et al.* (2017) (Theorem 7) include:

(*i*) $(\widetilde{Z}', \widetilde{W}')' \perp (\widetilde{A}', \widetilde{B}')'$; (*ii*) the support of $\widetilde{Z}$ contains an open interval; (*iii*) $\widetilde{A}_1 \perp \widetilde{B}$.

To map this setup into our model (with $T = 2$ and suppressing i), let

$$\widetilde{Y} \equiv Y_2,\ \widetilde{X} \equiv Y_1,\ \widetilde{Z} \equiv Y_0,\ \widetilde{W} \equiv X \equiv (X_1, X_2),$$
$$\widetilde{B}_1 \equiv \gamma,\ \widetilde{B}_2 \equiv (0, \beta_2)',\ \widetilde{B}_0 \equiv U_2,\ \widetilde{A}_1 \equiv \gamma,\ \widetilde{A}_2 \equiv (\beta_1, 0)',\ \widetilde{A}_0 \equiv U_1.$$

Our identification result differs from Hoderlein *et al.* (2017) in that we relax (i)–(iii) within the specific context of a random-coefficient dynamic panel data model, and also identify the joint distribution of random *intercepts*. First, we allow the random coefficients $(\widetilde{A}', \widetilde{B}')'$, which consist of $(\gamma, \beta, U_1, U_2)$ in our case, to be correlated with $\widetilde{Z}$, which is Y_0 in our case. Second, we allow $\widetilde{Z}$ (Y_0) to have limited or even discrete support. In fact, our identification strategy conditions on $\widetilde{Z}$ (Y_0) and only exploits strict exogeneity in $\widetilde{W}$ (X_i). Third, in our model, $\widetilde{A}_1 = \widetilde{B}_1 = \gamma$, which is allowed to be correlated with (β, U_1, U_2) in $(\widetilde{A}_0, \widetilde{B}_0, \widetilde{A}_2', \widetilde{B}_2')$. Last but not least, we also identify the *joint* distribution of $(\widetilde{A}_0, \widetilde{B}_0) = (U_1, U_2)$ and (γ, β) in $(\widetilde{A}_1, \widetilde{A}_2', \widetilde{B}_1, \widetilde{B}_2')$, whereas Theorem 7 in Hoderlein *et al.* (2017) identifies the joint distribution of $(\widetilde{B}_0, \widetilde{B}_1)$.[8]

To be clear, we are able to obtain these results without (i)–(iii) because of the specific structure in the random-coefficient dynamic panel data model.

Masten (2018, Proposition 4) studied a triangular system

$$Y_2 = \gamma Y_1 + \delta_2^{*\prime} W^* + U_2,$$
$$Y_1 = \beta_1 X_1 + \delta_1^{*\prime} W^* + U_1,$$

where $X_1 \perp (\gamma, \beta_1, \delta_1^*, \delta_2^*, U_1, U_2) \mid W^*$, and identified the distribution of $(\gamma, \beta_1) \mid W^*$.

[8]Theorem 5 in Hoderlein *et al.* (2017) identifies the joint distribution of $(\widetilde{B}_0, \widetilde{B}_1, \widetilde{B}_2')'$ under a stronger condition that the support of $(\widetilde{Z}', \widetilde{W}')'$ is $\mathbb{R}^{\dim(\widetilde{Z})+\dim(\widetilde{W})}$.

Our model in (1) can be partly reconciled with that of Masten (2018), with W^* consisting of $(W_{i1}, W_{i2}, Y_{i0})'$, $\delta_1^* \equiv (\delta_1, 0, \gamma)'$, $\delta_2^* \equiv (0, \delta_2, 0)'$, and the random coefficient for Y_0 in the structural form of Y_1 identical to that for Y_1 in the structural form of Y_2. Nonetheless, our model is non-nested with Masten (2018), because we include a strictly exogenous regressor X_2 with a random coefficient β_2 in the equation for Y_2 and identify the joint distribution of $(\gamma, \beta_1, \beta_2)$.

Yet an even more important distinction between our work and Masten (2018) is that we are able to identify the joint distribution of these random coefficients *and the random intercepts* (U_1, U_2), under the additional assumptions that δ_1, δ_2 are constants,[9] and that the random coefficients and intercepts are also conditionally independent from W_{i1}.

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[9] Masten (2018) does not provide results for identifying the distribution of these random coefficients (δ_1, δ_2) in his model.

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Online Appendix to "Estimation of Random-Coefficient Dynamic Panel Data Models with a Fixed T"

Xun Tang and Pei Yu

This Online Appendix contains additional proofs and Monte Carlo results to support the main text. Throughout, we maintain the identification conditions in Theorem 1.

S1 Auxiliary Lemmas for Proposition 2

Fix $d \in \mathbb{R}^2$ with $d_2 \neq 0$, and let $\mathcal{S} \subset \mathbb{R}^l$ be a compact subset of the joint support of $(C_i(d), S_i(d))$.

Lemma S.1 (Bias term of the RTE)**.** *Suppose Assumptions 5.2*(i) *and 5.4*(iv) *hold. Then there exist constants $M_T, M_A > 0$ such that*

$$\sup_{(c,s)\in\mathbb{R}^l} |(A_{\nu,n} - A_\nu)(Rf_{C_i,S_i})(c,s)| \leq M_T\, B\, \underline{q}_n^{1/2},$$
$$\sup_{(c,s)\in\mathbb{R}^l} |A_\nu(Rf_{C_i,S_i})(c,s) - f_{C_i,S_i}(c,s)| \leq M_A\, B\, \nu^{m-l/2-\varsigma},$$

where A_ν and $A_{\nu,n}$ are the operators defined in Lemma A.1.

Proof of Lemma S.1. Write $I_{3,1} \equiv (A_{\nu,n} - A_\nu)(Rf_{C_i,S_i})$ and $I_{3,2} \equiv A_\nu(Rf_{C_i,S_i}) - f_{C_i,S_i}$. *For the trimming bias $I_{3,1}$,*

$$\begin{aligned}
I_{3,1} &= -2 \int_{\{q\in\mathbb{S}_+^{l-1}:\, q_1<\underline{q}_n\}} \int_{-\infty}^{\infty} K_\nu\left(q'(c,s')' - v\right)\, (Rf_{C_i,S_i})(q,v)\, dv\, d\mu(q) \\
&= -2(2\pi)^{-1} \int_{\{q\in\mathbb{S}_+^{l-1}:\, q_1<\underline{q}_n\}} \int_{-\infty}^{\infty} \mathcal{F}K_\nu(t)\, e^{-it\, q'(c,s')'}\, \mathcal{F}_1(Rf_{C_i,S_i})(q,t)\, dt\, d\mu(q) \\
&= -(2\pi)^{-l} \int_{\{q\in\mathbb{S}_+^{l-1}:\, q_1<\underline{q}_n\}} \int_{-\infty}^{\infty} |t|^{l-1}\, \mathcal{L}(\nu|t|)\, e^{-it\, q'(c,s')'}\, \mathcal{F}f_{C_i,S_i}(tq)\, dt\, d\mu(q) \\
&= -(2\pi)^{-l} \int_{\{\xi\in\mathbb{R}^l:|\xi_1|<\underline{q}_n\|\xi\|\}} \mathcal{L}(\nu\|\xi\|)\, \mathcal{F}f_{C_i,S_i}(\xi)\, e^{-i\xi'(c,s')'}\, d\xi,
\end{aligned}$$

where $\mathcal{F}_1$ denotes the one-dimensional Fourier transform, the third equality uses the projection theorem $\mathcal{F}_1(Rf_{C_i,S_i})(q,t) = \mathcal{F}f_{C_i,S_i}(tq)$ and the definition of $\mathcal{F}K_\nu$ in Footnote 6, and the last is the change to polar coordinates $\xi = tq$ with $q \in \mathbb{S}_+^{l-1}$, under which $|t|^{l-1}\,dt\,d\mu(q) = d\xi$ and $\|\xi\| = |t|$. Hence, by the triangle inequality and $0 \le \mathcal{L} \le 1$,

$$\begin{aligned}
\sup_{(c,s)\in\mathbb{R}^l} |I_{3,1}| &\le (2\pi)^{-l} \int_{\{\xi\in\mathbb{R}^l : |\xi_1| < \underline{q}_n\|\xi\|\}} |\mathcal{F}f_{C_i,S_i}(\xi)|\,d\xi \\
&\le (2\pi)^{-l} \left(\int_{\{\xi\in\mathbb{R}^l : |\xi_1| < \underline{q}_n\|\xi\|\}} (1+\|\xi\|^2)^m \,|\mathcal{F}f_{C_i,S_i}(\xi)|^2\,d\xi \right)^{1/2} \\
&\quad \times \left(\int_{\{\xi\in\mathbb{R}^l : |\xi_1| < \underline{q}_n\|\xi\|\}} (1+\|\xi\|^2)^{-m} d\xi \right)^{1/2} \\
&\le (2\pi)^{-l} B \left(\int_{\{\xi\in\mathbb{R}^l : |\xi_1| < \underline{q}_n\|\xi\|\}} \left(1+\|\xi\|^2\right)^{-m} d\xi \right)^{1/2},
\end{aligned}$$

where the second inequality is by the Cauchy–Schwarz inequality and the third uses $\|f_{C_i,S_i}\|_m \le B$ in Assumption 5.2(i) together with $\{\xi \in \mathbb{R}^l : |\xi_1| < \underline{q}_n\|\xi\|\} \subset \mathbb{R}^l$. Using polar coordinates $\xi = tq$ with $q \in \mathbb{S}_+^{l-1}$ and $t \in \mathbb{R}$,

$$\begin{aligned}
\int_{\{\xi\in\mathbb{R}^l : |\xi_1| < \underline{q}_n\|\xi\|\}} (1+\|\xi\|^2)^{-m}\,d\xi &= 2\int_{\{q\in\mathbb{S}_+^{l-1} : q_1 < \underline{q}_n\}} \int_0^\infty t^{l-1}(1+t^2)^{-m}\,dt\,d\mu(q) \\
&= O(\underline{q}_n) \int_0^\infty t^{l-1}(1+t^2)^{-m}\,dt,
\end{aligned}$$

where the integral is finite under $m > l/2$, and $\mu(\{q \in \mathbb{S}_+^{l-1} : q_1 < \underline{q}_n\}) = \int_0^{\underline{q}_n} \omega_{l-1}(1-q_1^2)^{(l-3)/2}\,dq_1 = O(\underline{q}_n)$ as in (15). Hence, for some positive constant $M_T = M_T(l,m)$,

$$\sup_{(c,s)\in\mathbb{R}^l} |I_{3,1}| \le M_T\, B\, \underline{q}_n^{1/2}.$$

For the approximation bias $I_{3,2}$, the kernel K_ν is defined by Equations (7)–(8) of Bissantz *et al.* (2014), and the filter $\mathcal{L}$ satisfies their Assumption 1(i)–(iiia) with smoothness parameter $M = r$ (see their Example 1), where $r \ge m - l/2$ by Assumption 5.4(iv). By their Lemma 1 and its proof, there exists a positive constant $M_A = M_A(\varsigma, l, m)$, independent of B, such that, uniformly over $f_{C_i,S_i} \in \mathcal{W}^m(\mathbb{R}^l; B)$,

$$\sup_{(c,s)\in\mathbb{R}^l} |I_{3,2}| \le M_A\, B\, \nu^{m-l/2-\varsigma}.$$

■

Lemma S.2 (Variance term of the RTE)**.** *Suppose Assumptions 5.2*(ii)*, 5.3*(i) *and* (iii)*, and 5.4*(i)–(ii) *hold. Then, for any ν with $\nu \to 0$ and $\underline{q}_n^{-k}\sqrt{\ln n/(n\nu)} \to 0$ as $n \to \infty$,*

$$\sup_{(c,s)\in\mathcal{S}} \left|\bar{f}_{C_i(d),S_i(d)}(c,s) - E\left(\bar{f}_{C_i(d),S_i(d)}(c,s)\right)\right| = O_p\left(\Xi_{q,n}^{1/2}\,\nu^{-l+1/2}\sqrt{\ln n/n}\right),$$

where $\bar{f}_{C_i(d),S_i(d)}$ denotes the infeasible Radon transform estimator constructed with the true design density f_{Q_i} and $\Xi_{q,n} \equiv \int_{\mathbb{S}(\underline{q}_n)} f_{Q_i}^{-1}(q)\,d\mu(q)$.

Proof of Lemma S.2. The argument adapts the uniform convergence proof of Theorem 1.4 in Li and Racine (2007) to the Radon transform estimator. Since $\mathcal{S}$ is compact, it can be covered by a finite number of cubes $\{\mathcal{S}_k\}_{k=1}^{N_n}$ with centers $(\dot{c}_k, \dot{s}_k)$ and side length ℓ_n. We suppress d in $C_i(d)$ and $S_i(d)$. Then,

$$\begin{aligned}
&\sup_{(c,s)\in\mathcal{S}} \left|\bar{f}_{C_i,S_i}(c,s) - E(\bar{f}_{C_i,S_i}(c,s))\right| = \max_{1\le k\le N_n} \sup_{(c,s)\in\mathcal{S}\cap\mathcal{S}_k} \left|\bar{f}_{C_i,S_i}(c,s) - E(\bar{f}_{C_i,S_i}(c,s))\right| \\
&\le \underbrace{\max_{1\le k\le N_n} \sup_{(c,s)\in\mathcal{S}\cap\mathcal{S}_k} \left|\bar{f}_{C_i,S_i}(c,s) - \bar{f}_{C_i,S_i}(\dot{c}_k,\dot{s}_k)\right|}_{I_{2,1}} + \underbrace{\max_{1\le k\le N_n} \left|\bar{f}_{C_i,S_i}(\dot{c}_k,\dot{s}_k) - E(\bar{f}_{C_i,S_i}(\dot{c}_k,\dot{s}_k))\right|}_{I_{2,2}} \\
&+ \underbrace{\max_{1\le k\le N_n} \sup_{(c,s)\in\mathcal{S}\cap\mathcal{S}_k} \left|E(\bar{f}_{C_i,S_i}(\dot{c}_k,\dot{s}_k)) - E(\bar{f}_{C_i,S_i}(c,s))\right|}_{I_{2,3}}.
\end{aligned}$$

First, consider $I_{2,2}$. Define $T_n(c,s) \equiv \bar{f}_{C_i,S_i}(c,s) - E(\bar{f}_{C_i,S_i}(c,s)) = \sum_{i=1}^n T_{n,i}(c,s)$, with

$$T_{n,i}(c,s) \equiv \frac{2}{n}\left[\frac{\mathbf{1}\{Q_{i1}\ge \underline{q}_n\}}{f_{Q_i}(Q_i)}K_\nu(Q_i'(c,s')' - V_i) - E\left(\frac{\mathbf{1}\{Q_{i1}\ge \underline{q}_n\}}{f_{Q_i}(Q_i)}K_\nu(Q_i'(c,s')' - V_i)\right)\right].$$

For any sequence $\epsilon_n > 0$, we can write $\Pr(I_{2,2} > \epsilon_n)$ as

$$\Pr\left(\max_{1\le k\le N_n} |T_n(\dot{c}_k,\dot{s}_k)| > \epsilon_n\right) \le \sum_{k=1}^{N_n} \Pr\left(|T_n(\dot{c}_k,\dot{s}_k)| > \epsilon_n\right) \le N_n \sup_{(c,s)\in\mathcal{S}} \Pr(|T_n(c,s)| > \epsilon_n). \tag{23}$$

Applying the Chernoff bound, for any sequence $\chi_n > 0$,

$$\begin{aligned}\Pr(|T_n(c,s)| > \epsilon_n) &= \Pr\left(\sum_{i=1}^n T_{n,i}(c,s) > \epsilon_n\right) + \Pr\left(-\sum_{i=1}^n T_{n,i}(c,s) > \epsilon_n\right) \\ &\leq \frac{E[\exp(\sum_{i=1}^n \chi_n T_{n,i}(c,s))] + E[\exp(-\sum_{i=1}^n \chi_n T_{n,i}(c,s))]}{\exp(\chi_n \epsilon_n)}.\end{aligned} \tag{24}$$

Choose $\chi_n = \Xi_{q,n}^{-1/2} \nu^{l-1/2} \sqrt{n \ln n}$. By Assumption 5.3(iii), $M_{q,n} \equiv \inf_{q \in \mathbb{S}(\underline{q}_n)} f_{Q_i}(q) \geq M_q \underline{q}_n^k > 0$. Under Assumption 5.4(i),

$$\chi_n |T_{n,i}(c,s)| \leq 4 \sup|K| M_{q,n}^{-1} \Xi_{q,n}^{-1/2} \sqrt{\frac{\ln n}{n\nu}} \leq 4 \sup|K| M_q^{-1} \underline{q}_n^{-k} \Xi_{q,n}^{-1/2} \sqrt{\frac{\ln n}{n\nu}}.$$

By Assumption 5.3(i), there exists a constant $\bar{M}_q > 0$ such that $\sup_{q \in \mathbb{S}(\underline{q}_n)} f_{Q_i}(q) \leq \bar{M}_q < \infty$. Hence

$$\Xi_{q,n} = \int_{\mathbb{S}(\underline{q}_n)} f_{Q_i}^{-1}(q)\, d\mu(q) \geq \bar{M}_q^{-1} \mu(\mathbb{S}(\underline{q}_n)).$$

Since $\mathbb{S}(\underline{q}_n) \to \mathbb{S}_+^{l-1}$ as $\underline{q}_n \to 0$ and $\mu(\mathbb{S}_+^{l-1}) > 0$, we have $\mu(\mathbb{S}(\underline{q}_n)) > 0$ for all sufficiently large n. Thus $\Xi_{q,n}$ is bounded below by a positive constant. Then, $\chi_n |T_{n,i}(c,s)| \to 0$ under $\underline{q}_n^{-k} \sqrt{\ln n/(n\nu)} \to 0$. Hence, $\chi_n |T_{n,i}(c,s)| \leq 1/2$ for all i and n sufficiently large. Since $\exp(x) \leq 1 + x + x^2$ for $|x| \leq 1/2$, we have

$$\begin{aligned}E(\exp[\pm \chi_n T_{n,i}(c,s)]) &\leq 1 + \chi_n E(T_{n,i}(c,s)) + \chi_n^2 E(T_{n,i}^2(c,s)) \\ &= 1 + \chi_n^2 E(T_{n,i}^2(c,s)) \leq \exp[\chi_n^2 E(T_{n,i}^2(c,s))],\end{aligned} \tag{25}$$

where the equality follows from $E(T_{n,i}(c,s)) = 0$ by construction, and the second inequal-

ity uses $1 + x \leq \exp(x)$ for $x \geq 0$. Then, using $\mathrm{Var}(X) \leq E(X^2)$,

$$\begin{aligned}
E(T_{n,i}^2(c,s)) &= \frac{4}{n^2}\,\mathrm{Var}\left(\frac{\mathbf{1}\{Q_{i1} \geq \underline{q}_n\}}{f_{Q_i}(Q_i)} K_\nu(Q_i'(c,s')' - V_i)\right) \\
&\leq \frac{4}{n^2} E\left(\frac{\mathbf{1}\{Q_{i1} \geq \underline{q}_n\}}{f_{Q_i}^2(Q_i)} K_\nu^2(Q_i'(c,s')' - V_i)\right) \\
&= \frac{4}{n^2} E\left[\frac{\mathbf{1}\{Q_{i1} \geq \underline{q}_n\}}{f_{Q_i}^2(Q_i)}\, E\big(K_\nu^2(Q_i'(c,s')' - V_i) \mid Q_i\big)\right] \\
&= \frac{4}{n^2} \int_{\mathbb{S}(\underline{q}_n)} \frac{1}{f_{Q_i}(q)} \left[\int_{-\infty}^{\infty} K_\nu^2(q'(c,s')' - v)\,(Rf_{C_i,S_i})(q,v)\,dv\right] d\mu(q) \\
&\leq \frac{4M_r(d)}{n^2}\,\Xi_{q,n} \int_{-\infty}^{\infty} K_\nu^2(w)\,dw,
\end{aligned} \tag{26}$$

where the third equality follows from $f_{V_i|Q_i}(v \mid q) = (Rf_{C_i,S_i})(q,v)$ as in the proof of Lemma A.1, and $M_r(d) \equiv \sup_{(q,v)} (Rf_{C_i(d),S_i(d)})(q,v)$. For any $q \in \mathbb{S}^{l-1}$, define $\mathcal{Q}_q^\perp \equiv \{q^\perp \in \mathbb{R}^l : q'q^\perp = 0\}$. The hyperplane $\{z \in \mathbb{R}^l : q'z = v\}$ can be reparameterized as $\{qv + q^\perp \in \mathbb{R}^l : q^\perp \in \mathcal{Q}^\perp\}$. Then, by the definition of the Radon transform in Lemma A.1,

$$\begin{aligned}
(Rf_{C_i,S_i})(q,v) &= \int_{\mathcal{Q}_q^\perp} f_{C_i,S_i}(qv + q^\perp)\,dq^\perp \leq M_c \int_{\mathcal{Q}_q^\perp} (1 + \|q^\perp\|)^{-\kappa}\,dq^\perp \\
&= M_c\,\omega_{l-1} \int_0^\infty (1+\varrho)^{-\kappa} \varrho^{l-2}\,d\varrho,
\end{aligned}$$

where the inequality follows from Assumption 5.2(ii) and $\|vq + q^\perp\|^2 = v^2 + \|q^\perp\|^2 \geq \|q^\perp\|^2$ (since $\|q\| = 1$ and $q'q^\perp = 0$), the second equality uses polar coordinates on $\mathcal{Q}_q^\perp$, with ω_{l-1} the surface area of $\mathbb{S}^{l-2}$, and the integral is finite if and only if $\kappa > l - 1$.[10] Thus,

$$M_r(d) < \infty, \qquad \text{for any } d \in \mathbb{R}^2 \text{ with } d_2 \neq 0. \tag{27}$$

It remains to bound the integral in (26). By the Plancherel theorem and the Fourier

[10] Decompose the integral as

$$\begin{aligned}
\int_0^\infty (1+\varrho)^{-\kappa} \varrho^{l-2}\,d\varrho &= \int_0^1 (1+\varrho)^{-\kappa} \varrho^{l-2}\,d\varrho + \int_1^\infty (1+\varrho)^{-\kappa} \varrho^{l-2}\,d\varrho \\
&\leq \int_0^1 \varrho^{l-2}\,d\varrho + \int_1^\infty \varrho^{-(\kappa-l+2)}\,d\varrho = \frac{1}{l-1} + \frac{1}{\kappa - l + 1} < \infty,
\end{aligned}$$

where the second equality holds for $l \geq 2$ and $\kappa - l + 2 > 1$.

transform $\mathcal{F}K_\nu(t) = \frac{1}{2}(2\pi)^{-l+1}|t|^{l-1}\mathcal{L}(\nu|t|)$ defined in Footnote 6,

$$\int_{-\infty}^{\infty} K_\nu^2(w)\,dw = \frac{1}{2\pi}\int_{-\infty}^{\infty} |\mathcal{F}K_\nu(t)|^2\,dt = \frac{1}{4}(2\pi)^{-2l+1}\int_{-\infty}^{\infty} |t|^{2l-2}\mathcal{L}^2(\nu|t|)\,dt.$$

Using the definition and symmetry of $\mathcal{L}(\cdot)$ and applying the change of variables $w = \nu t$,

$$\begin{aligned}\int_{-\infty}^{\infty} |t|^{2l-2}\mathcal{L}^2(\nu|t|)\,dt &= 2\int_0^{1/\nu} t^{2l-2}(1-(\nu t)^r)^2\,dt \\ &= 2\nu^{-2l+1}\int_0^1 w^{2l-2}(1-w^r)^2\,dw = O(\nu^{-2l+1}).\end{aligned}$$

Hence $\int_{-\infty}^{\infty} K_\nu^2(w)\,dw = O(\nu^{-2l+1})$, which together with (27) yields

$$\sum_{i=1}^{n} E(T_{n,i}^2(c,s)) \le M_\sigma \Xi_{q,n} n^{-1}\nu^{-2l+1}, \tag{28}$$

for some constant $M_\sigma \propto M_r(d) > 0$.

Then, substituting (25) into (24) and combining with (28),

$$\Pr(|T_n(c,s)| > \epsilon_n) \le \frac{2\exp\left[\chi_n^2 \sum_{i=1}^n E(T_{n,i}^2(c,s))\right]}{\exp\left(\chi_n\epsilon_n\right)} \le 2\exp(-\chi_n\epsilon_n + M_\sigma \Xi_{q,n}\chi_n^2 n^{-1}\nu^{-2l+1}),$$

where the first inequality uses the independence of $\{T_{n,i}\}$ across i. The bound is independent of (c,s), and thus

$$\sup_{(c,s)\in\mathcal{S}} \Pr(|T_n(c,s)| > \epsilon_n) \le 2\exp(-\chi_n\epsilon_n + M_\sigma \Xi_{q,n}\chi_n^2 n^{-1}\nu^{-2l+1}). \tag{29}$$

To characterize the order of $T_n(c,s)$ uniformly over $\mathcal{S}$, we want to have $\epsilon_n \to 0$ while $\chi_n\epsilon_n \to \infty$ at a rate that ensures (29) is summable. To do so, set $\chi_n\epsilon_n = M_\chi \ln n$ for some constant $M_\chi > 0$, and choose $\chi_n = \Xi_{q,n}^{-1/2}\nu^{l-1/2}\sqrt{n\ln n}$ and $\epsilon_n = M_\chi \Xi_{q,n}^{1/2}\nu^{-l+1/2}\sqrt{\ln n/n}$. Substituting into (29) and using (23),

$$\Pr(I_{2,2} > \epsilon_n) \le 2N_n n^{M_\sigma - M_\chi}.$$

Let $\ell_n = M_{q,n}\Xi_{q,n}^{1/2}\nu^{3/2}\sqrt{\ln n/n}$, so that $N_n \asymp \ell_n^{-l}$. Since $\underline{q}_n^{-k}\sqrt{\ln n/(n\nu)} \to 0$ implies that

$M_{q,n}^{-1}\Xi_{q,n}^{-1/2}\sqrt{\ln n/(n\nu)} \to 0$ and $\sqrt{\ln n/(n\nu)} \to 0$, we have

$$\ell_n^{-1} = \left(M_{q,n}^{-1}\Xi_{q,n}^{-1/2}\sqrt{\tfrac{\ln n}{n\nu}}\right)\nu^{-1}\frac{n}{\ln n} = o\left((n/\ln n)^2\right).$$

Hence $N_n = o(n^{2l}(\ln n)^{-2l})$. Then $N_n n^{M_\sigma - M_\chi} = o\big(n^{2l+M_\sigma-M_\chi}(\ln n)^{-2l}\big)$, and choosing $M_\chi \geq M_\sigma + 2l + 1$ ensures $2l + M_\sigma - M_\chi \leq -1$, so $\sum_{n=1}^{\infty}\Pr(I_{2,2} > \epsilon_n) < \infty$. By the Borel–Cantelli lemma,

$$I_{2,2} = O(\epsilon_n) = O\Big(\Xi_{q,n}^{1/2}\nu^{-l+1/2}\sqrt{\ln n/n}\Big) \qquad \text{almost surely.} \tag{30}$$

For $I_{2,1}$, write $\left|\bar{f}_{C_i,S_i}(c,s) - \bar{f}_{C_i,S_i}(\dot{c}_k,\dot{s}_k)\right|$ as

$$\begin{aligned}
&\left|\frac{2}{n}\sum_{i=1}^{n}\frac{\mathbf{1}\{Q_{i1}\geq \underline{q}_n\}}{f_{Q_i}(Q_i)}\left[K_\nu(Q_i'(c,s')' - V_i) - K_\nu(Q_i'(\dot{c}_k,\dot{s}_k')' - V_i)\right]\right| \\
&\leq \frac{2}{n}\sum_{i=1}^{n}\left|\frac{\mathbf{1}\{Q_{i1}\geq \underline{q}_n\}}{f_{Q_i}(Q_i)}\right|\left|K_\nu(Q_i'(c,s')' - V_i) - K_\nu(Q_i'(\dot{c}_k,\dot{s}_k')' - V_i)\right| \qquad \text{(by the triangle inequality)} \\
&\leq \frac{2}{M_{q,n}\,n}\sum_{i=1}^{n}\left|K_\nu(Q_i'(c,s')' - V_i) - K_\nu(Q_i'(\dot{c}_k,\dot{s}_k')' - V_i)\right| \\
&= \frac{2}{M_{q,n}\,n}\sum_{i=1}^{n}\nu^{-l}\left|K\left(\frac{Q_i'(c,s')' - V_i}{\nu}\right) - K\left(\frac{Q_i'(\dot{c}_k,\dot{s}_k')' - V_i}{\nu}\right)\right| \qquad \text{(by } K_\nu(\cdot) = \nu^{-l}K(\cdot/\nu)\text{)} \\
&\leq \frac{2M_k}{M_{q,n}\,n}\sum_{i=1}^{n}\nu^{-l-1}\left\|Q_i'[(c,s')' - (\dot{c}_k,\dot{s}_k')']\right\| \qquad \text{(by Assumption 5.4(ii))} \\
&\leq \frac{2M_k}{M_{q,n}\,n}\sum_{i=1}^{n}\nu^{-l-1}\|Q_i\|\,\|(c,s) - (\dot{c}_k,\dot{s}_k)\| \qquad \text{(by the Cauchy–Schwarz inequality)} \\
&= \frac{2M_k}{M_{q,n}}\nu^{-l-1}\|(c,s) - (\dot{c}_k,\dot{s}_k)\| \qquad \text{(by } Q_i \in \mathbb{S}^{l-1}\text{)}.
\end{aligned} \tag{31}$$

Then, we have

$$I_{2,1} \leq \frac{2M_k}{M_{q,n}}\nu^{-l-1}\max_{1\leq k\leq N_n}\sup_{(c,s)\in\mathcal{S}\cap\mathcal{S}_k}\|(c,s) - (\dot{c}_k,\dot{s}_k)\| \leq \frac{2M_kM_\ell}{M_{q,n}}\nu^{-l-1}\ell_n, \tag{32}$$

for some constant $M_\ell > 0$. *For* $I_{2,3}$,

$$\begin{aligned}
&\max_{1\le k\le N_n} \sup_{(c,s)\in\mathcal{S}\cap\mathcal{S}_k} \left|E(\bar{f}_{C_i,S_i}(c,s) - \bar{f}_{C_i,S_i}(\dot{c}_k,\dot{s}_k))\right| \\
&\le \max_{1\le k\le N_n} \sup_{(c,s)\in\mathcal{S}\cap\mathcal{S}_k} E\left|\bar{f}_{C_i,S_i}(c,s) - \bar{f}_{C_i,S_i}(\dot{c}_k,\dot{s}_k)\right| \qquad \text{(by Jensen's inequality)} \\
&\le \max_{1\le k\le N_n} E\left(\sup_{(c,s)\in\mathcal{S}\cap\mathcal{S}_k} \left|\bar{f}_{C_i,S_i}(c,s) - \bar{f}_{C_i,S_i}(\dot{c}_k,\dot{s}_k)\right|\right) \qquad \text{(by monotonicity of expectation)} \\
&\le E\left(\max_{1\le k\le N_n} \sup_{(c,s)\in\mathcal{S}\cap\mathcal{S}_k} \left|\bar{f}_{C_i,S_i}(c,s) - \bar{f}_{C_i,S_i}(\dot{c}_k,\dot{s}_k)\right|\right) \le \frac{2M_kM_\ell}{M_{q,n}}\nu^{-l-1}\ell_n,
\end{aligned} \tag{33}$$

where the last inequality follows from the deterministic bound in (32). With $\ell_n = M_{q,n}\Xi_{q,n}^{1/2}\nu^{3/2}\sqrt{\ln n/n}$, we have

$$I_{2,1} = O\left(\Xi_{q,n}^{1/2}\nu^{-l+1/2}\sqrt{\ln n/n}\right), \qquad I_{2,3} = O\left(\Xi_{q,n}^{1/2}\nu^{-l+1/2}\sqrt{\ln n/n}\right),$$

which together with (30) yields

$$\sup_{(c,s)\in\mathcal{S}} |I_2(c,s)| = O\left(\Xi_{q,n}^{1/2}\nu^{-l+1/2}\sqrt{\ln n/n}\right) \qquad \text{almost surely.}$$

The same rate holds in probability, since almost sure convergence implies convergence in probability. ■

Lemma S.3 (Plug-in error of the RTE)**.** *Suppose Assumptions 5.2*(ii)*, 5.3, and 5.4*(i)*–(ii) and 5.6 hold. Then, for any* ν *with* $\nu \to 0$ *and* $\underline{q}_n^{-k}\sqrt{\ln n/(n\nu)} \to 0$ *as* $n \to \infty$*,*

$$\sup_{(c,s)\in\mathcal{S}} \left|\widehat{f}_{C_i(d),S_i(d)}(c,s) - \bar{f}_{C_i(d),S_i(d)}(c,s)\right| = O_p\big(\nu^{-l+1}\,\Xi_{q,n}\,r_{2n}\big)\,.$$

Proof of Lemma S.3. Write $K_{n,i}(c,s) \equiv K_\nu(Q_i'(c,s')' - V_i)$. By the triangle inequality, for any $(c,s) \in \mathcal{S}$,

$$|I_1| = \left|\widehat{f}_{C_i,S_i}(c,s) - \bar{f}_{C_i,S_i}(c,s)\right| \le \frac{2}{n}\sum_{i=1}^n \mathbf{1}\{Q_{i1} \ge \underline{q}_n\}\,\frac{\left|\widehat{f}_{Q_i}(Q_i) - f_{Q_i}(Q_i)\right|}{f_{Q_i}(Q_i)\,\widehat{f}_{Q_i}(Q_i)}\,|K_{n,i}(c,s)|.$$

For the spherical kernel estimator $\widehat{f}_{Q_i}$, balancing the bias and variance in Hoderlein *et al.*

(2010, Appendix B) at $\tau \asymp (\ln n/n)^{1/(2\zeta+l-1)}$, we have

$$\sup_{q\in\mathbb{S}^{l-1}} \left|\widehat{f}_{Q_i}(q) - f_{Q_i}(q)\right| = O_p(r_{2n}), \tag{34}$$

under Assumption 5.3. As $\underline{q}_n^{-k} r_{2n} \to 0$ under Assumption 5.6 and $M_{q,n} \geq M_q \underline{q}_n^k$ from Assumption 5.3(iii), $M_{q,n}^{-1} r_{2n} \to 0$. Then, by (34) and $\mathbb{S}(\underline{q}_n) \subset \mathbb{S}_+^{l-1}$, w.p.a.1,

$$\sup_{q\in\mathbb{S}(\underline{q}_n)} \left|\widehat{f}_{Q_i}(q) - f_{Q_i}(q)\right| \leq \frac{M_{q,n}}{2}.$$

Hence, by the reverse triangle inequality, w.p.a.1,

$$\widehat{f}_{Q_i}(Q_i) \geq f_{Q_i}(Q_i) - \sup_{q\in\mathbb{S}(\underline{q}_n)} |\widehat{f}_{Q_i}(q) - f_{Q_i}(q)| \geq f_{Q_i}(Q_i) - M_{q,n}/2 \geq f_{Q_i}(Q_i)/2, \tag{35}$$

where the last inequality uses $f_{Q_i}(Q_i) \geq M_{q,n}$. For any $(c,s) \in \mathcal{S}$, (35) gives

$$|I_1| \leq 4 \sup_{q\in\mathbb{S}(\underline{q}_n)} |\widehat{f}_{Q_i}(q) - f_{Q_i}(q)| \Bigg(\underbrace{\frac{1}{n}\sum_{i=1}^n \left(\frac{\mathbf{1}\{Q_{i1} \geq \underline{q}_n\}|K_{n,i}(c,s)|}{f_{Q_i}^2(Q_i)} - E\left[\frac{\mathbf{1}\{Q_{i1} \geq \underline{q}_n\}|K_{n,i}(c,s)|}{f_{Q_i}^2(Q_i)}\right]\right)}_{I_{1,1}}$$
$$+ \underbrace{\frac{1}{n}\sum_{i=1}^n E\left[\frac{\mathbf{1}\{Q_{i1} \geq \underline{q}_n\}|K_{n,i}(c,s)|}{f_{Q_i}^2(Q_i)}\right]}_{I_{1,2}} \Bigg).$$

For $I_{1,2}$, by the law of iterated expectations,

$$E\left[\frac{\mathbf{1}\{Q_{i1} \geq \underline{q}_n\}|K_{n,i}(c,s)|}{f_{Q_i}^2(Q_i)}\right] = \int_{\mathbb{S}(\underline{q}_n)} \left[\int_{-\infty}^{\infty} |K_\nu(q'(c,s')' - v)| (Rf_{C_i,S_i})(q,v)\, dv\right] \frac{1}{f_{Q_i}(q)}\, d\mu(q)$$
$$\leq M_r(d)\Xi_{q,n} \int_{-\infty}^{\infty} |K_\nu(w)|\, dw,$$

where $M_r(d) < \infty$ by (27). Recall from Footnote 6 that $K_\nu(w) = \nu^{-l} K(w/\nu)$. A change of variables $\tilde{w} = w/\nu$ gives $\int_{-\infty}^{\infty} |K_\nu(w)|\, dw = \nu^{-l+1} \int_{-\infty}^{\infty} |K(\tilde{w})|\, d\tilde{w}$. It therefore suffices to show $K \in L^1(\mathbb{R})$. Since $\mathcal{L}(t) = (1-t^r)\mathbf{1}_{[0,1]}(t)$ for $t \geq 0$, the kernel K can be rewritten as

$$K(\tilde{w}) = (2\pi)^{-l} \int_0^1 \cos(t\tilde{w})\, t^{l-1}(1-t^r)\, dt.$$

First, we have $|K(\tilde{w})| \leq (2\pi)^{-l} \int_0^1 t^{l-1}(1-t^r)\, dt < \infty$, so K is bounded. In addition,

since $l = 2J_x + 2 \geq 2$, $t^{l-1}(1-t^r)$ is continuously differentiable for $t \in [0,1]$, vanishes at $t = 0$ and $t = 1$, and has a second derivative in $L^1[0,1]$. Hence $K \in L^1(\mathbb{R})$, and $\int_{-\infty}^{\infty} |K_\nu(w)|\, dw = O\big(\nu^{-l+1}\big)$. It follows that, uniformly for $(c,s) \in \mathcal{S}$,

$$I_{1,2} = O\big(\nu^{-l+1}\Xi_{q,n}\big).$$

For $I_{1,1}$, write $I_{1,1} \equiv T_n^1(c,s) = \sum_{i=1}^n T_{n,i}^1(c,s)$ with

$$T_{n,i}^1(c,s) = \frac{1}{n}\left(\frac{\mathbf{1}\{Q_{i1} \geq \underline{q}_n\}|K_{n,i}(c,s)|}{f_{Q_i}^2(Q_i)} - E\left[\frac{\mathbf{1}\{Q_{i1} \geq \underline{q}_n\}|K_{n,i}(c,s)|}{f_{Q_i}^2(Q_i)}\right]\right),$$

so that $E(T_{n,i}^1) = 0$. By Assumption 5.3(iii) and Assumption 5.4(i), for all i,

$$|T_{n,i}^1| \leq 2n^{-1}M_{q,n}^{-2}\nu^{-l}\sup|K|, \qquad \sum_{i=1}^n E[(T_{n,i}^1)^2] \leq M_\sigma\, M_{q,n}^{-2}\,\Xi_{q,n}\, n^{-1}\nu^{-2l+1}.$$

Covering $\mathcal{S}$ by N_n cubes $\{\mathcal{S}_k\}_{k=1}^{N_n}$ with centers $(\dot{c}_k, \dot{s}_k)$ and side length ℓ_n as in Lemma S.2,

$$\sup_{(c,s)\in\mathcal{S}} T_n^1(c,s) \leq \max_{1\leq k\leq N_n} \sup_{(c,s)\in\mathcal{S}\cap\mathcal{S}_k} \big|T_n^1(c,s) - T_n^1(\dot{c}_k,\dot{s}_k)\big| + \max_{1\leq k\leq N_n} T_n^1(\dot{c}_k,\dot{s}_k).$$

The Chernoff and Borel–Cantelli argument for $I_{2,2}$ applies to the second term, taking $\chi_n = M_{q,n}\Xi_{q,n}^{-1/2}\nu^{l-1/2}\sqrt{n\ln n}$ and $\epsilon_n = M_{q,n}^{-1}\Xi_{q,n}^{1/2}\nu^{-l+1/2}\sqrt{\ln n/n}$, and choosing $\ell_n = M_{q,n}\Xi_{q,n}^{1/2}\nu^{3/2}\sqrt{\ln n/n}$, the second term is $O(\epsilon_n)$ almost surely. As for $I_{2,1}$ and $I_{2,3}$, the first term is $O(\epsilon_n)$. Hence

$$\sup_{(c,s)\in\mathcal{S}} I_{1,1} = O\Big(M_{q,n}^{-1}\Xi_{q,n}^{1/2}\nu^{-l+1/2}\sqrt{\ln n/n}\Big) \qquad \text{almost surely,}$$

which is dominated by $I_{1,2}$ since $\underline{q}_n^{-k}\sqrt{\ln n/(n\nu)} \to 0$ implies that $M_{q,n}^{-1}\Xi_{q,n}^{-1/2}\sqrt{\ln n/(n\nu)} \to 0$. Combining this with (34) gives $\sup_{(c,s)\in\mathcal{S}}|I_1| = O_p\big(\nu^{-l+1}\Xi_{q,n}\, r_{2n}\big)$. ■

S2 Auxiliary Lemmas for Theorem 2

For any γ on the support of γ_i and any $a \in \mathbb{R}^2$, write $\bar{d} = \bar{d}(a,\gamma) \equiv (a_1 - a_2\gamma, a_2)$.

Assumption S.1. *For a with $a_2 \neq 0$, write $\bar{a} \equiv a/\max(|a_1|,|a_2|)$, so that $\max(|\bar{a}_1|,|\bar{a}_2|) = 1$ and $\bar{a}_2 \neq 0$. Let λ_a be the diagonal matrix that scales $(C_i(\bar{d}), S_{i1}(\bar{d}))$ by $\max(|a_1|,|a_2|)$*

and $(S_{i2}(\bar{d}), S_{i3}(\bar{d}))$ *by* $|a_2|$*, and let* $\tilde{f}$ *denote the density of* $\lambda_a^{-1}(C_i(\bar{d}), S_i(\bar{d}))$*, which depends on* a *only through* $\bar{a}$*. There exist constants* $\tilde{M}_c < \infty$*,* $\tilde{\kappa} > l$*,* $\tilde{B} < \infty$*, and a smoothness order* $m > l/2$ *such that, uniformly over* $\bar{a}$ *with* $\max(|\bar{a}_1|, |\bar{a}_2|) = 1$ *and* $\bar{a}_2 \neq 0$*,*

$$\tilde{f}(z) \leq \tilde{M}_c(1 + \|z\|)^{-\tilde{\kappa}} \quad \text{for all } z \in \mathbb{R}^l, \qquad \|\tilde{f}\|_m \leq \tilde{B}.$$

Assumption S.2. *Let* $\varepsilon_n^* \equiv \bar{\vartheta}_n^{2J_x+1} r_{1n} + \bar{\vartheta}_n^{J_x} \underline{q}_n^{1/2} + \bar{\vartheta}_n^{(J_x-1)/2} \underline{\vartheta}_n^{(J_x+1)/2-m} \left(r_{1n} + \underline{q}_n^{1/2}\right) + \bar{\vartheta}_n^{2J_x} \underline{q}_n^{1-k} r_{2n} r_{1n}^{-\frac{2(l-1)}{2m-l-2\varsigma}} + \underline{\vartheta}_n^{-1} r_n^{p/(p+1)} + \underline{\vartheta}_n^{-2} r_{1n}^{\frac{2m+l+2-2\varsigma}{2m-l-2\varsigma}}$*. The tuning sequences* $\underline{\vartheta}_n$ *and* $\bar{\vartheta}_n$ *satisfy* $\underline{\vartheta}_n \to 0$*,* $\bar{\vartheta}_n \to \infty$*,* $\bar{\vartheta}_n = O(n^{C_v})$ *for some constant* $C_v > 0$*,* $\varepsilon_n^* \underline{\vartheta}_n^{-2J_x} \underline{q}_n^{-k} r_{1n}^{\frac{2(l-1)}{2m-l-2\varsigma}} \to 0$*,* $\bar{\vartheta}_n^{p+2} r_n^{p/(p+1)} \to 0$*, and* $\bar{\vartheta}_n^2 r_n^{-1/(p+1)} \varepsilon_n^* \to 0$*.*

The restrictions in Assumptions 5.6 and S.2 are mutually compatible. To see this, write $\underline{q}_n \asymp (\ln n/n)^{A_1}$, $\underline{\vartheta}_n \asymp (\ln n/n)^{A_2}$ and $\bar{\vartheta}_n \asymp (\ln n/n)^{-A_3}$ with $A_1, A_2, A_3 > 0$. Then, every sequence in Assumptions 5.6 and S.2 is a power of $\ln n/n$, so every convergence requirement becomes one strict linear inequality in the exponents.

Assumption 5.6 holds when A_1 lies strictly below the exponent there, and ties $\bar{c}_n$ to A_1: setting $\rho_0 \equiv \frac{2(l-1)}{2m-l-2\varsigma}$, $\rho_1 \equiv (1-(k-1)A_1)\frac{m-l/2-\varsigma}{2m+l-1-2\varsigma}$ and $\rho_r \equiv \min\{\rho_1,\ A_1/2,\ \frac{\zeta}{2\zeta+l-1} - (k-1)A_1 - \rho_0\rho_1\}$, one has $r_{1n} \asymp (\ln n/n)^{\rho_1}$, $r_n \asymp (\ln n/n)^{\rho_r}$ and $\bar{c}_n \asymp (\ln n/n)^{-\rho_r/(p+1)}$.

Expanding ε_n^* term by term, Assumption S.2 becomes a finite system of inequalities of the form $c_2 A_2 + c_3 A_3 < R(A_1)$, where the coefficients $c_2, c_3 \geq 0$ are constants and all dependence on A_1 is collected in $R(\cdot)$. Such a system is satisfied by all sufficiently small $A_2, A_3 > 0$ if and only if every $R(A_1)$ is positive. For a fixed A_1 satisfying the trimming restriction in Assumption 5.6, such $A_2, A_3 > 0$ exist if and only if $p > 1$ and $kA_1 < \frac{p}{p+1}\rho_r + \rho_0\rho_1$, both arising from the fifth term of ε_n^*.[11]

Lemma S.4 (Exponential inequality for the RTE)**.** *Let* $M_r(d) \equiv \sup_{(q,v)}(Rf_{C_i(d),S_i(d)})(q,v)$ *and* $B(d) \equiv \|f_{C_i(d),S_i(d)}\|_m$*. Suppose that Assumptions 5.2, 5.3*(i)*–(iii), (v), 5.4*(i)*, (ii), and (iv) hold. Let the bandwidths* (ν, τ) *satisfy* $\nu \to 0, \tau \to 0$*,* $\tau^{-1} = O(n^{C_\tau})$ *for some finite constant* $C_\tau > 0$*,* $M_{q,n}\tau^{-\zeta} \to \infty$*, and* $M_r(d)^{-1} M_{q,n}^{-1} \Xi_{q,n}^{-1/2} \sqrt{\ln n/(n\nu)} \to 0$ *as* $n \to \infty$*. Then there exist constants* $M_e, \tilde{M}_e, C_e > 0$ *and a sufficiently large constant* $C_0 > 0$ *such that, for any sequence* $\epsilon_n > 0$ *satisfying* $M_r(d) M_{q,n} \Xi_{q,n} \nu^{-l+1} \epsilon_n^{-1} \to \infty$ *and* $\epsilon_n \geq C_0[\Xi_{q,n}^{1/2} \nu^{-l+1/2} \sqrt{\ln n/n} + M_r(d) \Xi_{q,n} \nu^{-l+1} \tau^\zeta + B(d)(\underline{q}_n^{1/2} + \nu^{m-l/2-\varsigma})]$*, and for all* n

[11] For example, with $J_x = 2$, $m = 6$, $\varsigma = 0.5$, $\zeta = 20$, $k = 21/20$, and $p = 12$, the conditions are satisfied by $\underline{q}_n \asymp (\ln n/n)^{6/25}$, and the rate in Proposition 2 is then $(\ln n/n)^{36/325}$.

sufficiently large,

$$\Pr\left(\sup_{(c,s)\in\mathcal{S}}\left|\widehat{f}_{C_i(d),S_i(d)}(c,s)-f_{C_i(d),S_i(d)}(c,s)\right|>\epsilon_n\right)$$
$$\leq M_e n^{C_e}\exp\left(-\tilde{M}_e M_r(d)^{-1}\Xi_{q,n}^{-1}n\nu^{2l-1}\epsilon_n^2\right)+M_e n^{C_e}\exp\left(-\tilde{M}_e\, M_r(d)^{-2}\Xi_{q,n}^{-2}\nu^{2l-2}n\tau^{l-1}\epsilon_n^2\right).$$

Proof of Lemma S.4. Throughout this proof, $\eta_n>0$ denotes a generic sequence satisfying $M_r(d)M_{q,n}\Xi_{q,n}\nu^{-l+1}\eta_n^{-1}\to\infty$ and $\eta_n\geq C_0[\Xi_{q,n}^{1/2}\nu^{-l+1/2}\sqrt{\ln n/n}+M_r(d)\Xi_{q,n}\nu^{-l+1}\tau^{\zeta}+B(d)(\underline{q}_n^{1/2}+\nu^{m-l/2-\varsigma})]$, $\rho_n>0$ denotes a sequence with $\rho_n\geq C_0\tau^{\zeta}$, and C denotes a generic positive constant that may vary from line to line.

Decompose $\widehat{f}_{C_i(d),S_i(d)}-f_{C_i(d),S_i(d)}=I_1+I_2+I_3$ as in the proof of Lemma A.1. The bounds on I_1,I_2,I_3 used below hold for arbitrary (ν,τ) under the maintained assumptions.

Step 1: term I_2. Recall from Lemma S.2 that

$$\sup_{(c,s)\in\mathcal{S}}|I_2|\leq I_{2,1}+I_{2,2}+I_{2,3},$$

where $\mathcal{S}$ is covered by a collection of cubes $\{\mathcal{S}_k\}_{k=1}^{N_n}$ with length $\ell_n=M_{q,n}\Xi_{q,n}^{1/2}\nu^{3/2}\sqrt{\ln n/n}$.

For $I_{2,2}$, recall $T_n(c,s)=\sum_{i=1}^n T_{n,i}(c,s)$ with $E(T_{n,i})=0$, $|T_{n,i}|\leq 4\sup|K|/(nM_{q,n}\nu^l)$ under Assumption 5.4(i), and $\sum_{i=1}^n E(T_{n,i}^2)\leq M_\sigma\Xi_{q,n}n^{-1}\nu^{-2l+1}$ by (28), where $M_\sigma\propto M_r(d)$. Bernstein's inequality gives, for any $(c,s)\in\mathcal{S}$,

$$\Pr\left(|T_n(c,s)|>\eta_n\right)\leq 2\exp\left(-\frac{\eta_n^2/2}{M_\sigma\Xi_{q,n}n^{-1}\nu^{-2l+1}+\frac{4\sup|K|}{3nM_{q,n}\nu^l}\eta_n}\right).$$

In the denominator, the second term is negligible relative to the first, their ratio being proportional to $M_r(d)^{-1}(M_{q,n}\Xi_{q,n})^{-1}\nu^{l-1}\eta_n\to 0$ by the maintained condition, so the denominator is at most $2M_\sigma\Xi_{q,n}n^{-1}\nu^{-2l+1}$ for n large, and

$$\Pr\left(|T_n(c,s)|>\eta_n\right)\leq 2\exp\left(-\frac{n\nu^{2l-1}\eta_n^2}{4M_\sigma\Xi_{q,n}}\right).$$

Applying a union bound over the N_n centers at threshold $\eta_n/3$ gives

$$\Pr(I_{2,2}>\eta_n/3)\leq 2N_n\exp\left(-\frac{n\nu^{2l-1}(\eta_n/3)^2}{4M_\sigma\Xi_{q,n}}\right).$$

Since $\ell_n^{-1} = o((n/\ln n)^2)$ as in the proof of Lemma S.2, we have $N_n \leq n^{C_e}$ for some finite constant $C_e > 0$ and all sufficiently large n.

The terms $I_{2,1}$ and $I_{2,3}$ admit deterministic bounds of order $O\Big(\Xi_{q,n}^{1/2}\nu^{-l+1/2}\sqrt{\ln n/n}\Big)$, which fall below $\eta_n/3$ for n large by $\eta_n \geq C_0\Xi_{q,n}^{1/2}\nu^{-l+1/2}\sqrt{\ln n/n}$ with C_0 large. Collecting the bounds for $I_{2,1}$, $I_{2,2}$, and $I_{2,3}$, we have

$$\Pr\left(\sup_{(c,s)\in\mathcal{S}}|I_2| > \eta_n\right) \leq M_1 n^{C_e}\exp\Big(-\tilde{M}_1 M_r(d)^{-1}\Xi_{q,n}^{-1}n\nu^{2l-1}\eta_n^2\Big), \tag{36}$$

for some constants $M_1, \tilde{M}_1 > 0$ and all n sufficiently large.

Step 2: term I_1. Write $K_{n,i}(c,s) \equiv K_\nu(Q_i'(c,s')' - V_i)$ and $\Delta_n \equiv \sup_{q\in\mathbb{S}(\underline{q}_n)}|\widehat{f}_{Q_i}(q) - f_{Q_i}(q)|$. On $\mathcal{E}_n \equiv \{\Delta_n \leq M_{q,n}/2\}$ we have $\widehat{f}_{Q_i} \geq f_{Q_i}/2$, so, decomposing I_1 as in Lemma S.3,

$$\sup_{(c,s)\in\mathcal{S}}|I_1| \leq 4\Delta_n\left(\sup_{(c,s)\in\mathcal{S}}|T_n^1(c,s)| + \sup_{(c,s)\in\mathcal{S}}I_{1,2}\right).$$

As in Lemma S.3, $\sup_{(c,s)\in\mathcal{S}}I_{1,2} \leq M_r(d)M_K\nu^{-l+1}\Xi_{q,n}$ with $M_K \equiv \int_{-\infty}^{\infty}|K(w)|\,dw < \infty$, and $E(T_{n,i}^1) = 0$, $|T_{n,i}^1| \leq 2n^{-1}M_{q,n}^{-2}\nu^{-l}\sup|K|$, $\sum_{i=1}^n E[(T_{n,i}^1)^2] \leq M_\sigma M_{q,n}^{-2}\Xi_{q,n}n^{-1}\nu^{-2l+1}$. On $\mathcal{E}_n$ and on $\{\sup_{(c,s)}|T_n^1| \leq M_r(d)M_K\nu^{-l+1}\Xi_{q,n}\}$, we have

$$\sup_{(c,s)\in\mathcal{S}}|T_n^1(c,s)| + \sup_{(c,s)\in\mathcal{S}}I_{1,2} \leq 2M_r(d)M_K\nu^{-l+1}\Xi_{q,n}.$$

Then, $\sup_{(c,s)\in\mathcal{S}}|I_1| \leq 8M_r(d)M_K\nu^{-l+1}\Xi_{q,n}\Delta_n$. Hence

$$\Pr\left(\sup_{(c,s)\in\mathcal{S}}|I_1| > \eta_n\right) \leq \underbrace{\Pr\left(\mathcal{E}_n^c\right)}_{I_1(\mathrm{I})} + \underbrace{\Pr\left(\sup_{(c,s)\in\mathcal{S}}|T_n^1(c,s)| > M_r(d)M_K\nu^{-l+1}\Xi_{q,n}\right)}_{I_1(\mathrm{II})}$$
$$+ \underbrace{\Pr\left(\Delta_n > \frac{\eta_n\,\nu^{l-1}}{8M_r(d)M_K\Xi_{q,n}}\right)}_{I_1(\mathrm{III})}.$$

Term I_1(II). Cover $\mathcal{S}$ by the N_n cubes of Step 1, and set $t_n^* \equiv M_r(d)M_K\nu^{-l+1}\Xi_{q,n}$. As for $I_{2,1}$ in Lemma S.2, the deterministic fluctuation of T_n^1 between any point of a cube and its center is bounded, by Assumption 5.4(ii), by $CM_{q,n}^{-2}\nu^{-l-1}\ell_n = o(t_n^*)$ under

$M_{q,n}^{-1}\Xi_{q,n}^{-1/2}\sqrt{\ln n/(n\nu)} \to 0$. Hence, for all sufficiently large n, if $\sup_{(c,s)\in\mathcal{S}} |T_n^1(c,s)| > t_n^*$, then $|T_n^1(\dot{c}_k, \dot{s}_k)| > t_n^*/2$ for some cube center $(\dot{c}_k, \dot{s}_k)$. Applying Bernstein's inequality at each center with $|T_{n,i}^1| \leq 2n^{-1}M_{q,n}^{-2}\nu^{-l}\sup|K|$ and $\sum_{i=1}^n E[(T_{n,i}^1)^2] \leq M_\sigma M_{q,n}^{-2}\Xi_{q,n}n^{-1}\nu^{-2l+1}$, the Bernstein denominator at $t_n^*/2$ is bounded by a constant multiple of $M_r(d)M_{q,n}^{-2}\Xi_{q,n}n^{-1}\nu^{-2l+1}$. Taking the union bound over the N_n centers, with $N_n \leq n^{C_e}$, gives

$$I_1(\mathrm{II}) \leq 2n^{C_e}\exp\left(-\tilde{M}_2' M_r(d) M_{q,n}^2 \Xi_{q,n} n\nu\right),$$

for some constant $\tilde{M}_2' > 0$.

Terms $I_1(\mathrm{I})$ *and* $I_1(\mathrm{III})$. We bound them through the following inequality for the spherical kernel estimator $\widehat{f}_{Q_i}$. Under Assumption 5.3(v), the spherical kernel $\widetilde{K}$ is bounded, Hölder continuous, and of order at least ζ, as in Appendix B of Hoderlein *et al.* (2010). We then have

$$\sup_{q\in\mathbb{S}(\underline{q}_n)}\left|\widehat{f}_{Q_i}(q) - f_{Q_i}(q)\right| \leq \sup_{q\in\mathbb{S}(\underline{q}_n)}\left|\widehat{f}_{Q_i}(q) - E(\widehat{f}_{Q_i}(q))\right| + \sup_{q\in\mathbb{S}(\underline{q}_n)}\left|E(\widehat{f}_{Q_i}(q)) - f_{Q_i}(q)\right|.$$

Since $\mathbb{S}(\underline{q}_n) \subset \mathbb{S}_+^{l-1}$, the bias term satisfies $\sup_{q\in\mathbb{S}(\underline{q}_n)}|E(\widehat{f}_{Q_i}(q)) - f_{Q_i}(q)| = O(\tau^\zeta)$, which falls below $\rho_n/2$ for n large under $\rho_n \geq C_0\tau^\zeta$ with C_0 large. Write the stochastic term as $\widehat{f}_{Q_i}(q) - E(\widehat{f}_{Q_i}(q)) = \sum_{i=1}^n \xi_i(q)$, with

$$\xi_i(q) \equiv \frac{1}{n}\left[\varkappa(\tau)\widetilde{K}\left(\tau^{-2}(1-q'Q_i)\right) - E\left[\varkappa(\tau)\widetilde{K}\left(\tau^{-2}(1-q'Q_i)\right)\right]\right].$$

Then, we have $E(\xi_i(q)) = 0$ and $|\xi_i(q)| = O(n^{-1}\tau^{-l+1})$ since the kernel $\widetilde{K}$ is bounded and $\varkappa(\tau) \asymp \tau^{-(l-1)}$, and $\sum_{i=1}^n \mathrm{Var}(\xi_i(q)) = O(n^{-1}\tau^{-l+1})$, where the variance bound uses $\sup_{q\in\mathbb{S}_+^{l-1}} f_{Q_i}(q) < \infty$ (Assumption 5.3(i)). Bernstein's inequality gives

$$\Pr\left(\left|\widehat{f}_{Q_i}(q) - E(\widehat{f}_{Q_i}(q))\right| > \rho_n\right) \leq 2\exp\left(-\frac{Mn\tau^{l-1}\rho_n^2}{1+\rho_n}\right).$$

The grid argument of Hoderlein *et al.* (2010, Appendix B), together with the exponential inequality for the spherical kernel estimator, gives

$$\Pr\left(\sup_{q\in\mathbb{S}(\underline{q}_n)}\left|\widehat{f}_{Q_i}(q) - f_{Q_i}(q)\right| > \rho_n\right) \leq M_2 n^{C_e}\exp\left(-\frac{\tilde{M}_2 n\tau^{l-1}\rho_n^2}{1+\rho_n}\right), \tag{37}$$

for some constants $M_2, \tilde{M}_2 > 0$ and all n sufficiently large.

For I_1(I), apply (37) at $\rho_n = M_{q,n}/2$. Since $M_{q,n}\tau^{-\zeta} \to \infty$, this threshold dominates the bias term $O(\tau^{\zeta})$. For I_1(III), apply (37) at $\rho_n = \eta_n \nu^{l-1}/(8M_r(d)M_K\Xi_{q,n})$. The maintained lower bound on η_n makes this threshold at least $C_0\tau^{\zeta}$ for C_0 sufficiently large. Using that $1+\rho_n$ is bounded in both applications under the maintained conditions,

$$
\begin{aligned}
I_1(\mathrm{I}) &\le M_3 n^{C_e} \exp\Big(-\tilde{M}_3\, M_{q,n}^2\, n\tau^{l-1}\Big), \\
I_1(\mathrm{III}) &\le M_4 n^{C_e} \exp\Big(-\tilde{M}_4\, M_r(d)^{-2}\Xi_{q,n}^{-2}\nu^{2l-2}\, n\tau^{l-1}\eta_n^2\Big),
\end{aligned}
$$

for some constants $M_3, \tilde{M}_3, M_4, \tilde{M}_4 > 0$. The ratio of the exponent of I_1(I) to that of I_1(III), and the ratio of the exponent of I_1(II) to that of (36), are both of order $\big(M_r(d)M_{q,n}\Xi_{q,n}\nu^{-l+1}\eta_n^{-1}\big)^2 \to \infty$ by the maintained condition. Thus I_1(I) is dominated by the exponential rate of I_1(III), and I_1(II) by that of (36). Hence

$$
\begin{aligned}
\Pr\left(\sup_{(c,s)\in\mathcal{S}} |I_1| > \eta_n\right) \le{}& M_5 n^{C_e} \exp\Big(-\tilde{M}_5\, M_r(d)^{-1}\Xi_{q,n}^{-1} n\nu^{2l-1}\eta_n^2\Big) \\
&+ M_5 n^{C_e} \exp\Big(-\tilde{M}_5\, M_r(d)^{-2}\Xi_{q,n}^{-2}\nu^{2l-2} n\tau^{l-1}\eta_n^2\Big),
\end{aligned} \tag{38}
$$

for some constants $M_5, \tilde{M}_5 > 0$.

Step 3: term I_3. By Lemma S.1, there exist constants $M_T, M_A > 0$, such that

$$
\sup_{(c,s)\in\mathcal{S}} |I_3| \le M_T B(d) \underline{q}_n^{1/2} + M_A B(d)\nu^{m-l/2-\varsigma},
$$

which is at most $\epsilon_n/3$ for n large under $\epsilon_n \ge C_0 B(d)(\underline{q}_n^{1/2} + \nu^{m-l/2-\varsigma})$ with C_0 large.

Applying (36) and (38) at $\eta_n = \epsilon_n/3$, we obtain

$$
\begin{aligned}
\Pr\left(\sup_{(c,s)\in\mathcal{S}} \left|\widehat{f}_{C_i(d),S_i(d)} - f_{C_i(d),S_i(d)}\right| > \epsilon_n\right) \le{}& M_e n^{C_e} \exp\Big(-\tilde{M}_e M_r(d)^{-1}\Xi_{q,n}^{-1} n\nu^{2l-1}\epsilon_n^2\Big) \\
&+ M_e n^{C_e} \exp\Big(-\tilde{M}_e\, M_r(d)^{-2}\Xi_{q,n}^{-2}\nu^{2l-2} n\tau^{l-1}\epsilon_n^2\Big),
\end{aligned}
$$

for some constants $M_e, \tilde{M}_e > 0$ and all n sufficiently large. ■

Lemma S.5. *Suppose Assumption S.1 holds, with λ_a and $\tilde{f}$ defined there. Then, uniformly over $a \in \mathcal{A}_n$,*

(a) $M_r(\bar{d}) \asymp |a_2|^{-1}$*; in particular,* $\sup_{a\in\mathcal{A}_n} M_r(\bar{d}) \asymp \underline{\vartheta}_n^{-1}$*, and* $\inf_{a\in\mathcal{A}_n} M_r(\bar{d}) \asymp \bar{\vartheta}_n^{-1}$.

(b) For $|a_2| \le 1$, $B(\bar{d}) = O\big(\max(|a_1|,|a_2|)^{-(J_x+1)/2}\,|a_2|^{-(m+(J_x+1)/2)}\big)$*; for* $|a_2| > 1$, $B(\bar{d}) = O\left((\max(|a_1|,|a_2|)\,|a_2|)^{-(J_x+1)/2}\right)$.

Proof of Lemma S.5. Write $f \equiv f_{C_i(\bar{d}),S_i(\bar{d})}$, $\tilde{a}_1 \equiv \max(|a_1|,|a_2|)$, $\tilde{a}_2 \equiv |a_2|$, so that $\det\lambda_a = (\tilde{a}_1\tilde{a}_2)^{J_x+1}$ and $f(z) = \det(\lambda_a)^{-1}\tilde{f}(\lambda_a^{-1}z)$ for $z \in \mathbb{R}^l$.

Part (a). Let $Z \equiv \lambda_a^{-1}(C_i(\bar{d}),S_i(\bar{d}))$, with density $\tilde{f}$. For any unit q, $q'(C_i(\bar{d}),S_i(\bar{d})) = (\lambda_a q)'Z = \|\lambda_a q\|\,(\lambda_a q/\|\lambda_a q\|)'Z$, so by the definition of the Radon transform,

$$(Rf_{(C_i(\bar{d}),S_i(\bar{d}))})(q,v) = \frac{1}{\|\lambda_a q\|}\,(R\tilde{f})\left(\frac{\lambda_a q}{\|\lambda_a q\|},\ \frac{v}{\|\lambda_a q\|}\right). \tag{39}$$

Every diagonal entry of λ_a is at least $|a_2|$, so $\|\lambda_a q\| \ge |a_2|$. Let

$$\tilde{M}_r \equiv \sup_{\max(|\bar{a}_1|,|\bar{a}_2|)=1,\ \bar{a}_2\neq 0}\ \sup_{q,v}(R\tilde{f})(q,v) < \infty,$$

because the tail bound $\tilde{f}(z) \le \tilde{M}_c(1+\|z\|)^{-\tilde{\kappa}}$ with $\tilde{\kappa} > l$ bounds each Radon transform as in Lemma S.2, uniformly in $\bar{a}$ because $\tilde{M}_c$ and $\tilde{\kappa}$ do not depend on $\bar{a}$ (Assumption S.1), so that $\tilde{M}_r$ does not depend on n. Then (39) gives

$$M_r(\bar{d}) = \sup_{q,v}(Rf_{(C_i(\bar{d}),S_i(\bar{d}))})(q,v) \le \frac{\tilde{M}_r}{|a_2|}.$$

For the lower bound, let $q^\star$ be a unit vector such that $\lambda_a q^\star = |a_2|q^\star$. Then (39) at $q^\star$ gives

$$M_r(\bar{d}) \ge \sup_v(Rf_{(C_i(\bar{d}),S_i(\bar{d}))})(q^\star,v) = \frac{1}{|a_2|}\sup_v(R\tilde{f})(q^\star,v).$$

Now $(R\tilde{f})(q^\star,\cdot)$ is the density of $q^{\star\prime}Z$,[12] so it integrates to 1, and the tail bound $\tilde{f}(z) \le \tilde{M}_c(1+\|z\|)^{-\tilde{\kappa}}$ with $|q^{\star\prime}z| \le \|z\|$ gives

$$\int_{|v|>\bar{v}}(R\tilde{f})(q^\star,v)\,dv \le \tilde{M}_c\,\omega_l\int_{\bar{v}}^{\infty}(1+r)^{-\tilde{\kappa}}r^{l-1}\,dr \to 0,$$

as $\bar{v}\to\infty$ under $\tilde{\kappa} > l$, where ω_l is the surface area of $\mathbb{S}^{l-1}$; since this bound depends only on $\tilde{M}_c$, $\tilde{\kappa}$ and l, a $\bar{v}_0$ with $\int_{|v|\le\bar{v}_0}(R\tilde{f})(q^\star,v)\,dv \ge 1/2$ can be chosen independently of

[12]Since $q^\star$ is a unit vector, $(R\tilde{f})(q^\star,\cdot)$ is the density of the projection $q^{\star\prime}Z$. Write $z = v\,q^\star + q^\perp$ with $q^{\star\prime}q^\perp = 0$, as in the proof of Lemma A.1. Then $\frac{d}{dv}\Pr(q^{\star\prime}Z \le v) = \int_{\{z:\,q^{\star\prime}z=v\}}\tilde{f}(z)\,dz = (R\tilde{f})(q^\star,v)$.

$\bar{a}$. Then

$$\sup_v (R\tilde{f})(q^\star, v) \geq \sup_{|v|\leq \bar{v}_0} (R\tilde{f})(q^\star, v) \geq \frac{1}{2\bar{v}_0} \int_{|v|\leq \bar{v}_0} (R\tilde{f})(q^\star, v)\, dv \geq \frac{1}{4\bar{v}_0} > 0.$$

Combining the upper and lower bounds, $M_r(\bar{d}) \asymp |a_2|^{-1}$; since $\mathcal{A}_n$ is closed and attains both $|a_2| = \underline{\vartheta}_n$ and $|a_2| = \bar{\vartheta}_n$, $\sup_{a\in\mathcal{A}_n} M_r(\bar{d}) \asymp \underline{\vartheta}_n^{-1}$ and $\inf_{a\in\mathcal{A}_n} M_r(\bar{d}) \asymp \bar{\vartheta}_n^{-1}$.

Part (b). From $\mathcal{F}f(\omega) = \mathcal{F}\tilde{f}(\lambda_a\omega)$ and $\xi = \lambda_a\omega$,

$$B(\bar{d})^2 = \|f\|_m^2 = \det(\lambda_a)^{-1} \int_{\mathbb{R}^l} (1 + \|\lambda_a^{-1}\xi\|^2)^m \big|\mathcal{F}\tilde{f}(\xi)\big|^2 d\xi.$$

Every entry of λ_a is no less than $|a_2|$, so $\|\lambda_a^{-1}\xi\| \leq |a_2|^{-1}\|\xi\|$ and $(1 + \|\lambda_a^{-1}\xi\|^2)^m \leq \max(1, |a_2|^{-2m})(1 + \|\xi\|^2)^m$. Hence

$$B(\bar{d})^2 \leq \det(\lambda_a)^{-1} \max(1, |a_2|^{-2m}) \|\tilde{f}\|_m^2 \leq \tilde{B}^2\, \tilde{a}_1^{-(J_x+1)} \tilde{a}_2^{-(J_x+1)} \max(1, |a_2|^{-2m}).$$

For $|a_2| \leq 1$, this gives $B(\bar{d}) = O\big(\tilde{a}_1^{-(J_x+1)/2}\, |a_2|^{-(m+(J_x+1)/2)}\big)$; for $|a_2| > 1$, $B(\bar{d}) = O\left((\max(|a_1|, |a_2|)\, |a_2|)^{-(J_x+1)/2}\right)$. ■

Lemma S.6. *Let $M_f(d) \equiv \sup_{c\in\mathcal{C}_n} f_{C_i(d),\gamma_i,\beta_i}(c, \gamma, b)$. Under Assumptions 5.1*(iii) *and 5.5*(i)*–(ii), for any (γ, b) on the joint support of (γ_i, β_i), $\sup_{a\in\mathcal{A}_n} M_f(\bar{d}) \asymp \underline{\vartheta}_n^{-1}$. Under $\bar{c}_n/\bar{\vartheta}_n \to \infty$, $M_f(\bar{d}) \asymp \max(|a_1|, |a_2|)^{-1}$ uniformly over $a \in \mathcal{A}_n$, so $\inf_{a\in\mathcal{A}_n} M_f(\bar{d}) \asymp \bar{\vartheta}_n^{-1}$.*

Proof of Lemma S.6. By (10), conditional on $\gamma_i = \gamma, \beta_i = b$, $C_i(\bar{d}) = \widetilde{U}_i(a) + a_1\gamma y_0$, where $\widetilde{U}_i(a) \equiv a_1 U_{i1} + a_2 U_{i2}$ and $y_0 \neq 0$ is fixed. Recall $\mathcal{U}_n \equiv \{\tilde{u} \in \mathbb{R} : |\tilde{u} + a_1\gamma y_0| \leq \bar{c}_n\}$. Using $c = \tilde{u} + a_1\gamma y_0$,

$$M_f(\bar{d}) = \sup_{c\in\mathcal{C}_n} f_{\gamma_i,\beta_i}(\gamma, b)\, f_{\widetilde{U}_i(a)|\gamma,b}(c - a_1\gamma y_0) = f_{\gamma_i,\beta_i}(\gamma, b) \sup_{\tilde{u}\in\mathcal{U}_n} f_{\widetilde{U}_i(a)|\gamma,b}(\tilde{u}). \tag{40}$$

Upper bound. By the change of variable $u_1 = (\tilde{u} - a_2u_2)/a_1$ and Assumption 5.5(ii),

$$f_{\widetilde{U}_i(a)|\gamma,b}(\tilde{u}) = \frac{1}{|a_1|} \int_{-\infty}^{\infty} f_{U_{i1},U_{i2}|\gamma,b}\Big(\tfrac{\tilde{u}-a_2u_2}{a_1}, u_2\Big)\, du_2 \leq \frac{M_U}{|a_1|}, \qquad |a_1| \geq |a_2|,$$

with the symmetric bound $M_U/|a_2|$ when $|a_2| \geq |a_1|$, where

$$M_U \equiv \max\left\{ \int_{-\infty}^{\infty} \sup_{u_2} f_{U_{i1},U_{i2}|\gamma,b}(u_1, u_2)\, du_1,\ \int_{-\infty}^{\infty} \sup_{u_1} f_{U_{i1},U_{i2}|\gamma,b}(u_1, u_2)\, du_2 \right\} < \infty.$$

Hence, by (40) and $\max(|a_1|, |a_2|) \geq \underline{\vartheta}_n$ on $\mathcal{A}_n$,

$$M_f(\bar{d}) \leq f_{\gamma_i,\beta_i}(\gamma, b) \frac{M_U}{\max(|a_1|, |a_2|)} \leq f_{\gamma_i,\beta_i}(\gamma, b) \frac{M_U}{\underline{\vartheta}_n},$$

for any $a \in \mathcal{A}_n$, so that $\sup_{a \in \mathcal{A}_n} M_f(\bar{d}) = O(\underline{\vartheta}_n^{-1})$.

Lower bound. Let $\mathcal{T}$ be a compact subset of the joint support of (γ_i, β_i) that contains (γ, b). By Assumption 5.5(i), $\bar{\mu} \equiv \sum_{j=1,2} \sup_{(\gamma,b) \in \mathcal{T}} E(|U_{ij}| \mid \gamma, b) < \infty$, so $E(|\widetilde{U}_i(a)| \mid \gamma, b) \leq \bar{\mu} \max(|a_1|, |a_2|)$. Markov's inequality with $t_0 \equiv 2\bar{\mu} \max(|a_1|, |a_2|)$ gives

$$\Pr\Big(|\widetilde{U}_i(a)| > t_0 \mid \gamma, b\Big) \leq \frac{E\Big(|\widetilde{U}_i(a)| \mid \gamma, b\Big)}{t_0} \leq \frac{1}{2}.$$

Hence

$$\int_{-t_0}^{t_0} f_{\widetilde{U}_i(a)|\gamma,b}(\tilde{u})\, d\tilde{u} = \Pr\Big(|\widetilde{U}_i(a)| \leq t_0 \mid \gamma, b\Big) \geq \frac{1}{2}.$$

Since $f_{\widetilde{U}_i(a)|\gamma,b}$ is nonnegative and integrates to at least $1/2$ over $[-t_0, t_0]$, its supremum on that interval is at least the average. Then, we have

$$\sup_{|\tilde{u}| \leq t_0} f_{\widetilde{U}_i(a)|\gamma,b}(\tilde{u}) \geq \frac{1}{2t_0} \int_{-t_0}^{t_0} f_{\widetilde{U}_i(a)|\gamma,b}(\tilde{u})\, d\tilde{u} \geq \frac{1}{4t_0}.$$

We have $[-t_0, t_0] \subset \mathcal{U}_n$ whenever $t_0 + |a_1 \gamma y_0| \leq \bar{c}_n$. Substituting $t_0 = 2\bar{\mu} \max(|a_1|, |a_2|)$ and $|a_1 \gamma y_0| \leq |\gamma y_0| \max(|a_1|, |a_2|)$, this is ensured by $(2\bar{\mu} + |\gamma y_0|) \max(|a_1|, |a_2|) \leq \bar{c}_n$. When this holds, (40) and $f_{\gamma_i,\beta_i}(\gamma, b) \geq M_b$ on $\mathcal{T}$ (Assumption 5.1(iii)) give

$$M_f(\bar{d}) \geq M_b \sup_{\tilde{u} \in \mathcal{U}_n} f_{\widetilde{U}_i(a)|\gamma,b}(\tilde{u}) \geq M_b \sup_{|\tilde{u}| \leq t_0} f_{\widetilde{U}_i(a)|\gamma,b}(\tilde{u}) \geq \frac{M_b}{4t_0} = \frac{M_b}{8\bar{\mu} \max(|a_1|, |a_2|)}.$$

At $a = (\underline{\vartheta}_n, \underline{\vartheta}_n) \in \mathcal{A}_n$, $(2\bar{\mu} + |\gamma y_0|)\underline{\vartheta}_n \leq \bar{c}_n$ for large n, so $\sup_{a \in \mathcal{A}_n} M_f(\bar{d}) \geq \frac{M_b}{8\bar{\mu}} \underline{\vartheta}_n^{-1}$, which with the upper bound gives $\sup_{a \in \mathcal{A}_n} M_f(\bar{d}) \asymp \underline{\vartheta}_n^{-1}$.

If $\bar{c}_n / \bar{\vartheta}_n \to \infty$, then $(2\bar{\mu} + |\gamma y_0|) \max(|a_1|, |a_2|) \leq (2\bar{\mu} + |\gamma y_0|)\bar{\vartheta}_n \leq \bar{c}_n$ for any $a \in \mathcal{A}_n$ at large n, so the lower bound $M_f(\bar{d}) \geq M_b / (8\bar{\mu} \max(|a_1|, |a_2|))$ holds uniformly over $a \in \mathcal{A}_n$. Combined with the upper bound, $M_f(\bar{d}) \asymp \max(|a_1|, |a_2|)^{-1}$ uniformly over $a \in \mathcal{A}_n$; in particular, $\inf_{a \in \mathcal{A}_n} M_f(\bar{d}) \asymp \bar{\vartheta}_n^{-1}$. ■

Lemma S.7 (Lipschitz continuity of the RTE in a)**.** *Suppose Assumptions 5.3, 5.4*(ii)*, and 5.6 hold, and that $E(|\widetilde{Y}_{i1}|) < \infty$ and $E(|\widetilde{Y}_{i2}|) < \infty$. Then for any $a, \tilde{a} \in \mathcal{A}_n$ with*

$\|a-\tilde{a}\| \leq \underline{\vartheta}_n$, *any* $c \in \mathbb{R}$, *and any* (γ, b) *on the joint support of* (γ_i, β_i),

$$\begin{aligned}
&\left|\widehat{f}_{C_i(a,\gamma),\gamma_i,\beta_i}(c,\gamma,b) - \widehat{f}_{C_i(\tilde{a},\gamma),\gamma_i,\beta_i}(c,\gamma,b)\right| \\
&\leq \left(M_{q,n}^{-1}\mathcal{R}_n\,\lambda_n\,\nu^{-l-1} + M_x\,\underline{\vartheta}_n^{-1}\left|\widehat{f}_{C_i(\tilde{a},\gamma),\gamma_i,\beta_i}(c,\gamma,b)\right|\right)\|a-\tilde{a}\|,
\end{aligned}$$

where $\mathcal{R}_n = O_p(1)$ *does not depend on* $(a, \tilde{a}, c)$, $\lambda_n = \sup_{a\in\mathcal{A}_n}|J_i(\bar{d})|$, *and* $M_x = (2J_x + 1)\exp(2J_x+1)$.

Proof of Lemma S.7. Write $V_i(a,\gamma) = \|(1,H_i')\|^{-1}[(a_1 - a_2\gamma)\widetilde{Y}_{i1} + a_2\widetilde{Y}_{i2}]$. For any $a, \tilde{a} \in \mathbb{R}^2$ and any $(c,s) \in \mathbb{R}^l$, the two estimators differ only through $V_i(a,\gamma)$ and $V_i(\tilde{a},\gamma)$. As for $I_{2,1}$ in Lemma S.2, under Assumption 5.4(ii),

$$\begin{aligned}
&\left|\widehat{f}_{C_i(a,\gamma),S_i(a,\gamma)}(c,s) - \widehat{f}_{C_i(\tilde{a},\gamma),S_i(\tilde{a},\gamma)}(c,s)\right| \\
&\leq \frac{2}{n}\sum_{i=1}^{n}\frac{\mathbf{1}\{Q_{i1}\geq \underline{q}_n\}}{|\widehat{f}_{Q_i}(q_i)|}\left|K_\nu(q_i'(c,s')' - V_i(a,\gamma)) - K_\nu(q_i'(c,s')' - V_i(\tilde{a},\gamma))\right| \\
&\leq \nu^{-l-1}\frac{4M_k}{M_{q,n}}\frac{1}{n}\sum_{i=1}^{n}\left|q_i'(c,s')' - V_i(a,\gamma) - (q_i'(c,s')' - V_i(\tilde{a},\gamma))\right| \\
&\leq \nu^{-l-1}\frac{4M_k}{M_{q,n}}\frac{1}{n}\sum_{i=1}^{n}\|(1,H_i')\|^{-1}\left(|\widetilde{Y}_{i1}|\,|a_1-\tilde{a}_1| + \left(|\widetilde{Y}_{i1}|\,|\gamma| + |\widetilde{Y}_{i2}|\right)\,|a_2-\tilde{a}_2|\right) \\
&\leq \nu^{-l-1}M_{q,n}^{-1}\underbrace{4M_k\frac{1}{n}\sum_{i=1}^{n}\|(1,H_i')\|^{-1}\left(|\widetilde{Y}_{i1}|(1+|\gamma|) + |\widetilde{Y}_{i2}|\right)}_{\tilde{\mathcal{R}}_n}\|a-\tilde{a}\|,
\end{aligned} \tag{41}$$

where the second inequality follows from $\inf_{q\in\mathbb{S}(\underline{q}_n)}\widehat{f}_{Q_i}(q) \geq M_{q,n}/2$ w.p.a.1 under Assumption 5.6, as in the proof of Lemma S.3, and the last inequality from $|a_1-\tilde{a}_1|, |a_2-\tilde{a}_2| \leq \|a-\tilde{a}\|$. Since the sample is i.i.d. across i, $\|(1,H_i')\|^{-1} \leq 1$, and $E(|\widetilde{Y}_{it}|) < \infty$, the weak law of large numbers gives $\tilde{\mathcal{R}}_n = O_p(1)$.

For any γ on the support of γ_i and $a \in \mathbb{R}^2$, let $s_a \equiv (b_1a_1, b_2a_2, \delta_1a_2\gamma)$ and $|J_a| \equiv |J_i(\bar{d}(a,\gamma))| = |\delta_1|\,|a_1|^{J_x}|a_2|^{J_x+1}$. By $\widehat{f}_{C_i(a,\gamma),\gamma_i,\beta_i}(c,\gamma,b) = \widehat{f}_{C_i(a,\gamma),S_i(a,\gamma)}(c,s_a)\,|J_a|$ and the

triangle inequality,

$$\begin{aligned}
&\left|\widehat{f}_{C_i(a,\gamma),\gamma_i,\beta_i}(c,\gamma,b) - \widehat{f}_{C_i(\tilde{a},\gamma),\gamma_i,\beta_i}(c,\gamma,b)\right| \\
&\leq \underbrace{\left|\widehat{f}_{C_i(a,\gamma),S_i(a,\gamma)}(c,s_a) - \widehat{f}_{C_i(\tilde{a},\gamma),S_i(\tilde{a},\gamma)}(c,s_a)\right| |J_a|}_{\text{(I)}} + \underbrace{\left|\widehat{f}_{C_i(\tilde{a},\gamma),S_i(\tilde{a},\gamma)}(c,s_a) - \widehat{f}_{C_i(\tilde{a},\gamma),S_i(\tilde{a},\gamma)}(c,s_{\tilde{a}})\right| |J_a|}_{\text{(II)}} \\
&\quad + \underbrace{\left|\widehat{f}_{C_i(\tilde{a},\gamma),S_i(\tilde{a},\gamma)}(c,s_{\tilde{a}})\right| \,|\,|J_a| - |J_{\tilde{a}}|\,|}_{\text{(III)}}.
\end{aligned}$$

(I) compares the two estimated densities at common (c, s_a). By (41) and $|J_a| \leq \lambda_n$,

$$\text{(I)} \leq M_{q,n}^{-1}\tilde{\mathcal{R}}_n\, \nu^{-l-1}\, \lambda_n\, \|a - \tilde{a}\|.$$

(II) is the difference of the same estimated density at (c, s_a) and $(c, s_{\tilde{a}})$. By Assumption 5.4(ii),

$$\text{(II)} \leq \nu^{-l-1} M_{q,n}^{-1} 4M_k \|(c, s_a) - (c, s_{\tilde{a}})\|\, |J_a| \leq M_{q,n}^{-1} C\, \nu^{-l-1}\, \lambda_n\, \|a - \tilde{a}\|,$$

for some finite constant C that collects $4M_k$, $|b_1|, |b_2|$, and $|\delta_1\gamma|$.
For (III), write

$$\text{(III)} = \left|\widehat{f}_{C_i(\tilde{a},\gamma),\gamma_i,\beta_i}(c,\gamma,b)\right| \,\frac{|\,|J_a| - |J_{\tilde{a}}|\,|}{|J_{\tilde{a}}|}.$$

We bound the relative change of the Jacobian via the derivative of $\ln|J_a|$. Since $\ln|J_a| = \ln|\delta_1| + J_x \ln|a_1| + (J_x+1)\ln|a_2|$, $\|\nabla_a \ln|J_a|\,\| \leq (2J_x+1)\underline{\vartheta}_n^{-1}$ for $a \in \mathcal{A}_n$. Since $\|a-\tilde{a}\| \leq \underline{\vartheta}_n$ and $|a_j|, |\tilde{a}_j| \geq \underline{\vartheta}_n$, a_j and $\tilde{a}_j$ have the same sign for each j (opposite signs imply $|a_j - \tilde{a}_j| \geq 2\underline{\vartheta}_n$); the segment $\{(1-\chi)\tilde{a} + \chi a : \chi \in [0,1]\}$ therefore stays in $[\underline{\vartheta}_n, \bar{\vartheta}_n]$ coordinatewise, hence in $\mathcal{A}_n$. By the mean value theorem,

$$|\ln|J_a| - \ln|J_{\tilde{a}}|\,| \leq (2J_x+1)\underline{\vartheta}_n^{-1}\|a - \tilde{a}\| \leq 2J_x + 1.$$

It follows that

$$\begin{aligned}
\frac{|\,|J_a| - |J_{\tilde{a}}|\,|}{|J_{\tilde{a}}|} &= |\exp(\ln|J_a| - \ln|J_{\tilde{a}}|) - 1| \\
&\leq |\ln|J_a| - \ln|J_{\tilde{a}}|\,|\, \exp(|\ln|J_a| - \ln|J_{\tilde{a}}||) \leq M_x\, \underline{\vartheta}_n^{-1}\|a - \tilde{a}\|,
\end{aligned}$$

with $M_x = (2J_x+1)\exp(2J_x+1)$, where the first inequality uses $|\exp(x)-1| \leq |x|\exp(|x|)$.

Hence

$$\text{(III)} \leq M_x \, \underline{\vartheta}_n^{-1} \left| \widehat{f}_{C_i(\tilde{a},\gamma),\gamma_i,\beta_i}(c,\gamma,b) \right| \, \|a - \tilde{a}\|.$$

Collecting (I)–(III) and setting $\mathcal{R}_n \equiv \tilde{\mathcal{R}}_n + C = O_p(1)$ yields the stated bound. As $\tilde{\mathcal{R}}_n$ and C do not depend on $a, \tilde{a}, c$, the bound holds with a common $\mathcal{R}_n$. ■

Lemma S.8. *Fix $a \in \mathbb{R}^2$ with $a_1, a_2 \neq 0$, and suppose Assumptions 5.1, 5.2, 5.3, 5.4, 5.5*(i)–(ii)*, and 5.6 hold. For any (γ, b) on the joint support of (γ_i, β_i), there exist constants $M_1, M_2, M_3, C_e > 0$ such that, for any sequence $t_n > 0$ satisfying $t_n \to 0$, $t_n \geq C_0 M_f(\bar{d}) \bar{c}_n^{-p}$, and*

(i) $t_n \geq C_0 |J_i(\bar{d})| \Big((1 + M_r(\bar{d})^{1/2}) \Xi_{q,n}^{1/2} \nu^{-l+1/2} \sqrt{\ln n/n} + M_r(\bar{d}) \Xi_{q,n} \nu^{-l+1} r_{2n} + B(\bar{d}) (\underline{q}_n^{1/2} + \nu^{m-l/2-\varsigma}) \Big)$, *and* $|J_i(\bar{d})| M_r(\bar{d}) M_{q,n} \Xi_{q,n} \nu^{-l+1} t_n^{-1} \to \infty$.

(ii) $t_n \geq C_0 M_f(\bar{d}) \bar{c}_n \left(\Xi_{q,n}^{1/2} \nu^{-l+1/2} \sqrt{\ln n/n} + \Xi_{q,n} \nu^{-l+1} r_{2n} + \underline{q}_n^{1/2} + \nu^{m-l/2-\varsigma} \right)$, *and* $M_f(\bar{d}) \bar{c}_n M_{q,n} \Xi_{q,n} \nu^{-l+1} t_n^{-1} \to \infty$,

for some sufficiently large constant $C_0 > 0$. Then, for all n large,

$$\begin{aligned} &\Pr\left(\sup_{c \in \mathcal{C}_n} \left| \widehat{f}_{C_i(a,\gamma)|\gamma_i=\gamma,\beta_i=b}(c) - f_{C_i(a,\gamma)|\gamma_i=\gamma,\beta_i=b}(c) \right| > t_n \right) \\ &\leq M_1 n^{C_e} \exp\left(-\tilde{M}_1(\bar{d}) \, \Xi_{q,n}^{-1} n \nu^{2l-1} t_n^2 \right) + M_1 n^{C_e} \exp\left(-\tilde{M}_1'(\bar{d}) \, \Xi_{q,n}^{-2} \nu^{2l-2} n \tau^{l-1} t_n^2 \right), \end{aligned}$$

where $C_i(a,\gamma) = C_i(\bar{d})$, $\tilde{M}_1(\bar{d}) \equiv \min\left\{ M_2 |J_i(\bar{d})|^{-2} M_r(\bar{d})^{-1}, M_3 M_f(\bar{d})^{-2} \bar{c}_n^{-2} \right\} > 0$, and $\tilde{M}_1'(\bar{d}) \equiv \min\left\{ M_2 |J_i(\bar{d})|^{-2} M_r(\bar{d})^{-2}, M_3 M_f(\bar{d})^{-2} \bar{c}_n^{-2} \right\} > 0$, with $M_f(d)$ defined in Lemma S.6.

Proof of Lemma S.8. Let $\mathcal{T}$ be a compact subset of the joint support of (γ_i, β_i) that contains (γ, b). By the triangle inequality,

$$\begin{aligned} &\left| \widehat{f}_{C_i(a,\gamma)|\gamma,b}(c) - f_{C_i(a,\gamma)|\gamma,b}(c) \right| = \left| \frac{\widehat{f}_{C_i(a,\gamma),\gamma_i,\beta_i}(c,\gamma,b)}{\widehat{f}_{\gamma_i,\beta_i}(\gamma,b)} - \frac{f_{C_i(a,\gamma),\gamma_i,\beta_i}(c,\gamma,b)}{f_{\gamma_i,\beta_i}(\gamma,b)} \right| \\ &\leq \frac{\left| \widehat{f}_{C_i(a,\gamma),\gamma_i,\beta_i}(c,\gamma,b) - f_{C_i(a,\gamma),\gamma_i,\beta_i}(c,\gamma,b) \right|}{\left| \widehat{f}_{\gamma_i,\beta_i}(\gamma,b) \right|} + \frac{f_{C_i(a,\gamma),\gamma_i,\beta_i}(c,\gamma,b)}{f_{\gamma_i,\beta_i}(\gamma,b)} \cdot \frac{\left| \widehat{f}_{\gamma_i,\beta_i}(\gamma,b) - f_{\gamma_i,\beta_i}(\gamma,b) \right|}{\left| \widehat{f}_{\gamma_i,\beta_i}(\gamma,b) \right|}. \end{aligned}$$

Define $\tilde{\mathcal{E}}_n \equiv \{ \inf_{(\gamma,b)\in\mathcal{T}} \widehat{f}_{\gamma_i,\beta_i}(\gamma,b) > M_b/2 \}$. Under Assumption 5.1(iii) and the uniform consistency $\sup_{(\gamma,b)\in\mathcal{T}} \left| \widehat{f}_{\gamma_i,\beta_i}(\gamma,b) - f_{\gamma_i,\beta_i}(\gamma,b) \right| = o_p(1)$ by Proposition 2, we have

$\Pr(\tilde{\mathcal{E}}_n) \to 1$. On $\tilde{\mathcal{E}}_n$, using $f_{\gamma_i,\beta_i}(\gamma, b) \geq M_b$ on $\mathcal{T}$ and $M_f(\bar{d}) \equiv \sup_{c\in\mathcal{C}_n} f_{C_i(a,\gamma),\gamma_i,\beta_i}(c,\gamma,b)$,

$$\sup_{c\in\mathcal{C}_n}\left|\widehat{f}_{C_i(a,\gamma)|\gamma,b}(c) - f_{C_i(a,\gamma)|\gamma,b}(c)\right| \leq \underbrace{\frac{2}{M_b}\sup_{c\in\mathcal{C}_n}\left|\widehat{f}_{C_i(a,\gamma),\gamma_i,\beta_i}(c,\gamma,b) - f_{C_i(a,\gamma),\gamma_i,\beta_i}(c,\gamma,b)\right|}_{\text{(I)}}$$
$$+ \underbrace{\frac{2M_f(\bar{d})}{M_b^2}\sup_{(\gamma,b)\in\mathcal{T}}\left|\widehat{f}_{\gamma_i,\beta_i}(\gamma,b) - f_{\gamma_i,\beta_i}(\gamma,b)\right|}_{\text{(II)}}.$$

Then, for any $t_n > 0$,

$$\Pr\left(\sup_{c\in\mathcal{C}_n}\left|\widehat{f}_{C_i(a,\gamma)|\gamma,b}(c) - f_{C_i(a,\gamma)|\gamma,b}(c)\right| > t_n\right) \leq \Pr\left(\text{(I)} > \frac{t_n}{2}\right) + \Pr\left(\text{(II)} > \frac{t_n}{2}\right) + \Pr\left(\tilde{\mathcal{E}}_n^c\right).$$

Lemma S.4 establishes Bernstein's inequality for the RTE over a compact set $\mathcal{S}$. As in the proof of Proposition 2, the same argument extends to $\mathcal{S}_n \equiv \mathcal{C}_n \times \{\tilde{s}(d,\gamma,b) : (\gamma,b) \in \mathcal{T}\}$, where $\tilde{s}(d,\gamma,b) \equiv (b_1(d_1 + d_2\gamma), b_2 d_2, d_2\delta_1\gamma)$. Because $\tau^\zeta \asymp r_{2n}$ by Assumption 5.3(iv), the term involving r_{2n} in Condition (i) is of the same order as the term involving τ^ζ required by Lemma S.4. Therefore, for any $\epsilon_n > 0$ that satisfies the conditions in Lemma S.4, and all n sufficiently large,

$$\begin{aligned}&\Pr\left(\sup_{(c,s)\in\mathcal{S}_n}|\widehat{f}_{C_i(d),S_i(d)}(c,s) - f_{C_i(d),S_i(d)}(c,s)| > \epsilon_n\right)\\ &\leq M_e n^{C_e}\exp\left(-\tilde{M}_e M_r(d)^{-1}\Xi_{q,n}^{-1} n\nu^{2l-1}\epsilon_n^2\right) + M_e\, n^{C_e}\exp\left(-\tilde{M}_e\, M_r(d)^{-2}\Xi_{q,n}^{-2}\nu^{2l-2}n\tau^{l-1}\epsilon_n^2\right),\end{aligned} \tag{42}$$

for some constants $M_e, \tilde{M}_e, C_e > 0$.

Term (I). Since $\widehat{f}_{C_i(a,\gamma),\gamma_i,\beta_i}(c,\gamma,b) = \widehat{f}_{C_i(\bar{d}),S_i(\bar{d})}(c,s_a)\,|J_i(\bar{d})|$ with $s_a \equiv (b_1a_1, b_2a_2, a_2\delta_1\gamma)$, applying (42) with $d = \bar{d}$ and threshold $\epsilon_n = M_b t_n/(4|J_i(\bar{d})|)$ yields

$$\begin{aligned}\Pr\left(\text{(I)} > \frac{t_n}{2}\right) &\leq M_e n^{C_e}\exp\left(-M_2\,|J_i(\bar{d})|^{-2}M_r(\bar{d})^{-1}\Xi_{q,n}^{-1}\, n\nu^{2l-1}t_n^2\right)\\ &\quad + M_e\, n^{C_e}\exp\left(-M_2\,|J_i(\bar{d})|^{-2}M_r(\bar{d})^{-2}\Xi_{q,n}^{-2}\nu^{2l-2}n\tau^{l-1}t_n^2\right),\end{aligned} \tag{43}$$

where $M_2 \equiv (M_b/4)^2\,\tilde{M}_e > 0$ is independent of $\bar{d}$. The required conditions hold by the

maintained condition (i). *Term* (II). Applying (16) with $d = (0,1)$, we obtain

$$\text{(II)} \leq \underbrace{\frac{4M_f(\bar{d})}{M_b^2}\bar{c}_n\,|J^*| \sup_{(c,\gamma,b)\in\mathcal{C}_n\times\mathcal{T}}\left|\widehat{f}_{C_i^*,S_i^*}(c,s^*) - f_{C_i^*,S_i^*}(c,s^*)\right|}_{\tilde{T}_1}$$
$$+ \underbrace{\frac{2M_f(\bar{d})}{M_b^2}\sup_{(\gamma,b)\in\mathcal{T}}\int_{|c|>\bar{c}_n} f_{C_i^*,\gamma_i,\beta_i}(c,\gamma,b)\,dc}_{\tilde{T}_2},$$

where $s^* \equiv (b_1\gamma, b_2, \delta_1\gamma)$ and $|J^*| = \sup_{(\gamma,b)\in\mathcal{T}}|\delta_1\gamma^{J_x}| < \infty$. For $\tilde{T}_1$, (42) applies with $d = (0,1)$ and $\epsilon_n = M_b^2 t_n/(16M_f(\bar{d})\bar{c}_n|J^*|)$ by (ii),

$$\Pr\left(\tilde{T}_1 > \frac{t_n}{4}\right) \leq M_e n^{C_e}\exp\left(-M_3 M_f(\bar{d})^{-2}\bar{c}_n^{-2}\Xi_{q,n}^{-1}\,n\nu^{2l-1}t_n^2\right)$$
$$+ M_e n^{C_e}\exp\left(-M_3 M_f(\bar{d})^{-2}\bar{c}_n^{-2}\Xi_{q,n}^{-2}\nu^{2l-2}n\tau^{l-1}t_n^2\right),$$

for some constant $M_3 > 0$. By (19), we have $\tilde{T}_2 = O(M_f(\bar{d})\bar{c}_n^{-p})$. Then $\tilde{T}_2 \leq t_n/4$ for n sufficiently large under $t_n \geq C_0 M_f(\bar{d})\bar{c}_n^{-p}$ with C_0 large. Hence

$$\Pr\left(\text{(II)} > \frac{t_n}{2}\right) \leq M_e n^{C_e}\exp\left(-M_3\,M_f(\bar{d})^{-2}\,\bar{c}_n^{-2}\Xi_{q,n}^{-1}\,n\nu^{2l-1}t_n^2\right)$$
$$+ M_e\,n^{C_e}\exp\left(-M_3\,M_f(\bar{d})^{-2}\,\bar{c}_n^{-2}\Xi_{q,n}^{-2}\nu^{2l-2}n\tau^{l-1}t_n^2\right). \tag{44}$$

Term $\Pr(\tilde{\mathcal{E}}_n^c)$. We have $\tilde{\mathcal{E}}_n^c \subseteq \{\sup_{(\gamma,b)\in\mathcal{T}}|\widehat{f}_{\gamma_i,\beta_i} - f_{\gamma_i,\beta_i}| > M_b/2\}$. On this event, $\text{(II)} \geq (2M_f(\bar{d})/M_b^2)(M_b/2) = M_f(\bar{d})/M_b$. Since $M_f(\bar{d}) > 0$ and $t_n \to 0$, $t_n/2 < M_f(\bar{d})/M_b$ for n large, so $\tilde{\mathcal{E}}_n^c \subseteq \{\text{(II)} > t_n/2\}$ and $\Pr(\tilde{\mathcal{E}}_n^c) \leq \Pr((\text{II}) > t_n/2)$, bounded by (44). Combining (43) and (44) and collecting the first and second exponentials separately, with rates $\tilde{M}_1(\bar{d})$ and $\tilde{M}_1'(\bar{d})$ respectively, completes the proof. ■

Lemma A.2. *Suppose Assumptions 5.1, 5.2, 5.3, 5.4, 5.5*(i)*–(iii), 5.6, S.1, and S.2 hold, and that* $E(|\widetilde{Y}_{i1}|) < \infty$ *and* $E(|\widetilde{Y}_{i2}|) < \infty$. *Then, for any* (γ, b) *on the joint support of* (γ_i, β_i),

$$\sup_{a\in\mathcal{A}_n}\sup_{c\in\mathcal{C}_n}\left|\widehat{f}_{C_i(a,\gamma)|\gamma_i=\gamma,\beta_i=b}(c) - f_{C_i(a,\gamma)|\gamma_i=\gamma,\beta_i=b}(c)\right| = O_p\left(\varepsilon_n^*\right),$$

where ε_n^* *is defined in Assumption S.2.*

Proof of Lemma A.2. For $a \in \mathcal{A}_n$, define

$$D(a) \equiv \sup_{c \in \mathcal{C}_n} |\widehat{f}_{C_i(a,\gamma)|\gamma,b}(c) - f_{C_i(a,\gamma)|\gamma,b}(c)|.$$

Cover $\mathcal{A}_n$ by a finite collection of balls $\{\mathcal{B}_k\}_{k=1}^{N_n^{\mathcal{A}}}$ of radius $\ell_n = M_{q,n}\Xi_{q,n}^{1/2}\nu^{3/2}\sqrt{\ln n/n}$ centered at $a_k = (a_{1k}, a_{2k})$. Then

$$\sup_{a \in \mathcal{A}_n} D(a) \leq \underbrace{\max_{1 \leq k \leq N_n^{\mathcal{A}}} \sup_{a \in \mathcal{A}_n \cap \mathcal{B}_k} |D(a) - D(a_k)|}_{D_1} + \underbrace{\max_{1 \leq k \leq N_n^{\mathcal{A}}} |D(a_k)|}_{D_2}. \tag{45}$$

We first consider D_2**.** Applying Lemma S.8 to each a_k, we obtain for any $\varepsilon_n > 0$ that satisfies the conditions of Lemma S.8, and all n sufficiently large,

$$\begin{aligned}\Pr(|D(a_k)| > \varepsilon_n) \leq{}& M_1\, n^{C_e} \exp\Big(-\tilde{M}_1\big(\bar{d}(a_k,\gamma)\big)\, \Xi_{q,n}^{-1}\, n\nu^{2l-1}\varepsilon_n^2\Big) \\ &+ M_1\, n^{C_e} \exp\Big(-\tilde{M}_1'\big(\bar{d}(a_k,\gamma)\big)\, \Xi_{q,n}^{-2}\nu^{2l-2} n\tau^{l-1}\varepsilon_n^2\Big),\end{aligned} \tag{46}$$

where $\tilde{M}_1(\bar{d}(a_k,\gamma)) = \min\big\{M_2\,|J_i(\bar{d}(a_k,\gamma))|^{-2} M_r(\bar{d}(a_k,\gamma))^{-1},\, M_3\, M_f(\bar{d}(a_k,\gamma))^{-2}\bar{c}_n^{-2}\big\}$, as defined in Lemma S.8, and $\tilde{M}_1'(\bar{d}(a_k,\gamma))$ analogously with M_r^{-1} replaced by M_r^{-2}. By (46) and a union bound over the $N_n^{\mathcal{A}}$ centers,

$$\begin{aligned}\Pr\Big(\max_{1 \leq k \leq N_n^{\mathcal{A}}} |D(a_k)| > \varepsilon_n\Big) \leq{}& M_1 N_n^{\mathcal{A}}\, n^{C_e} \exp\Big(-\min_k \tilde{M}_1(\bar{d}(a_k,\gamma))\Xi_{q,n}^{-1} n\nu^{2l-1}\varepsilon_n^2\Big) \\ &+ M_1 N_n^{\mathcal{A}}\, n^{C_e} \exp\Big(-\min_k \tilde{M}_1'(\bar{d}(a_k,\gamma))\Xi_{q,n}^{-2}\nu^{2l-2} n\tau^{l-1}\varepsilon_n^2\Big).\end{aligned} \tag{47}$$

We suppress $\bar{d}(a_k,\gamma)$ in $|J_i|$, M_r, and M_f. By Lemma S.5(a), $M_r \asymp |a_{2k}|^{-1}$; by Lemma S.6, $M_f \asymp \max(|a_{1k}|, |a_{2k}|)^{-1}$, where $\bar{c}_n/\bar{\vartheta}_n \to \infty$ of that lemma is implied by Assumptions 5.6 and S.2. Combining with $|J_i| = |\delta_1|\,|a_{2k}|^{J_x+1}|a_{1k}|^{J_x}$, we have

$$\inf_{a_k \in \mathcal{A}_n} |J_i|^{-2} M_r^{-1} \asymp \bar{\vartheta}_n^{-(4J_x+1)}, \qquad \inf_{a_k \in \mathcal{A}_n} M_f^{-2}\bar{c}_n^{-2} \asymp \underline{\vartheta}_n^2\, \bar{c}_n^{-2}.$$

Hence, for some constants $M_5, M_5' > 0$,

$$\min_{1 \leq k \leq N_n^{\mathcal{A}}} \tilde{M}_1(\bar{d}(a_k,\gamma)) \geq M_5 \min\big\{\bar{\vartheta}_n^{-(4J_x+1)}, \underline{\vartheta}_n^2\bar{c}_n^{-2}\big\}, \tag{48}$$

and the same steps give $\min_k \tilde{M}_1'(\bar{d}(a_k,\gamma)) \geq M_5' \min\{\bar{\vartheta}_n^{-4J_x}, \underline{\vartheta}_n^2 \bar{c}_n^{-2}\}$.

Since $\mathcal{A}_n \subset [-\bar{\vartheta}_n, \bar{\vartheta}_n]^2$, the number of balls of radius ℓ_n required to cover $\mathcal{A}_n$ satisfies $N_n^{\mathcal{A}} = O\big(\bar{\vartheta}_n^2 \ell_n^{-2}\big)$. Using $\ell_n^{-1} = o(n^2(\ln n)^{-2})$ in the proof of Lemma S.2 and $\bar{\vartheta}_n = O(n^{C_v})$ in Assumption S.2, we have $N_n^{\mathcal{A}} = o\big(n^{2C_v+4}(\ln n)^{-4}\big)$. Take $\varepsilon_n = M_6\, \varepsilon_n^*$ for some constant $M_6 > 0$, which satisfies the conditions of Lemma S.8 uniformly over $\mathcal{A}_n$. By the union bound and (48), and using $\Xi_{q,n} = O(\underline{q}_n^{1-k})$, $n\nu^{2l-1} r_{1n}^2 \asymp \underline{q}_n^{1-k} \ln n$, and $n\tau^{l-1} r_{2n}^2 \asymp \ln n$, both exponents in (47) are bounded below by $M_7 M_6^2 \ln n$ for some constant $M_7 > 0$. Hence

$$\Pr(D_2 > \varepsilon_n) \leq 2M_1 N_n^{\mathcal{A}} n^{C_e - M_7 M_6^2} = o\Big(n^{2C_v+4+C_e-M_7M_6^2}(\ln n)^{-4}\Big) = o(1),$$

by taking M_6 large enough. Therefore,

$$D_2 = O_p(\varepsilon_n^*)\,. \tag{49}$$

We next turn to D_1. For any $a \in \mathcal{A}_n \cap \mathcal{B}_k$, since $|\sup_{c\in\mathcal{C}_n} g(c) - \sup_{c\in\mathcal{C}_n} h(c)| \leq \sup_{c\in\mathcal{C}_n} |g(c) - h(c)|$,

$$\begin{aligned}
|D(a) - D(a_k)| &\leq \sup_{c\in\mathcal{C}_n} \Big| \Big|\widehat{f}_{C_i(a,\gamma)|\gamma,b}(c) - f_{C_i(a,\gamma)|\gamma,b}(c)\Big| - \Big|\widehat{f}_{C_i(a_k,\gamma)|\gamma,b}(c) - f_{C_i(a_k,\gamma)|\gamma,b}(c)\Big| \Big| \\
&\leq \underbrace{\sup_{c\in\mathcal{C}_n} \Big|\widehat{f}_{C_i(a,\gamma)|\gamma,b}(c) - \widehat{f}_{C_i(a_k,\gamma)|\gamma,b}(c)\Big|}_{D_{1,1}} + \underbrace{\sup_{c\in\mathcal{C}_n} \big|f_{C_i(a,\gamma)|\gamma,b}(c) - f_{C_i(a_k,\gamma)|\gamma,b}(c)\big|}_{D_{1,2}},
\end{aligned}$$

where the second inequality uses the reverse and triangle inequalities.

For $D_{1,1}$, for $a \in \mathcal{A}_n \cap \mathcal{B}_k$, $\|a - a_k\| \leq \ell_n \leq \underline{\vartheta}_n/2$ for large n because $\underline{\vartheta}_n^{-1}\nu^{l+1} r_{1n} \to 0$, which follows because the last term of ε_n^* converges to zero. Then, by Lemma S.7, uniformly over $a \in \mathcal{A}_n \cap \mathcal{B}_k$, and on $\tilde{\mathcal{E}}_n$ of the proof of Lemma S.8, which has probability tending to one and on which $|\widehat{f}_{\gamma_i,\beta_i}(\gamma,b)| \geq M_b/2$,

$$\begin{aligned}
D_{1,1} &= \sup_{c\in\mathcal{C}_n} \frac{1}{|\widehat{f}_{\gamma_i,\beta_i}(\gamma,b)|} \Big|\widehat{f}_{C_i(a,\gamma),\gamma_i,\beta_i}(c,\gamma,b) - \widehat{f}_{C_i(a_k,\gamma),\gamma_i,\beta_i}(c,\gamma,b)\Big| \\
&\leq \frac{2}{M_b}\left(M_{q,n}^{-1}\mathcal{R}_n\, \lambda_n \nu^{-l-1} + M_x\, \underline{\vartheta}_n^{-1} \sup_{c\in\mathcal{C}_n} \Big|\widehat{f}_{C_i(a_k,\gamma),\gamma_i,\beta_i}(c,\gamma,b)\Big|\right) \ell_n,
\end{aligned} \tag{50}$$

where $\mathcal{R}_n = O_p(1)$, $\lambda_n = \sup_{a\in\mathcal{A}_n} |J_i(\bar{d}(a,\gamma))|$, and $M_x = O(1)$ as defined in Lemma S.7. By uniform consistency of $\widehat{f}_{\gamma_i,\beta_i}$ in Proposition 2 and Assumption 5.1(ii), $\widehat{f}_{\gamma_i,\beta_i}(\gamma,b) =$

$O_p(1)$. By Lemma S.6 and $f_{\gamma_i,\beta_i}(\gamma,b) \geq M_b$, $\max_k \sup_c f_{C_i(a_k,\gamma)|\gamma,b}(c) = O(\underline{\vartheta}_n^{-1})$. By the triangle inequality and (49), we have

$$\begin{aligned}
&\max_{1\leq k\leq N_n^{\mathcal{A}}} \sup_{c\in\mathcal{C}_n} \left|\widehat{f}_{C_i(a_k,\gamma),\gamma_i,\beta_i}(c,\gamma,b)\right| \\
&\leq \max_{1\leq k\leq N_n^{\mathcal{A}}} \sup_{c\in\mathcal{C}_n} \widehat{f}_{\gamma_i,\beta_i}(\gamma,b)\left(\left|\widehat{f}_{C_i(a_k,\gamma)|\gamma,b}(c) - f_{C_i(a_k,\gamma)|\gamma,b}(c)\right| + \left|f_{C_i(a_k,\gamma)|\gamma,b}(c)\right|\right) \\
&= O_p(\varepsilon_n^*) + O_p(\underline{\vartheta}_n^{-1}) = O_p(\underline{\vartheta}_n^{-1}),
\end{aligned}$$

under $\varepsilon_n^* \to 0$. Then, substituting back into (50),

$$\max_{1\leq k\leq N_n^{\mathcal{A}}} \sup_{a\in\mathcal{A}_n\cap\mathcal{B}_k} D_{1,1} = O_p\left(M_{q,n}^{-1}\bar{\vartheta}_n^{2J_x+1}\nu^{-l-1}\ell_n + \underline{\vartheta}_n^{-2}\ell_n\right). \tag{51}$$

For $D_{1,2}$, using $f_{C_i(a,\gamma)|\gamma,b}(c) = f_{\widetilde{U}_i(a)|\gamma,b}(c - a_1\gamma y_0)$, decompose

$$\begin{aligned}
D_{1,2} \leq \sup_{c\in\mathcal{C}_n} \underbrace{\left|f_{\widetilde{U}_i(a)|\gamma,b}(c - a_1\gamma y_0) - f_{\widetilde{U}_i(a_k)|\gamma,b}(c - a_1\gamma y_0)\right|}_{D_{1,2}(\mathrm{I})} \\
+ \sup_{c\in\mathcal{C}_n} \underbrace{\left|f_{\widetilde{U}_i(a_k)|\gamma,b}(c - a_1\gamma y_0) - f_{\widetilde{U}_i(a_k)|\gamma,b}(c - a_{1k}\gamma y_0)\right|}_{D_{1,2}(\mathrm{II})}.
\end{aligned}$$

For $D_{1,2}(\mathrm{I})$, by a change of variables $u_1 = (\tilde{u} - a_2u_2)/a_1$,

$$f_{\widetilde{U}_i(a)|\gamma,b}(\tilde{u}) = |a_1|^{-1}\int f_{U_{i1},U_{i2}|\gamma,b}\left(\tfrac{\tilde{u}-a_2u_2}{a_1}, u_2\right) du_2.$$

By the mean value theorem,

$$D_{1,2}(\mathrm{I}) \leq \sup_{\tilde{a}\in[a_k,a]} \left\|\nabla_a f_{\widetilde{U}_i(\tilde{a})|\gamma,b}(c - a_1\gamma y_0)\right\| \, \|a - a_k\|,$$

where $[a_k, a]$ denotes the line segment between a_k and a. It remains to bound the gradient along the segment. Under Assumption 5.5(iii),

$$\begin{aligned}
\left|\partial_{a_2} f_{\widetilde{U}_i(a)|\gamma,b}(\tilde{u})\right| &= |a_1|^{-2}\left|\int u_2\, \partial_{u_1} f_{U_{i1},U_{i2}|\gamma,b}\left(\tfrac{\tilde{u}-a_2u_2}{a_1}, u_2\right) du_2\right| \\
&\leq |a_1|^{-2}\int \sup_{u_1}|u_2|\, \left|\partial_{u_1} f_{U_{i1},U_{i2}|\gamma,b}(u_1,u_2)\right| du_2 = O\big(|a_1|^{-2}\big),
\end{aligned}$$

and, using instead the representation $f_{\widetilde{U}_i(a)|\gamma,b}(\tilde{u}) = |a_2|^{-1} \int f_{U_{i1},U_{i2}|\gamma,b}\left(u_1, \frac{\tilde{u}-a_1u_1}{a_2}\right) du_1$, in which a_1 enters only through the argument,

$$\left|\partial_{a_1} f_{\widetilde{U}_i(a)|\gamma,b}(\tilde{u})\right| \leq |a_2|^{-2} \int \sup_{u_2} |u_1| \left|\partial_{u_2} f_{U_{i1},U_{i2}|\gamma,b}(u_1,u_2)\right| du_1 = O\big(|a_2|^{-2}\big)\,,$$

with the bounds holding uniformly in $\tilde{u}$. Since $\|a - a_k\| \leq \ell_n$ and $|a_j|, |a_{jk}| \geq \underline{\vartheta}_n$, a_j and a_{jk} share a sign for $j = 1, 2$, as in the proof of Lemma S.7, so the segment $[a_k, a]$ stays in $\mathcal{A}_n$ and $|\tilde{a}_1|, |\tilde{a}_2|$ lie in $[\underline{\vartheta}_n - \ell_n, \bar{\vartheta}_n]$ with $\underline{\vartheta}_n - \ell_n \geq \underline{\vartheta}_n/2$ for n large. Hence $\sup_{\tilde{a}\in[a_k,a]} \|\nabla_a f_{\widetilde{U}_i(\tilde{a})|\gamma,b}(\tilde{u})\| = O(\underline{\vartheta}_n^{-2})$ uniformly in $\tilde{u}$, and then uniformly over $a \in \mathcal{A}_n \cap \mathcal{B}_k$,

$$D_{1,2}(\mathrm{I}) = O\big(\underline{\vartheta}_n^{-2}\, \ell_n\big)\,.$$

For $D_{1,2}$(II),

$$\begin{aligned}\left|\partial_{\tilde{u}} f_{\widetilde{U}_i(a_k)|\gamma,b}(\tilde{u})\right| &= |a_{1k}|^{-2} \left|\int \partial_{u_1} f_{U_{i1},U_{i2}|\gamma,b}\Big(\tfrac{\tilde{u}-a_{2k}u_2}{a_{1k}}, u_2\Big)\, du_2\right| \\ &\leq |a_{1k}|^{-2} \int \sup_{u_1} \left|\partial_{u_1} f_{U_{i1},U_{i2}|\gamma,b}(u_1,u_2)\right| du_2 = O(\underline{\vartheta}_n^{-2}),\end{aligned}$$

uniformly in $\tilde{u}$ by Assumption 5.5(iii). Since $|(c-a_1\gamma y_0)-(c-a_{1k}\gamma y_0)| = |\gamma y_0|\,|a_1-a_{1k}| \leq |\gamma y_0|\,\ell_n$, the mean value theorem gives $D_{1,2}(\mathrm{II}) = O(\underline{\vartheta}_n^{-2}\ell_n)$. Therefore,

$$\max_{1\leq k\leq N_n^{\mathcal{A}}} \sup_{a\in\mathcal{A}_n\cap\mathcal{B}_k} D_{1,2} = O\big(\underline{\vartheta}_n^{-2}\ell_n\big)\,,$$

which together with the bound for $D_{1,1}$ in (51) gives

$$D_1 = O_p\left(M_{q,n}^{-1}\bar{\vartheta}_n^{2J_x+1}\nu^{-l-1}\ell_n + \underline{\vartheta}_n^{-2}\ell_n\right) = O_p(\bar{\vartheta}_n^{2J_x+1}r_{1n} + \underline{\vartheta}_n^{-2}\nu^{l+1}r_{1n}) = O_p\left(\varepsilon_n^*\right).$$

Combining this bound with (45) and (49),

$$\sup_{a\in\mathcal{A}_n} D(a) = O_p(\varepsilon_n^*)\,.$$

■

S3 Additional Monte Carlo Results

This section presents two robustness checks for the conditional density estimator $\widehat{f}_{U_i|\gamma_i,\beta_i}$. Table 3 reports the MISE as the values of (γ_i, β_i) range over their 20th, 50th, and 80th percentiles. For all designs, the MISE falls with n at each reported value, so consistency holds across the range of (γ_i, β_i).

Table 3: Estimator for the conditional density of U_i, varying the percentiles of (γ_i, β_i)

	n	$(P^{\gamma}_{50}, P^{\beta}_{20})$	$(P^{\gamma}_{50}, P^{\beta}_{80})$	$(P^{\gamma}_{20}, P^{\beta}_{50})$	$(P^{\gamma}_{80}, P^{\beta}_{50})$
Baseline	500	0.0871	0.0873	0.0863	0.0872
	2000	0.0303	0.0311	0.0326	0.0333
	8000	0.0063	0.0066	0.0129	0.0134
Correlated	500	0.0545	0.0547	0.0658	0.0441
	2000	0.0188	0.0195	0.0258	0.0155
	8000	0.0049	0.0054	0.0086	0.0100
Scale	500	0.0862	0.0861	0.1048	0.0718
	2000	0.0297	0.0302	0.0433	0.0248
	8000	0.0065	0.0066	0.0149	0.0134
Bimodal	500	0.0558	0.0554	0.0632	0.0528
	2000	0.0460	0.0449	0.0549	0.0413
	8000	0.0389	0.0378	0.0452	0.0391

Notes: The table reports the mean integrated squared error (MISE) of the estimated conditional density $\widehat{f}_{U_{i1},U_{i2}|\gamma_i,\beta_i}$ across $N_{\text{rep}} = 500$ Monte Carlo replications. Columns 3–4 hold γ_i at its median (P^{γ}_{50}) and move β_i to the 20th and 80th percentiles; Columns 5–6 hold β_i at its median (P^{β}_{50}) and move γ_i to the 20th and 80th percentiles.

The RTE can perform better with heavier-tailed regressors, as Masten (2018, Tables 1 and 2) finds. We replace the Gaussian regressors X_i and W_{i1} with Student-t draws with five degrees of freedom. Table 4 reports the MISE together with the bias, SD, and MSE of the estimated IQR of each marginal, all evaluated at the median of (γ_i, β_i). Figures 3 and 4 show the estimated densities under the heavy-tailed design. The results are comparable to those in Table 2, with a lower MISE for all four designs at all n.

Table 4: Estimator for the conditional density of U_i, heavy-tailed regressors

			$\widehat{\text{IQR}}(U_{i1} \mid \gamma_i, \beta_i)$			$\widehat{\text{IQR}}(U_{i2} \mid \gamma_i, \beta_i)$		
	n	MISE	Bias	SD	MSE	Bias	SD	MSE
Baseline	500	0.0743	0.4239	0.0278	0.1804	0.4039	0.0278	0.1639
	2000	0.0247	0.2230	0.0152	0.0499	0.2056	0.0153	0.0425
	8000	0.0052	0.1093	0.0086	0.0120	0.0936	0.0088	0.0088
Correlated	500	0.0411	0.3892	0.0313	0.1525	0.3639	0.0284	0.1332
	2000	0.0119	0.1826	0.0171	0.0336	0.1633	0.0176	0.0270
	8000	0.0041	0.0506	0.0108	0.0027	0.0324	0.0120	0.0012
Scale	500	0.0737	0.4222	0.0268	0.1790	0.4012	0.0253	0.1616
	2000	0.0242	0.2220	0.0155	0.0495	0.2037	0.0148	0.0417
	8000	0.0052	0.1076	0.0086	0.0117	0.0902	0.0086	0.0082
Bimodal	500	0.0527	−0.2119	0.0766	0.0508	−0.2201	0.0761	0.0542
	2000	0.0384	−0.1852	0.0591	0.0378	−0.1852	0.0698	0.0392
	8000	0.0305	−0.0703	0.0451	0.0070	−0.0882	0.0518	0.0105

Notes: The table reports, across $N_{\text{rep}} = 500$ Monte Carlo replications, the mean integrated squared error (MISE) of the estimated conditional density $\widehat{f}_{U_{i1},U_{i2}|\gamma_i,\beta_i}$ and the bias, standard deviation (SD), and mean squared error (MSE) of the estimated interquartile range $\widehat{\text{IQR}}(U_{ij} \mid \gamma_i, \beta_i)$, $j = 1, 2$, of the two marginal distributions, all evaluated at the median value of (γ_i, β_i). The regressors X_i and W_{i1} are drawn from a Student-t distribution with five degrees of freedom.

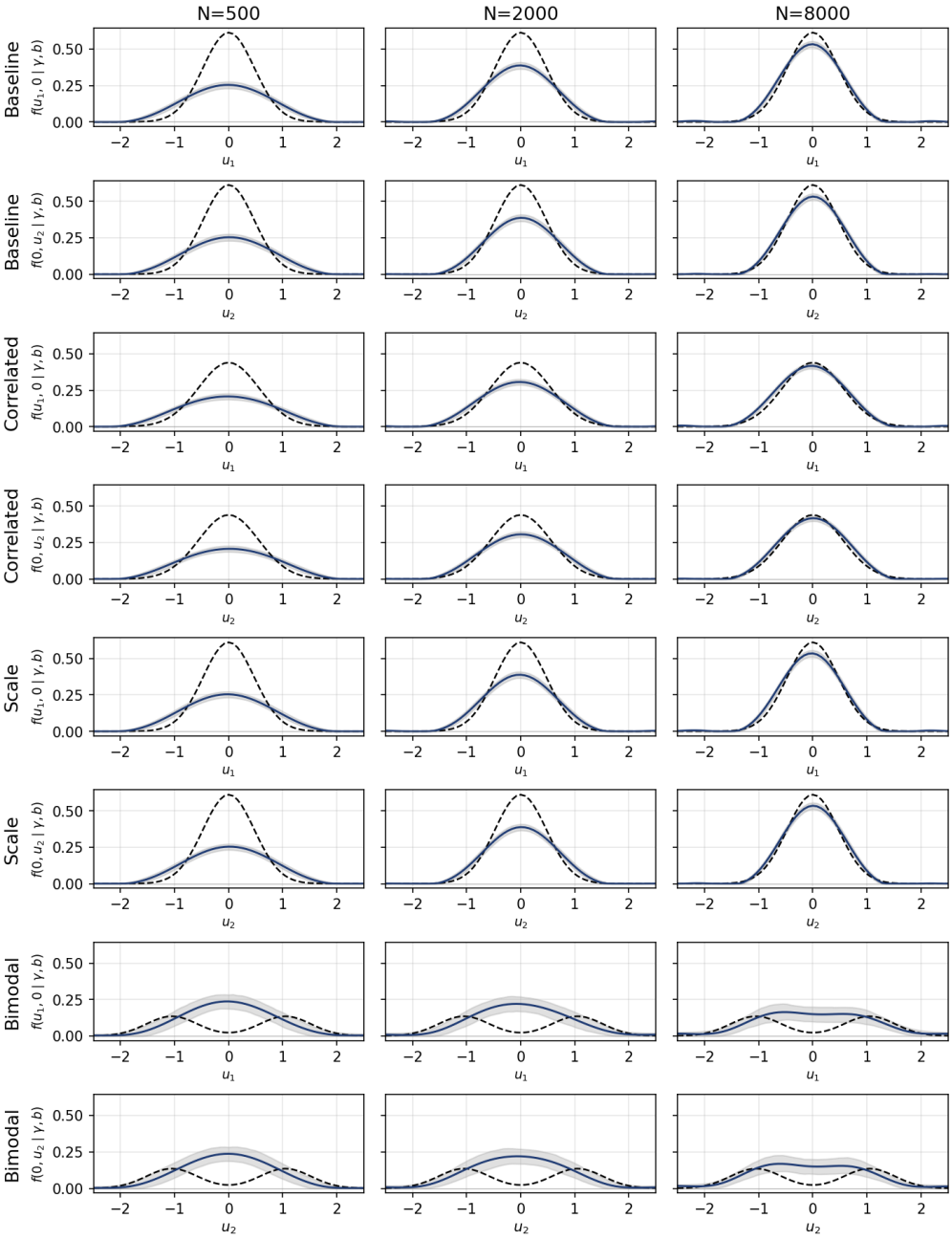


Figure 3: Estimated conditional density of U_i along each axis, heavy-tailed regressors

Notes: $\widehat{f}_{U_{i1},U_{i2}|\gamma_i,\beta_i}$ along u_1 (at $u_2 = 0$) and along u_2 (at $u_1 = 0$), at the median value of (γ_i, β_i), for the heavy-tailed design in which the regressors X_i and W_{i1} are drawn from a Student-t distribution with five degrees of freedom. Each row shows these two curves for one design; columns are the sample sizes $n \in \{500, 2000, 8000\}$. Black dashed: the true density; navy solid: the Monte Carlo mean across $N_{\text{rep}} = 500$ replications; shaded band: the 95% pointwise Monte Carlo confidence band.

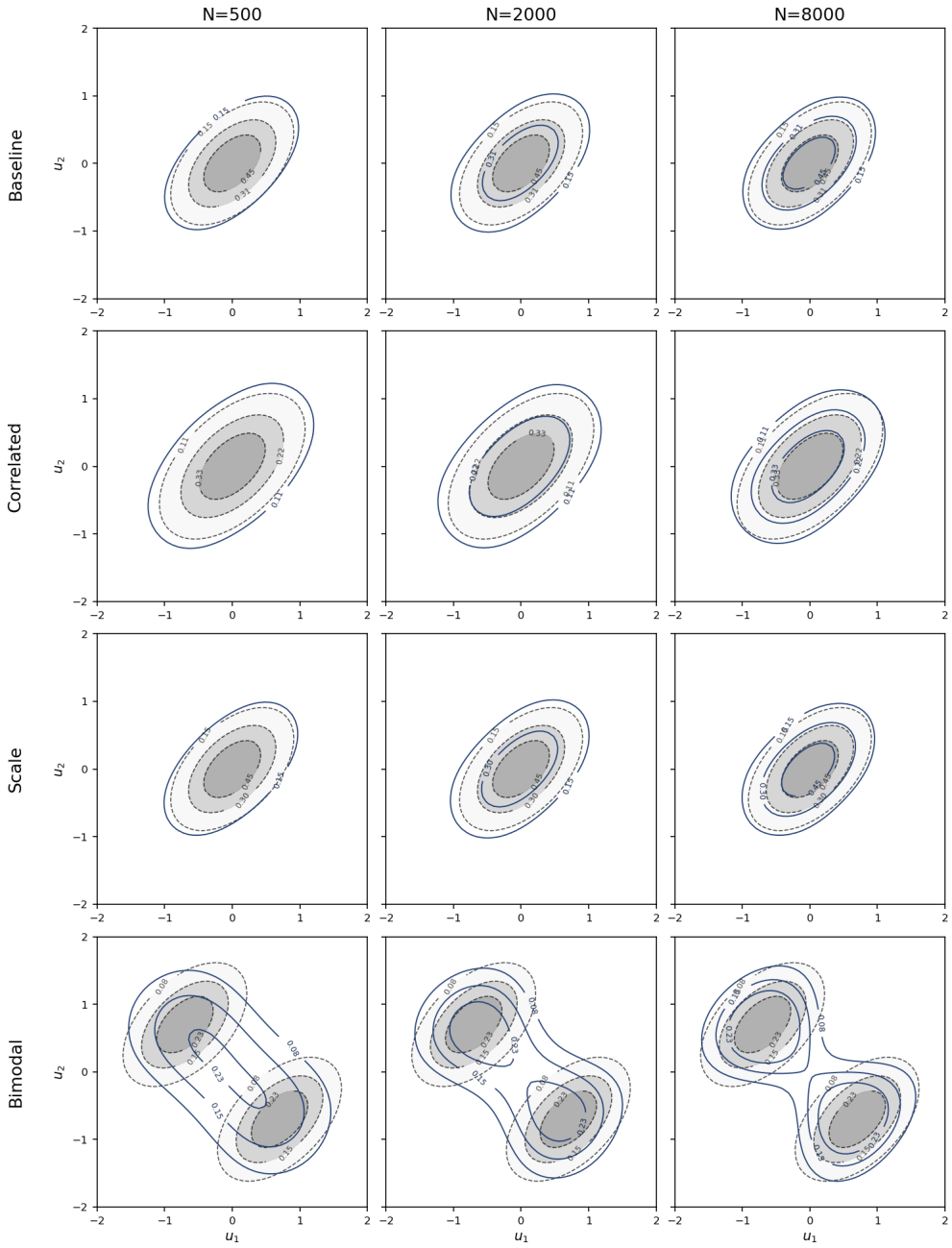


Figure 4: Contour plots of the estimated conditional density of U_i, heavy-tailed regressors

Notes: $\widehat{f}_{U_{i1},U_{i2}|\gamma_i,\beta_i}$ for the four designs (rows) at sample sizes $n \in \{500, 2000, 8000\}$ (columns), conditioning on the median value of (γ_i, β_i), for the heavy-tailed design in which the regressors X_i and W_{i1} are drawn from a Student-t distribution with five degrees of freedom. Gray fill with black dashed contours: the true density; navy solid contours: the Monte Carlo mean across $N_{\text{rep}} = 500$ replications. The contours of the true density enclose 25%, 50%, and 75% of its probability mass, and the Monte Carlo contours use the same levels.